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THE ULTRASPHERICAL BACKGROUND OF THE JACOBI PHASE EXPANSION, IN CLOSED FORM

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ABSTRACT. The large-degree phase of the Jacobi polynomial $P_n^{(\alpha,\beta)}(\cos\theta)$ contains, at each order in $\mathcal{N}^{-1}$, $\mathcal{N} = n + (\alpha+\beta+1)/2$, an additive constant κ_r that the phase equation does not determine. These constants are antisymmetric in (α^2, β^2) and vanish for $\alpha = \beta$. We obtain three results. First, on the diagonal $\alpha = \beta$ the logarithm of the midpoint amplitude is computed in closed form: if L_n denotes the coefficient of $(2\mathcal{N})^{-n}$ in that logarithm, then $\Upsilon_A(t) = \sum_{n\geq 1} L_n t^n/(n-1)!$ equals

$$\Upsilon_A(t) = \frac{e^t\big(\cosh 2\alpha t - \cosh t\big)}{\sinh 2t}.$$

The proof is algebraic: the identity is established for half-odd-integer α, where the expansion terminates, by means of two classical recurrences for Bessel polynomials, and is extended to all α by a degree argument. Second, the derivative of the logarithm of the amplitude with respect to α^2 at the Legendre point $\alpha = \beta = 0$ is computed in closed form to all orders. An exact reduction of an Appell F_3 function to a Gauss function shows that it is a parameter derivative of a zero-balanced ${}_2F_1$ at argument 2; the real part of its expansion is a trigamma series, and the imaginary part is the unique formal solution of an explicit central-difference equation. Third, we deduce that the coefficients w_m of an explicit even power series attached to the imaginary part satisfy $\nu_2(w_m) = -2\nu_2((2m)!)$ for every $m \geq 0$ (ν_2 the 2-adic valuation). This determines, for every r, the exact power of 2 in the denominator of the coefficient of α^2 in the polynomial that represents κ_r. We also give a factorisation of the amplitude into products of pairs of Gauss functions, further valuation formulas, and a numerical illustration for the zeros.

1. INTRODUCTION

We use the standard normalisation and notation for Jacobi polynomials in [1, 2]; general background on hypergeometric and classical special functions may be found in [3]. For $\alpha, \beta > -1$ the Jacobi polynomial $P_n^{(\alpha,\beta)}$ is orthogonal on $[-1,1]$ with respect to the weight $(1-x)^\alpha(1+x)^\beta$, and its n zeros are real, simple and lie in $(-1,1)$; they are the nodes of Gauss–Jacobi quadrature. The ultraspherical case $\alpha = \beta$ contains the Legendre polynomials ($\alpha = \beta = 0$), the Chebyshev polynomials ($\alpha = \beta = \pm\frac{1}{2}$) and the Gegenbauer polynomials. Whenever the zeros of $P_n^{(\alpha,\beta)}$ are considered as an ordered set in $(-1,1)$ (here and in Section 11) we assume $\alpha, \beta > -1$. The formal and hypergeometric identities proved below are polynomial or meromorphic in the parameters and, where explicitly stated, hold beyond this range. Write the zeros as $x_{n,k} = \cos\theta_{n,k}$, $0 < \theta_{n,1} < \cdots < \theta_{n,n} < \pi$, and put

$$\mathcal{N} = n + \frac{\alpha+\beta+1}{2}, \qquad A = \alpha^2, \qquad B = \beta^2.$$

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The starting point is the elementary observation that the zeros are almost equally spaced in the angle θ. For the Chebyshev polynomials this is exact: the zeros of T_n are $\cos\big((k-\frac12)\pi/n\big)$. In general one has, for the zeros away from the endpoints,

$$\theta_{n,k} = \frac{(k-\frac12)\pi + (\frac{\alpha}{2}+\frac14)\pi}{\mathcal{N}} + O(\mathcal{N}^{-2}), \tag{1}$$

and the question addressed by the phase expansion is what the corrections look like, to all orders in $\mathcal{N}^{-1}$. The source of (1) is the Liouville normal form of the Jacobi differential equation: the function $u(\theta) = \sin^{\alpha+\frac12}\frac{\theta}{2}\,\cos^{\beta+\frac12}\frac{\theta}{2}\,P_n^{(\alpha,\beta)}(\cos\theta)$ satisfies

$$u''(\theta) + \Big[\mathcal{N}^2 + \frac{\frac14 - A}{4\sin^2\frac{\theta}{2}} + \frac{\frac14 - B}{4\cos^2\frac{\theta}{2}}\Big]u(\theta) = 0 \tag{2}$$

[1, Eq. (4.24.2)]. For $A = B = \frac14$ this is the equation of a pure sinusoid of frequency $\mathcal{N}$; in general u is a sinusoid of frequency $\mathcal{N}$ whose phase is slowly deformed by the potential, and the zeros of $P_n^{(\alpha,\beta)}$ are the points where that phase crosses the values $(k-\frac12)\pi$. Note that the equation (2) depends on the parameters only through A and B; this is why all the coefficients considered in this paper are polynomials in A and B.

Solving the phase equation associated with (2) recursively in powers of $\mathcal{N}^{-2}$, one obtains a phase function

$$\Xi(\theta) = \mathcal{N}\theta - \Big(\frac{\alpha}{2}+\frac14\Big)\pi + \sum_{r\ge1}\frac{\Psi_r(\theta)}{\mathcal{N}^{2r-1}}, \tag{3}$$

such that the zeros are, to all orders in $1/\mathcal{N}$, the solutions of $\Xi(\theta_{n,k}) = (k-\frac12)\pi$. The first term is elementary,

$$\Psi_1(\theta) = \frac{(1-4B)\tan\frac{\theta}{2} - (1-4A)\cot\frac{\theta}{2}}{16},$$

and already very effective: for $n = 20$, $\alpha = \frac32$, $\beta = \frac14$, the maximal error of (1) on the ten central zeros is $2.2\cdot10^{-3}$, and adding the term $\Psi_1(\theta)/\mathcal{N}$ reduces it to $3.9\cdot10^{-6}$. Each Ψ_r is obtained from the recursion by an integration and is therefore determined only up to an additive constant. We normalise Ψ_r to be an odd Laurent polynomial in $\cot(\theta/2)$, without constant term; the information not fixed by the differential equation is then carried by the value at the midpoint,

$$\kappa_r = \Psi_r(\pi/2).$$

The numbers κ_r, which are polynomials in A and B with rational coefficients, are the only part of (3) that the phase equation does not produce. We call them the *phase constants*. The first three are

$$\kappa_1 = \frac{A-B}{4}, \qquad \kappa_2 = \frac{(A-B)(2A+2B-7)}{96}, \tag{4}$$
$$\kappa_3 = \frac{(A-B)(28A^2+88AB+28B^2-380A-380B+781)}{7680}.$$

Two features of (4) persist for all r and motivate the paper. The first is antisymmetry. Since $P_n^{(\alpha,\beta)}(-x) = (-1)^n P_n^{(\beta,\alpha)}(x)$, reflection at $\theta = \pi/2$ interchanges α and β and reverses the sign of the phase, so

$$\kappa_r(B,A) = -\kappa_r(A,B), \qquad \text{hence} \qquad (A-B)\mid\kappa_r. \tag{5}$$

In particular κ_r vanishes identically on the diagonal $A = B$, which contains the ultraspherical case $\alpha = \beta$: the Legendre and Gegenbauer cases need no phase constants. The nontrivial content of (3) is therefore transverse to the diagonal. On the diagonal itself the phase constants vanish but the series from which they are extracted (the amplitude, introduced below) does not; it is a nontrivial asymptotic series. Our first question is whether this series can be computed in closed form. The second feature is arithmetic: the denominators $4 = 2^2$, $96 = 2^5 \cdot 3$, $7680 = 2^9 \cdot 15$ and, for κ_4, $86016 = 2^{12} \cdot 21$ grow mainly through powers of 2. In [17] it is showed that the power of 2 in the denominator of κ_r is at most 2^{E_r} with $E_r = 3r - 1 + \nu_2((r-1)!)$, where ν_2 denotes the 2-adic valuation. Our second question is whether this bound is attained for every r. Both questions are answered in this paper: the diagonal series is the Stirling series of a quotient of Gamma functions (Theorem 1.1 and Corollary 1.2), and the bound 2^{E_r} is sharp for every r (Theorem 1.5). The proof of the second statement uses the first: the closed form on the diagonal is the background over which the first-order variation transverse to the diagonal is computed.

Large-degree asymptotics for Jacobi polynomials and functions have a long history, from the classical trigonometric and uniform expansions of Hahn [9] and Frenzen–Wong [10] to Riemann–Hilbert treatments of Jacobi-type weights [12], fast computation of Gauss–Jacobi nodes and weights [11, 14], nonoscillatory phase-function methods [13], explicit error bounds [16] and, closest to our starting point, the inverse-factorial large-degree expansions of Nemes [15] for Jacobi and related functions with fixed parameters, which carry explicit computable error bounds, are built from the same Hankel coefficients as (6) below, and terminate at half-odd-integer parameters. These works develop the expansions, their error bounds and the associated algorithms. The present paper addresses a different question: the exact algebraic and arithmetic structure of the coefficients of the phase expansion, and in particular of the constants κ_r.

We now state the main results, which are proved in the following sections. Let

$$a_k(\nu) = \frac{(\frac{1}{2}+\nu)_k(\frac{1}{2}-\nu)_k}{(-2)^k k!} = \frac{\prod_{j=0}^{k-1}\bigl(4\nu^2 - (2j+1)^2\bigr)}{8^k k!}$$

be the Hankel coefficients, and put $Z = 2\mathcal{N}$. The large-degree behaviour of the Jacobi polynomials at the midpoint $\theta = \pi/2$ is carried by the inverse-factorial series

$$F(Z) = \sum_{n\ge 0} \frac{G_n}{(Z-1)(Z-2)\cdots(Z-n)}, \tag{6}$$

$$G_n = \sum_{l=0}^{n} a_l(\alpha) a_{n-l}(\beta)\, p^l q^{n-l}, \qquad p = 1+i, \quad q = 1-i,$$

which we call the *amplitude*. Write $\log F = \sum_{n\ge 1} L_n Z^{-n}$; the coefficients L_n are polynomials in A and B. The phase constants are the imaginary parts of the odd coefficients,

$$\kappa_r = 2^{-(2r-1)} \operatorname{Im} L_{2r-1}. \tag{7}$$

Formula (7) is the definition of κ_r used in this paper; its agreement with the constants $\Psi_r(\pi/2)$ of (3) is discussed in Proposition 7.37.

Our first result concerns the diagonal $A = B = \alpha^2$. Let L_n^{diag} be the value of L_n there, and let

$$\Upsilon_A(t) = \sum_{n\ge 1} \frac{L_n^{\mathrm{diag}}}{(n-1)!}\, t^n$$

be the associated generating function. We write $\mathcal{B}_k(x)$ for the Bernoulli polynomials and $\mathcal{B}_k = \mathcal{B}_k(0)$ for the Bernoulli numbers, so that $\mathcal{B}_1 = -\frac{1}{2}$, and E_{2k} for the Euler numbers. As usual, $(a)_k = a(a+1)\cdots(a+k-1)$ is the Pochhammer symbol, $\psi = \Gamma'/\Gamma$ is the digamma function, Li_k is the polylogarithm, and $[x^k]f$ denotes the coefficient of x^k in a power series f.

Theorem 1.1. *For every α,*

$$\Upsilon_A(t) = \frac{e^t\big(\cosh 2\alpha t - \cosh t\big)}{\sinh 2t} = \tfrac{1}{2}\big(\operatorname{csch} t + \operatorname{sech} t\big)\big(\cosh 2\alpha t - \cosh t\big), \qquad A = \alpha^2. \tag{8}$$

Equivalently, coefficientwise,

$$L_n^{\mathrm{diag}} = \frac{4^n}{n(n+1)}\Big[\mathcal{B}_{n+1}\big(\tfrac{3+2\alpha}{4}\big) + \mathcal{B}_{n+1}\big(\tfrac{3-2\alpha}{4}\big) - \mathcal{B}_{n+1}(1) - \mathcal{B}_{n+1}\big(\tfrac{1}{2}\big)\Big]. \tag{9}$$

The proof occupies Sections 2 and 3. It does not use any asymptotic representation of the Jacobi polynomials. The identity is first established for $\alpha = \ell + \frac{1}{2}$ with ℓ a non-negative integer, where (6) terminates, by an induction based on two classical recurrences for Bessel polynomials; a degree bound in A then extends it to all α.

Corollary 1.2. *On the diagonal, as an identity of formal Laurent series at $Z = \infty$,*

$$F(Z+4)\,Z(Z+2) = F(Z)\big((Z+1)^2 - 4A\big).$$

Equivalently, $\mathcal{G}(Z) := \Gamma(Z)F(Z)$ satisfies, in the same formal sense,

$$\mathcal{G}(Z+4) = (Z+1)(Z+3)\big((Z+1)^2 - 4A\big)\mathcal{G}(Z).$$

For half-odd-integer α the first identity is an identity of rational functions of Z and the second one of meromorphic functions, by Theorem 2.10.

Corollary 1.3.

(i) $\Upsilon_A = \upsilon_1 \cosh 2\alpha t + \upsilon_0$ *with* $\upsilon_0(t)$, $\upsilon_1(t)$ *independent of* α*; coefficientwise, for* $k \ge 1$, $[A^k]\Upsilon_A = \frac{2^{2k-1}}{(2k)!} t^{2k-2}[A^1]\Upsilon_A$.

(ii) $\Upsilon_{1/4} \equiv 0$*: there is no correction in the Chebyshev cases* $\alpha = \beta = \pm\frac{1}{2}$.

(iii) In the Legendre case, $\Upsilon_0(t) = \frac{1}{2}\big(\operatorname{sech} t - \tanh\frac{t}{2} - 1\big)$.

The second group of results concerns the first-order variation of the amplitude at the Legendre point $A = B = 0$. Let

$$g_n = \frac{\partial L_n}{\partial A}\Big|_{A=B=0} \in \mathbb{Q}(i), \qquad \Phi(t) = \sum_{n\ge 1} \frac{g_n}{(n-1)!}\, t^n. \tag{10}$$

By (7), the coefficient of the monomial A in κ_r is

$$[A^1B^0]\kappa_r = 2^{-(2r-1)}\,\mathrm{Im}\, g_{2r-1} \qquad (r \ge 1). \tag{11}$$

We call the sequence (g_n), or its generating series $\sum_n g_n Z^{-n}$, the *Legendre tangent* of the amplitude: it is the derivative of $\log F$ in the direction of A at the Legendre point. At the midpoint, the derivative of $\log F$ along the diagonal is real and the derivative in the direction of $A - B$, transverse to the diagonal, is purely imaginary (Section 9). Thus

$\operatorname{Re} g_n$ is the diagonal part of the tangent and $\operatorname{Im} g_n$ its transverse part. The real parts are determined by Theorem 1.1:

Corollary 1.4. *With g_n and Φ as in (10),*

$$\operatorname{Re}\Phi(t) = \frac{t^2 e^t}{\sinh 2t},$$

that is, $\operatorname{Re} g_{2k+1} = (1-2^{2k-1})\mathcal{B}_{2k}$ *and* $\operatorname{Re} g_{2k} = \frac{1}{2}(2k-1)\mathsf{E}_{2k-2}$. *In particular* $\nu_2(\operatorname{Re} g_n) = -1$ *for every* $n \geq 1$.

Here and below $\mathbb{Z}_2$ denotes the ring of 2-adic integers, $\mathbb{Z}_2^\times$ its group of units, and $s_2(m)$ the number of digits 1 in the binary expansion of m, so that $\nu_2(m!) = m - s_2(m)$. The imaginary parts of the g_n are the substantial part. Put

$$W(t) = \frac{\sinh 2t}{t^2}\operatorname{Im}\Phi(t), \qquad w_m = [t^{2m}]\,W(t), \tag{12}$$

so that, by Corollary 1.4, $\sinh(2t)\,\Phi(t)/t^2 = e^t + i\,W(t)$, and

$$H_r = r + \nu_2\big((r-1)!\big), \qquad E_r = (2r-1) + H_r = 3r - 1 + \nu_2\big((r-1)!\big).$$

Theorem 1.5. *For every $k \geq 1$, $\operatorname{Im} g_{2k} = 0$, so that W is an even power series. For every $m \geq 0$,*

$$\nu_2(w_m) = -2\,\nu_2\big((2m)!\big), \qquad \textit{that is,} \qquad \big((2m)!\big)^2 w_m \in \mathbb{Z}_2^\times.$$

Equivalently, for every $r \geq 1$,

$$\nu_2\big(\operatorname{Im} g_{2r-1}\big) = -H_r, \qquad \nu_2\big([A^1B^0]\kappa_r\big) = -E_r,$$

with κ_r defined by (7).

Theorem 1.5 is proved in Subsection 7.7. The equivalence between the statement for w_m and the statement for $[A^1B^0]\kappa_r$ was established in [17], where the property $((2m)!)^2 w_m \in \mathbb{Z}_2^\times$ was verified in a finite range; the argument is repeated in the proof of Theorem 7.36. The proof of Theorem 1.5 rests on a closed form for the whole sequence (g_n):

Theorem 1.6. *As formal series in Z^{-1},*

$$\sum_{n\geq 1} g_n\, Z^{-n} = \frac{1}{16}\,\psi'\Big(\frac{Z+1}{4}\Big) - \frac{1}{16}\,\psi'\Big(\frac{Z+3}{4}\Big) + \frac{1}{4}\,\psi'\Big(\frac{Z+1}{2}\Big) - \frac{i}{2}\,\widehat{Q}_\Gamma(Z), \tag{13}$$

where each trigamma function ψ' stands for its asymptotic (Bernoulli) expansion, and $\widehat{Q}_\Gamma$ is the unique formal series without constant term that solves the central-difference equation

$$\widehat{Q}_\Gamma(s+1) - \widehat{Q}_\Gamma(s-1) = -\frac{(2s-1)}{4s(s-1)}\left[\frac{\Gamma(s/2)}{\Gamma((1+s)/2)}\right]^2, \qquad s = \frac{1-Z}{2},$$

the right-hand side being replaced by its asymptotic expansion in powers of s^{-1}. In particular

$$\operatorname{Im} g_n = -\tfrac{1}{2}\,\widehat{q}_n \qquad (n \geq 1), \tag{14}$$

where $\widehat{q}_n$ is the coefficient of Z^{-n} in $\widehat{Q}_\Gamma$.

The proof of Theorem 1.6 proceeds as follows. The amplitude (6) is a Kampé de Fériet double series (Lemma 7.2), in fact an Appell F_3 function evaluated on a curve on which it reduces to a single Gauss function (Theorem 7.3). Consequently $\sum_n g_n Z^{-n}$ is the expansion of a second parameter derivative of a zero-balanced ${}_2F_1$ at the boundary value $2+i0$ (Corollary 7.4). The classical connection formula splits this boundary value into two blocks which are quotients of Gamma functions at the Legendre point (Proposition 7.7); the variation of one of them satisfies a first-order difference equation which is solved by quadratures (Theorem 7.9); and the passage from the resulting convergent expressions to the formal series in Z^{-1} is made by an asymptotic matching along a vertical line (Subsection 7.6), whose estimates are given in Appendix A. Theorem 1.5 then follows from the 2-adic analysis of the explicit series $\widehat{Q}_\Gamma$ (Subsection 7.7): the exponent of the Gamma quotient has Genocchi-number coefficients, and in the expansion of its exponential exactly one term has minimal valuation.

Besides these results the paper contains further exact structure of the amplitude. The recurrences for Bessel polynomials used in the proof of Theorem 1.1 do not require the special relation $p^2+q^2=0$ satisfied at the midpoint (Theorem 4.1); on the whole curve described by the weights p, q as θ varies, where $p+q=pq$, the half-odd-integer amplitudes satisfy a three-term contiguity whose numerators are given in closed form in terms of ballot numbers (Theorems 5.2 and 5.4), and the derivatives of all orders normal to that curve are determined recursively (Theorem 6.4). For independent half-odd-integer parameters α, β the amplitudes satisfy a similar recurrence off the diagonal (Theorem 7.1). The amplitude is an explicit sum of products of pairs of Gauss functions, the r-th term being of order Z^{-3r} (Corollary 8.2); from this expansion we derive four valuation formulas for the numerator and the denominator of $\sum_n g_n Z^{-n}$ (Theorem 10.3 and Proposition 10.4). Section 11 illustrates the use of (3) for the numerical computation of zeros.

Two statements of the paper are not proved in full generality, and neither is used in the proofs of Theorems 1.1, 1.5 and 1.6. The first is the agreement of (7) with the constants $\Psi_r(\pi/2)$ of (3), which is proved for $r \le 8$ (Proposition 7.37); for $r > 8$ the last assertion of Theorem 1.5 is a statement about the polynomials defined by (7). The second is a periodicity hypothesis under which the function whose expansion is $\sum_n g_n Z^{-n}$ is evaluated in closed form on the real axis (Proposition 7.28).

Throughout the paper, identities in Z^{-1} (equivalently in $v = Z^{-1}$ or in s^{-1}, $s = (1-Z)/2$) are identities of formal Laurent series unless convergence is explicitly asserted. When a meromorphic function is introduced, its expansion for large Z is always understood in the stated sector. For half-odd-integer parameters the identities are identities of rational functions or of quotients of Gamma functions.

The paper is organised as follows. Sections 2 and 3 prove Theorem 1.1 and its corollaries. Sections 4, 5 and 6 treat the half-odd-integer amplitudes away from the midpoint. Section 7 contains the proofs of Theorems 1.6 and 1.5: the Appell reduction (Subsection 7.1), the boundary split and the quadratures (Subsections 7.2 and 7.3), the asymptotic matching (Subsection 7.6) and the arithmetic (Subsection 7.7); Subsections 7.4 and 7.5 evaluate the boundary value exactly on the real axis and are used only in Proposition 7.28. The proofs of Theorems 1.6 and 1.5 are contained in Section 7 and Appendix A; apart from the notation of the Introduction and Lemma 3.1, they do not depend on Sections 4–6 or 8–11. Section 8 gives the expansion of the amplitude into products of Gauss functions, Section 9 the linear system satisfied by the transverse variation, and Section 10

the valuation formulas. Section 11 contains the numerical illustration, and Appendix A the estimates used in Subsection 7.6.

2. The half-odd-integer family

2.1. **Bessel polynomials at the midpoint.** Write y_n for the Bessel polynomials

$$y_n(x) = \sum_{k=0}^{n} \frac{(n+k)!}{(n-k)!\,k!\,2^k}\,x^k.$$

Lemma 2.1. *For $\alpha = \ell + \frac{1}{2}$,*

$$a_k\big(\ell+\tfrac12\big) = \frac{(\ell+k)!}{(\ell-k)!\,k!\,2^k} \quad (0 \le k \le \ell), \qquad a_k\big(\ell+\tfrac12\big) = 0 \quad (k > \ell),$$

so that $\sum_k a_k(\ell+\frac12)\xi^k = y_\ell(\xi)$.

Proof. Since $4\alpha^2 = (2\ell+1)^2$, the product in the definition of a_k is

$$\prod_{j<k}\big((2\ell+1)^2 - (2j+1)^2\big) = \prod_{j<k}(2\ell-2j)(2\ell+2j+2) = 4^k \frac{\ell!}{(\ell-k)!}\frac{(\ell+k)!}{\ell!}$$

for $k \le \ell$, and it vanishes for $k > \ell$. Dividing by $8^k k!$ gives the stated value. □

We write $F_\ell(Z)$ for the series (6) on the diagonal at $\alpha = \beta = \ell + \frac12$, and $G_n^{(\ell)}$ for its coefficients. By Lemma 2.1, the generating function of these coefficients is the *polynomial*

$$\Pi_\ell(\xi) := \sum_n G_n^{(\ell)}\xi^n = y_\ell(p\xi)\,y_\ell(q\xi), \qquad p = 1+i,\ q = 1-i,$$

of degree 2ℓ; in particular $G_n^{(\ell)} = 0$ for $n > 2\ell$ and $F_\ell(Z)$ is a rational function of Z.

The four relations we use among the midpoint constants are

$$pq = 2, \qquad p+q = 2, \qquad p^2+q^2 = 0, \qquad \frac1p + \frac1q = 1. \tag{15}$$

The third relation is the one used in the proof of Proposition 2.5.

2.2. **Two classical recurrences, and their consequence.**

Lemma 2.2. *For $n \ge 0$, with the convention $y_{-1}(x) = 1$,*

(a) $x^2 y_n'(x) = (nx-1)y_n(x) + y_{n-1}(x)$, *(b)* $y_{n+1}(x) = (2n+1)x\,y_n(x) + y_{n-1}(x)$.

Proof. Both identities follow by comparing coefficients in the defining sum; they are classical [4, 5, 6]. □

Definition 2.3. $C_\ell(\xi) = \frac12\Big(q\,y_{\ell-1}(p\xi)y_\ell(q\xi) + p\,y_\ell(p\xi)y_{\ell-1}(q\xi)\Big)$, which is real because the two terms are complex conjugate.

Proposition 2.4. $\xi^2\Pi_\ell'(\xi) + \big(1-2\ell\xi\big)\Pi_\ell(\xi) = C_\ell(\xi)$.

Proof. Put $U = y_\ell(p\xi)$, $V = y_\ell(q\xi)$, $U_- = y_{\ell-1}(p\xi)$, $V_- = y_{\ell-1}(q\xi)$. Lemma 2.2(a) at $x = p\xi$ reads $p\,\xi^2 U' = (\ell p\xi - 1)U + U_-$, and similarly for V. Hence

$$\xi^2\Pi'_\ell = \xi^2(U'V + UV') = \frac{(\ell p\xi - 1)U + U_-}{p}\,V + U\,\frac{(\ell q\xi - 1)V + V_-}{q}$$
$$= \Big(2\ell\xi - \frac{1}{p} - \frac{1}{q}\Big)UV + \frac{U_-V}{p} + \frac{UV_-}{q}.$$

By (15), $\frac{1}{p} + \frac{1}{q} = 1$, $\frac{1}{p} = \frac{q}{2}$ and $\frac{1}{q} = \frac{p}{2}$, so the right-hand side is $(2\ell\xi - 1)\Pi_\ell + C_\ell$. □

Proposition 2.5. $C_{\ell+2} = C_\ell + (4\ell + 2)\,\xi\,\Pi_\ell$.

Proof. With $U_+ = y_{\ell+1}(p\xi)$, $U_{++} = y_{\ell+2}(p\xi)$ and similarly for V, Lemma 2.2(b) gives $U_{++} = (2\ell + 3)p\xi U_+ + U$ and $V_{++} = (2\ell + 3)q\xi V_+ + V$. Therefore

$$2C_{\ell+2} = qU_+V_{++} + pU_{++}V_+ = (2\ell + 3)\xi\,(p^2 + q^2)\,U_+V_+ \;+\; qU_+V + pUV_+,$$

and the first term vanishes by (15). Applying Lemma 2.2(b) once more, $U_+ = (2\ell + 1)p\xi U + U_-$ and $V_+ = (2\ell + 1)q\xi V + V_-$, so

$$qU_+V + pUV_+ = 2(2\ell + 1)pq\,\xi\,UV + \big(qU_-V + pUV_-\big) = 4(2\ell + 1)\xi\,\Pi_\ell + 2C_\ell,$$

using $pq = 2$. Dividing by 2 gives the claim. □

Corollary 2.6. $\xi^2\Pi'_{\ell+2} + \big[1 - (2\ell + 4)\xi\big]\Pi_{\ell+2} = \xi^2\Pi'_\ell + \big[1 + (2\ell + 2)\xi\big]\Pi_\ell$.

Proof. By Proposition 2.4 the left side is $C_{\ell+2}$ and the right side is $C_\ell + (4\ell + 2)\xi\Pi_\ell$; apply Proposition 2.5. □

2.3. **From the generating function to the inverse-factorial series.** Let $e_n(Z) = 1/\big((Z - 1)\cdots(Z - n)\big)$, $e_0 = 1$.

Lemma 2.7. $Z\,e_n = e_{n-1} + n\,e_n$ *for* $n \geq 1$.

Proof. $Ze_n = \big((Z - n) + n\big)e_n$ and $(Z - n)e_n = e_{n-1}$. □

Proposition 2.8. $F_{\ell+2}(Z)\,(Z - 2\ell - 4) = F_\ell(Z)\,(Z + 2\ell + 2)$.

Proof. Write $F_\ell = \sum_n G_n^{(\ell)} e_n$ with $G_0^{(\ell)} = 1$. By Lemma 2.7, for any constant c,

$$(Z - c)F_\ell = Z - c + G_1^{(\ell)} + \sum_{m\geq 1}\Big(G_{m+1}^{(\ell)} + (m - c)G_m^{(\ell)}\Big)e_m.$$

Since $G_0^{(\ell)} = 1$ for every ℓ, the terms in Z agree on the two sides of the claimed identity, and comparing coefficients of e_m (and the constant, which is the case $m = 0$) shows that the identity is equivalent to

$$G_{m+1}^{(\ell+2)} + (m - 2\ell - 4)G_m^{(\ell+2)} = G_{m+1}^{(\ell)} + (m + 2\ell + 2)G_m^{(\ell)} \qquad (m \geq 0).$$

Multiplying by ξ^m and summing, and using $\sum_m G_{m+1}\xi^m = (\Pi - 1)/\xi$ and $\sum_m mG_m\xi^m = \xi\Pi'$, this is exactly Corollary 2.6 after multiplication by ξ. □

Lemma 2.9. $F_0 = 1$ *and* $F_1(Z) = \dfrac{Z}{Z - 2}$.

Proof. $y_0(x) = 1$ gives $\Pi_0 = 1$, so $G_0 = 1$ and $G_n = 0$ for $n \geq 1$. $y_1(x) = 1 + x$ gives $\Pi_1(\xi) = (1 + p\xi)(1 + q\xi) = 1 + 2\xi + 2\xi^2$, so

$$F_1 = 1 + \frac{2}{Z-1} + \frac{2}{(Z-1)(Z-2)} = \frac{Z^2 - Z}{(Z-1)(Z-2)} = \frac{Z}{Z-2}.$$

□

Theorem 2.10. *For every $\ell \geq 0$, with $\zeta = Z/4$,*

$$F_\ell(Z) = \mathcal{M}_\ell(Z) := \frac{\Gamma\big(\zeta - \frac{\ell}{2}\big)\,\Gamma\big(\zeta + \frac{\ell+1}{2}\big)}{\Gamma(\zeta)\,\Gamma\big(\zeta + \frac{1}{2}\big)}.$$

Proof. By the functional equation of Γ,

$$\frac{\mathcal{M}_{\ell+2}}{\mathcal{M}_\ell} = \frac{\Gamma(\zeta - \frac{\ell}{2} - 1)}{\Gamma(\zeta - \frac{\ell}{2})} \cdot \frac{\Gamma(\zeta + \frac{\ell+3}{2})}{\Gamma(\zeta + \frac{\ell+1}{2})} = \frac{\zeta + \frac{\ell+1}{2}}{\zeta - \frac{\ell+2}{2}} = \frac{Z + 2\ell + 2}{Z - 2\ell - 4},$$

which is Proposition 2.8. For the base cases, $\mathcal{M}_0 = 1$ and

$$\mathcal{M}_1 = \frac{\Gamma(\zeta - \frac{1}{2})}{\Gamma(\zeta + \frac{1}{2})} \frac{\Gamma(\zeta + 1)}{\Gamma(\zeta)} = \frac{\zeta}{\zeta - \frac{1}{2}} = \frac{Z}{Z-2},$$

matching Lemma 2.9. Induction in steps of two along each parity class completes the proof. □

3. From half-integers to all α

Lemma 3.1. *For arbitrary weights p, q, each coefficient L_n of $\log F$ is a polynomial in (A, B) of total degree at most n. In particular $\deg_A L_n^{\mathrm{diag}} \leq n$. Moreover the right-hand side of (9) is a polynomial in A of degree at most $\lfloor (n+1)/2 \rfloor \leq n$.*

Proof. $a_k(\nu)$ is a polynomial in ν^2 of degree k, so $a_l(\alpha) a_{n-l}(\beta)$, and hence G_n, is a polynomial in (A, B) of total degree at most n. Since

$$e_n = \sum_{m \geq n} \left\{ {m \atop n} \right\} Z^{-m}$$

with $\left\{ {m \atop n} \right\}$ the Stirling numbers of the second kind, the coefficients

$$c_m = \sum_{n \leq m} G_n \left\{ {m \atop n} \right\}$$

of $F = 1 + \sum_m c_m Z^{-m}$ have total degree at most m, and each L_n is an isobaric polynomial of weight n in the c_m, whence the first two assertions. On the other side $\mathcal{B}_{n+1}\big(\frac{3+2\alpha}{4}\big) + \mathcal{B}_{n+1}\big(\frac{3-2\alpha}{4}\big)$ is even in α of degree at most $n+1$, hence a polynomial in $A = \alpha^2$ of degree at most $\lfloor (n+1)/2 \rfloor$. □

Lemma 3.2. *Put*

$$\mathcal{M}(\zeta; \alpha) = \frac{\Gamma\big(\zeta + \frac{1-2\alpha}{4}\big)\Gamma\big(\zeta + \frac{1+2\alpha}{4}\big)}{\Gamma(\zeta)\,\Gamma\big(\zeta + \frac{1}{2}\big)}.$$

Then, for all α and $\zeta \to \infty$,

$$\log \mathcal{M}(\zeta; \alpha) \sim \sum_{n \geq 1} \frac{\chi_n(\alpha)}{n(n+1)}\, \zeta^{-n},$$

$$\chi_n(\alpha) = \mathcal{B}_{n+1}\big(\tfrac{3+2\alpha}{4}\big) + \mathcal{B}_{n+1}\big(\tfrac{3-2\alpha}{4}\big) - \mathcal{B}_{n+1}(1) - \mathcal{B}_{n+1}\big(\tfrac{1}{2}\big).$$

Moreover $\mathcal{M}_\ell$ of Theorem 2.10 is the value of the left-hand quotient at $\alpha = \ell + \frac{1}{2}$.

Proof. The classical gamma-quotient expansion, obtained by subtracting [2, Eq. 5.11.8] at $h = a$ and $h = b$ (see also [2, Sec. 24.2] for the Bernoulli polynomials),

$$\log\frac{\Gamma(\zeta+a)}{\Gamma(\zeta+b)} - (a-b)\log\zeta \sim \sum_{n\ge1}\frac{(-1)^{n+1}}{n(n+1)}\big(\mathcal{B}_{n+1}(a) - \mathcal{B}_{n+1}(b)\big)\zeta^{-n}$$

applies with $a = \frac{1\mp2\alpha}{4}$ and $b \in \{0, \frac{1}{2}\}$; the logarithmic terms cancel because $\frac{1-2\alpha}{4} + \frac{1+2\alpha}{4} = \frac{1}{2} = 0 + \frac{1}{2}$. Using $(-1)^{n+1}\mathcal{B}_{n+1}(a) = \mathcal{B}_{n+1}(1-a)$ and $1 - \frac{1\mp2\alpha}{4} = \frac{3\pm2\alpha}{4}$, $1-0=1$, $1-\frac{1}{2}=\frac{1}{2}$ gives the stated form. For the second assertion, at $\alpha = \ell + \frac{1}{2}$ one has $\frac{1-2\alpha}{4} = -\frac{\ell}{2}$ and $\frac{1+2\alpha}{4} = \frac{\ell+1}{2}$. □

Proof of Theorem 1.1. Fix n. By Lemma 3.1 both sides of (9) are polynomials in A of degree at most n. By Theorem 2.10, $F = \mathcal{M}_\ell$ at $\alpha = \ell + \frac{1}{2}$ for every $\ell \ge 0$; both sides being rational functions of Z there, their expansions at $Z = \infty$ coincide, and by Lemma 3.2 the coefficients of $\log \mathcal{M}_\ell$ are the right-hand side of (9) evaluated at $A = (\ell + \frac{1}{2})^2$. Hence the two polynomials agree at the infinitely many distinct points $A = (\ell + \frac{1}{2})^2$, $\ell = 0, 1, 2, \dots$, and therefore coincide. This proves (9).

For (8), sum (9) with the generating function $\sum_k \mathcal{B}_k(x)u^k/k! = ue^{xu}/(e^u-1)$ at $u = 4t$. The constant and linear corrections cancel because $\frac{3+2\alpha}{4} + \frac{3-2\alpha}{4} - 1 - \frac{1}{2} = 0$, and one obtains

$$\Upsilon_A(t) = \frac{e^{(3+2\alpha)t} + e^{(3-2\alpha)t} - e^{4t} - e^{2t}}{e^{4t}-1} = \frac{2e^{3t}\big(\cosh 2\alpha t - \cosh t\big)}{2e^{2t}\sinh 2t},$$

which is (8). □

Proof of Corollary 1.2. By Theorem 1.1 and Lemma 3.2, the formal Laurent series $\log F$ on the diagonal is the Stirling series of $\Gamma(\zeta + \frac{1-2\alpha}{4})\Gamma(\zeta + \frac{1+2\alpha}{4})/\big(\Gamma(\zeta)\Gamma(\zeta+\frac{1}{2})\big)$ with $\zeta = Z/4$, whose ratio at $\zeta \mapsto \zeta+1$ is $\big(\zeta+\frac{1-2\alpha}{4}\big)\big(\zeta+\frac{1+2\alpha}{4}\big)/\big(\zeta(\zeta+\frac{1}{2})\big) = \big((Z+1)^2-4A\big)/\big(Z(Z+2)\big)$. Thus the ratio identity is first obtained coefficientwise in Z^{-1}. Exponentiating is legitimate because both formal logarithms have zero constant term, and gives the first displayed contiguity identity. Multiplying by the exact quotient $\Gamma(Z+4)/\Gamma(Z) = Z(Z+1)(Z+2)(Z+3)$ gives the second identity. □

Remark 3.3. Conversely, the contiguity of Corollary 1.2 determines (8). For $f = \sum_{n\ge1} c_n Z^{-n}$ put $\mathfrak{B}f(t) = \sum_n c_n t^{n-1}/(n-1)!$; then $\mathfrak{B}[f(Z+c)] = e^{-ct}\mathfrak{B}f$ and $\mathfrak{B}\big[\log\frac{Z+a}{Z+b}\big] = (e^{-bt}-e^{-at})/t$, both term by term. Applying $\mathfrak{B}$ to $\log F(Z+4) - \log F(Z) = \log\frac{Z+1-2\alpha}{Z} + \log\frac{Z+1+2\alpha}{Z+2}$ gives $(e^{-4t}-1)\Upsilon_A(t)/t = \big[1 + e^{-2t} - 2e^{-t}\cosh 2\alpha t\big]/t$, and multiplying numerator and denominator by e^{2t} yields (8).

Proof of Corollaries 1.3 and 1.4. (i) is $\cosh 2\alpha t = \sum_k (4A)^k t^{2k}/(2k)!$ read off (8); (ii) is $\cosh 2\alpha t = \cosh t$ at $\alpha = \pm\frac{1}{2}$; (iii) is $\alpha = 0$, where $\frac{1}{2}(\operatorname{csch} t + \operatorname{sech} t)(1 - \cosh t) = \frac{1}{2}(\operatorname{sech} t - \tanh\frac{t}{2} - 1)$. For Corollary 1.4: exchanging p and q, that is, complex conjugation, exchanges the roles of α and β in (6), so $\overline{L_n(A,B)} = L_n(B,A)$ for real A, B. Hence $\operatorname{Re} g_n = \frac{1}{2}(\partial_A + \partial_B)L_n|_{A=B=0}$, i.e. half the derivative along the diagonal, so $2\operatorname{Re}\Phi = \partial_A\Upsilon_A|_{A=0}$; and $\partial_A \cosh 2\alpha t|_{A=0} = 2t^2$ gives $\operatorname{Re}\Phi = t^2e^t/\sinh 2t = \frac{t^2}{2}(\operatorname{csch} t + \operatorname{sech} t)$. Expanding $\operatorname{sech} t = \sum_j \mathsf{E}_{2j}t^{2j}/(2j)!$ and $\operatorname{csch} t = \frac{1}{t} + \sum_{k\ge1} 2(1-2^{2k-1})\mathcal{B}_{2k}t^{2k-1}/(2k)!$

gives the coefficient formulas. Finally Euler numbers are odd integers and, by von Staudt–Clausen, $\nu_2(\mathcal{B}_{2k}) = -1$ because $2-1=1$ divides $2k$; since $1-2^{2k-1}$ is odd, $\nu_2(\operatorname{Re} g_n) = -1$ for every n. □

4. The ladder off the midpoint

The proof of Proposition 2.5 used the midpoint identity $p^2+q^2=0$. For arbitrary weights, the two recurrences of Lemma 2.2 give instead the following system, which couples the two parities of ℓ.

Fix arbitrary nonzero p, q and set, for $\ell \ge 0$ (with $y_{-1}=1$),

$$U = y_\ell(p\xi), \quad V = y_\ell(q\xi), \quad U_- = y_{\ell-1}(p\xi), \quad V_- = y_{\ell-1}(q\xi), \qquad \Pi_\ell(\xi) = UV,$$
$$\widetilde{C}_\ell = \frac{U_-V}{p} + \frac{UV_-}{q}, \qquad \widetilde{D}_\ell = \frac{UV_-}{p} + \frac{U_-V}{q}, \qquad \sigma = \frac{1}{p}+\frac{1}{q}, \qquad \rho = \frac{p}{q}+\frac{q}{p}.$$

At the midpoint $\sigma=1$, $\rho=0$, and $\widetilde{C}_\ell = C_\ell$ because $pq=2$.

Theorem 4.1. *For every $\ell \ge 0$ and arbitrary p, q,*

$$\xi^2\Pi_\ell' + (\sigma - 2\ell\xi)\,\Pi_\ell = \widetilde{C}_\ell, \tag{16}$$ (i)

$$\widetilde{C}_{\ell+1} = \widetilde{D}_\ell + \rho\,(2\ell+1)\,\xi\,\Pi_\ell, \qquad \widetilde{D}_{\ell+1} = \widetilde{C}_\ell + 2(2\ell+1)\,\xi\,\Pi_\ell, \tag{17}$$ (ii)

$$\widetilde{C}_{\ell+2} = \widetilde{C}_\ell + 2(2\ell+1)\,\xi\,\Pi_\ell + \rho\,(2\ell+3)\,\xi\,\Pi_{\ell+1}. \tag{18}$$ (iii)

Proof. (i) Lemma 2.2(a) at $x=p\xi$ reads $p\,\xi^2U' = (\ell p\xi - 1)U + U_-$, and similarly at $x=q\xi$; hence $\xi^2\Pi_\ell' = \big[(\ell p\xi-1)U + U_-\big]V/p + U\big[(\ell q\xi-1)V + V_-\big]/q = (2\ell\xi-\sigma)\Pi_\ell + \widetilde{C}_\ell$. (ii) With $U_+ = y_{\ell+1}(p\xi)$, $V_+ = y_{\ell+1}(q\xi)$, Lemma 2.2(b) gives $U_+ = (2\ell+1)p\xi U + U_-$, $V_+ = (2\ell+1)q\xi V + V_-$, so

$$\widetilde{C}_{\ell+1} = \frac{UV_+}{p} + \frac{U_+V}{q} = (2\ell+1)\xi\Big(\frac{q}{p}+\frac{p}{q}\Big)UV + \frac{UV_-}{p} + \frac{U_-V}{q} = \rho(2\ell+1)\xi\Pi_\ell + \widetilde{D}_\ell,$$

and likewise $\widetilde{D}_{\ell+1} = 2(2\ell+1)\xi\Pi_\ell + \widetilde{C}_\ell$. (iii) Compose the two identities of (ii). □

At the midpoint, (18) is Proposition 2.5; away from it, the correction is the single term $\rho(2\ell+3)\xi\Pi_{\ell+1}$, which couples the two parities.

5. The physical curve

Lemma 5.1. *For a general angle $0<\theta<\pi$ the series (6) is taken with the weights $p = 2/(1-e^{i\theta})$, $q = 2/(1+e^{i\theta})$. They satisfy*

$$p+q = pq = \frac{4}{1-e^{2i\theta}} =: \varpi, \qquad \textit{hence} \qquad \sigma = 1, \qquad \rho = \varpi - 2$$

identically in θ, *not only at the midpoint $\theta=\pi/2$ (where $\varpi=2$). Equivalently, in the variables $x=p/2$, $y=q/2$, the weights lie on the curve $x+y=2xy$, which we call the* physical curve*; x and y are the* Horn variables.

Proof. $p+q = 2\big[(1+e^{i\theta})+(1-e^{i\theta})\big]/(1-e^{2i\theta}) = 4/(1-e^{2i\theta}) = pq$. □

Because the translation to the basis e_n of Lemma 2.7 requires precisely $\sigma=1$, the analysis of Section 2 for half-odd-integer parameters applies on the whole curve. For $\alpha=\beta=\ell+\frac12$, $\Pi_\ell(\xi) = y_\ell(p\xi)y_\ell(q\xi)$ again generates the coefficients of $F_\ell(Z) = \sum_n G_n^{(\ell)} e_n(Z)$.

Theorem 5.2. *On the physical curve, for every $\ell \geq 0$, as rational functions in $\mathbb{Q}(\varpi)(Z)$,*

$$F_{\ell+2}(Z)\,(Z-2\ell-4) = F_\ell(Z)\,(Z+2\ell+2) + (\varpi-2)(2\ell+3)\,F_{\ell+1}(Z). \tag{19}$$

Proof. By Theorem 4.1(i) with $\sigma = 1$, comparing coefficients of ξ^{m+1}, $G^{(\ell)}_{m+1} + (m-2\ell)G^{(\ell)}_m = [\xi^{m+1}]\widetilde{C}_\ell$ for $m \geq 0$; by Theorem 4.1(iii) with $\rho = \varpi - 2$, $[\xi^{m+1}]\widetilde{C}_{\ell+2} = [\xi^{m+1}]\widetilde{C}_\ell + 2(2\ell+1)G^{(\ell)}_m + (\varpi-2)(2\ell+3)G^{(\ell+1)}_m$. Eliminating $\widetilde{C}$,

$$G^{(\ell+2)}_{m+1} + (m-2\ell-4)G^{(\ell+2)}_m = G^{(\ell)}_{m+1} + (m+2\ell+2)G^{(\ell)}_m + (\varpi-2)(2\ell+3)G^{(\ell+1)}_m \qquad (m \geq 0),$$

and the translation argument of Proposition 2.8 (the case $m = 0$ supplies the constants, and the coefficients of Z agree because $G^{(\ell)}_0 = 1$) yields (19). □

Corollary 5.3. *For every $\ell \geq 0$, $F_\ell(Z) = P_\ell(Z;\varpi)/\big((Z-2)(Z-4)\cdots(Z-2\ell)\big)$ with $P_\ell \in \mathbb{Z}[\varpi][Z]$ monic of degree ℓ and*

$$P_0 = 1, \qquad P_1 = Z+\varpi-2, \qquad P_{\ell+2}(Z) = \big(Z^2-(2\ell+2)^2\big)P_\ell(Z) + (\varpi-2)(2\ell+3)\,P_{\ell+1}(Z). \tag{20}$$

At $\varpi = 2$ this collapses to $P_{\ell+2} = (Z^2-(2\ell+2)^2)P_\ell$, whose solutions are the numerators of the Gamma quotient of Theorem 2.10: $S_{2m} = \prod_{j=1}^m (Z^2-(4j-2)^2)$, $S_{2m+1} = Z\prod_{j=1}^m (Z^2-16j^2)$, so that uniformly $S_{\nu+2} = (Z^2-(2\nu+2)^2)S_\nu$.

Proof. $\Pi_1 = (1+p\xi)(1+q\xi) = 1+\varpi\xi+\varpi\xi^2$ gives $F_1 = 1+\varpi e_1+\varpi e_2 = (Z^2+(\varpi-3)Z+2-\varpi)/((Z-1)(Z-2)) = (Z+\varpi-2)/(Z-2)$, the numerator vanishing at $Z=1$. Solving (19) for $F_{\ell+2}$ and clearing denominators gives $P_{\ell+2} = (Z+2\ell+2)(Z-2\ell-2)P_\ell + (\varpi-2)(2\ell+3)P_{\ell+1}$ by induction. □

The recurrence (20) has a closed solution. Write $\varpi_0 = \varpi - 2$ and let

$$\mathfrak{b}_j(k) = \frac{k}{2j+k}\binom{2j+k}{j} \qquad (\mathfrak{b}_0(k) = 1)$$

be the ballot (Catalan-triangle) numbers.

Theorem 5.4. *For every $\ell \geq 0$, with $d = \ell - k$ in the summand indexed by k,*

$$P_\ell(Z;2+\varpi_0) = \sum_{k=0}^{\ell} (2k-1)!!\binom{\ell+k}{2k}\varpi_0^k \sum_{j\geq 0}(-1)^j\,\frac{d!}{(d-2j)!}\,\mathfrak{b}_j(k)\,S_{d-2j}(Z), \tag{21}$$

the inner sum running over $j \leq d/2$.

Proof. The proof consists in verifying that the closed coefficients (21) satisfy the same recurrence and the same initial data as the curve numerators: the interior identity is checked first, then the boundary patterns excluded by the interior divisions, then the base cases; induction closes the argument. The polynomials S_ν have degree ν, so the expansion of any polynomial in the basis $\{\varpi_0^k S_\nu\}$ is unique. Write $c(\ell,k,j) = (2k-1)!!\binom{\ell+k}{2k}(-1)^j\frac{d!}{(d-2j)!}\mathfrak{b}_j(k)$ for the coefficient of $\varpi_0^k S_{\ell-k-2j}$ in (21), with $c(\ell,k,j) = 0$ outside $0 \leq k \leq \ell$, $0 \leq 2j \leq d$. Since $Z^2S_\nu = S_{\nu+2} + (2\nu+2)^2 S_\nu$, substituting (21) into (20) and matching the coefficient of $\varpi_0^k S_\nu$ with $\nu = (\ell+2)-k-2J$ reduces the claim, by uniqueness, to the four-term identity

$$c(\ell+2,k,J) = c(\ell,k,J) - 4(k+2J-2)(2\ell+4-k-2J)\,c(\ell,k,J-1) + (2\ell+3)\,c(\ell+1,k-1,J). \tag{22}$$

We verify (22) for all indices, including the boundary patterns. First suppose $k \geq 1$. The coefficient admits the factorial form

$$c(\ell,k,j) = (-1)^j \frac{(\ell+k)!\,(2j+k-1)!}{2^k(k-1)!\,j!\,(j+k)!\,(\ell-k-2j)!}, \tag{23}$$

whenever $\ell - k - 2j \geq 0$; outside this range it is zero by convention. This follows immediately from $(2k-1)!! = (2k)!/(2^k k!)$ and $\mathfrak{b}_j(k) = k(2j+k-1)!/(j!(j+k)!)$.

For the interior range $J \geq 1$ and $n := \ell - k - 2J \geq 0$, division by the nonzero coefficient $c(\ell,k,J)$ gives

$$\frac{c(\ell+2,k,J)}{c(\ell,k,J)} = \frac{(\ell+k+1)(\ell+k+2)}{(n+1)(n+2)},$$
$$\frac{c(\ell+1,k-1,J)}{c(\ell,k,J)} = \frac{2(k-1)(J+k)}{(n+1)(n+2)(2J+k-1)},$$
$$\frac{c(\ell,k,J-1)}{c(\ell,k,J)} = \frac{-J(J+k)}{(n+1)(n+2)(2J+k-2)(2J+k-1)}.$$

Substitution in (22), followed by clearing denominators, reduces the recurrence to

$$\begin{aligned}(\ell+k+1)(\ell+k+2)(2J+k-1) = {}& (n+1)(n+2)(2J+k-1)\\ &+ 4J(J+k)(2\ell+4-k-2J)\\ &+ 2(2\ell+3)(k-1)(J+k),\end{aligned} \tag{24}$$

and after $\ell = n+k+2J$ both sides expand to the same cubic polynomial.

It remains to treat the boundary cases excluded by that division. If $k = 0$, then $c(\ell,0,0) = 1$ and $c(\ell,0,j) = 0$ for $j \geq 1$. Hence (22) is immediate: for $J = 0$ it is $1 = 1$, for $J = 1$ the only potential term $c(\ell,0,0)$ is multiplied by $k+2J-2 = 0$, and for $J \geq 2$ every term is zero.

Now let $k \geq 1$ and $J = 0$. If $k \leq \ell$, the middle term of (22) vanishes and (23) gives

$$\frac{c(\ell+2,k,0)}{c(\ell,k,0)} = \frac{(\ell+k+1)(\ell+k+2)}{(\ell-k+1)(\ell-k+2)}, \qquad \frac{c(\ell+1,k-1,0)}{c(\ell,k,0)} = \frac{2k}{(\ell-k+1)(\ell-k+2)}.$$

(For $k = 1$ one uses $c(\ell+1,0,0) = 1$ directly.) Their substitution gives $c(\ell+2,k,0) = c(\ell,k,0) + (2\ell+3)c(\ell+1,k-1,0)$ exactly. The only remaining $J = 0$ targets have $k = \ell+1$ or $k = \ell+2$; then $c(\ell,k,0) = 0$ and the middle term is again zero, while direct evaluation of the double-factorial formula gives

$$c(\ell+2,k,0) = (2\ell+3)c(\ell+1,k-1,0) = (2\ell+3)!!.$$

Finally take $J \geq 1$. When the target coefficient $c(\ell+2,k,J)$ exists but $c(\ell,k,J)$ does not, necessarily $n = -2$ or $n = -1$; these are the only remaining possibilities because the target basis index is $n+2 \geq 0$. For $n = -2$,

$$\frac{c(\ell,k,J-1)}{c(\ell+2,k,J)} = -\frac{J}{2(2J+k-2)(2J+k-1)(2J+2k-1)},$$
$$\frac{c(\ell+1,k-1,J)}{c(\ell+2,k,J)} = \frac{k-1}{(2J+k-1)(2J+2k-1)}.$$

Since here $\ell = k+2J-2$, insertion in the right-hand side of (22) gives exactly 1 after division by $c(\ell+2,k,J)$. For $n = -1$ the corresponding ratios are

$$\frac{c(\ell,k,J-1)}{c(\ell+2,k,J)} = -\frac{J}{2(2J+k-2)(2J+k-1)(2J+2k+1)},$$

$$\frac{c(\ell+1,k-1,J)}{c(\ell+2,k,J)}=\frac{k-1}{(2J+k-1)(2J+2k+1)},$$

and with $\ell=k+2J-1$ the same substitution again gives 1. These formulas also cover $k=1$, because then $c(\ell+1,0,J)=0$ for $J\geq 1$ and the factor $k-1$ vanishes. Thus (22) holds for every admissible coefficient.

With the base cases $P_0=1=c(0,0,0)S_0$ and $P_1=Z+\varpi_0=S_1+\varpi_0 S_0$, induction on ℓ through (20) completes the proof. □

Remark 5.5. The curve $\sigma=1$ passes through the midpoint in the plane of the Horn variables (x,y). On it, Theorem 5.2, the degree bound of Lemma 3.1 (valid at any fixed (x,y)) and interpolation at $A=(\ell+\frac{1}{2})^2$ determine the diagonal amplitude for all α. The derivatives normal to the curve are not determined in this way: for $\sigma\neq 1$ the coefficient relations take the form $\sigma G_{m+1}+(m-c)G_m$, and their translation to the basis e_m involves, in addition, the operation $\sum_m G_m e_m\mapsto\sum_m mG_m e_m$, which is not expressible through rational operations and shifts in Z. This operation is studied in Section 6.

6. The index operator and the derivatives normal to the curve

By Remark 5.5, extending the curve contiguity away from $\sigma=1$ requires the multiplication of the coefficient G_m by its index m. We show that this operation is a first-order differential operator in the normal variable, and deduce a recurrence which determines every derivative normal to the curve of the half-odd-integer diagonal amplitudes.

For arbitrary p,q with $pq\neq 0$ and $p+q\neq 0$, put $\sigma=\frac{1}{p}+\frac{1}{q}$ and $\rho=\frac{p}{q}+\frac{q}{p}$, and let

$$F_\ell(Z;\sigma,\rho)=\sum_{m\geq 0}G_m^{(\ell)}e_m(Z),\qquad e_m(Z)=\frac{1}{(Z-1)\cdots(Z-m)}.$$

The two invariants recover the symmetric functions of p,q:

$$pq=\frac{\rho+2}{\sigma^2},\qquad p+q=\frac{\rho+2}{\sigma}. \tag{25}$$

Consequently F_ℓ is symmetric in p,q and is a rational function of (Z,σ,ρ) for every integer $\ell\geq 0$.

Definition 6.1. On an inverse-factorial series define the *index operator*

$$\mathcal{I}\Big(\sum_{m\geq 0}G_m e_m\Big)=\sum_{m\geq 0}mG_m e_m.$$

Lemma 6.2. *If every $G_m(x,y)$ is homogeneous of total degree m, then $\mathcal{I}F=(\Theta_x+\Theta_y)F$, where $\Theta_x=x\partial_x$ and $\Theta_y=y\partial_y$. Moreover, if $G_0=1$ and $\mathcal{R}F=\sum_{m\geq 0}G_{m+1}e_m$, then*

$$\mathcal{R}F=Z(F-1)-\mathcal{I}F, \tag{26}$$

and if F is symmetric in p,q, hence a function of (σ,ρ), then at fixed ρ, $\mathcal{I}=-\sigma\partial_\sigma$.

Proof. Homogeneity gives the first assertion. Lemma 2.7 gives $ZF=Z+\sum_{m\geq 1}G_m e_{m-1}+\sum_{m\geq 1}mG_m e_m=Z+\mathcal{R}F+\mathcal{I}F$, which is (26). Under a common scaling $(p,q)\mapsto(rp,rq)$, ρ is fixed and $\sigma\mapsto r^{-1}\sigma$; hence $\Theta_x+\Theta_y=-\sigma\partial_\sigma$ at fixed ρ. □

Define the first-order normal operator

$$\mathcal{Q}_c^{(\sigma)}F:=\sigma Z(F-1)-\sigma(1-\sigma)\partial_\sigma F-cF=\sum_{m\geq 0}\big(\sigma G_{m+1}+(m-c)G_m\big)e_m. \tag{27}$$

Thus the operation of Remark 5.5 is a first-order derivative in the coordinate σ.

Theorem 6.3. *For every $\ell \geq 0$ and arbitrary (σ, ρ),*

$$\mathcal{Q}^{(\sigma)}_{2\ell+4} F_{\ell+2} = \mathcal{Q}^{(\sigma)}_{-2\ell-2} F_\ell + \rho(2\ell+3) F_{\ell+1}. \tag{28}$$

At $\sigma = 1$ this reduces to the curve contiguity (19).

Proof. From Theorem 4.1(i), comparison of the coefficient of ξ^{m+1} gives $\sigma G^{(\ell)}_{m+1} + (m - 2\ell) G^{(\ell)}_m = [\xi^{m+1}]\widetilde{C}_\ell$. Use Theorem 4.1(iii) to eliminate $\widetilde{C}_{\ell+2}$ and $\widetilde{C}_\ell$. One obtains, for every $m \geq 0$,

$$\sigma G^{(\ell+2)}_{m+1} + (m - 2\ell - 4) G^{(\ell+2)}_m = \sigma G^{(\ell)}_{m+1} + (m + 2\ell + 2) G^{(\ell)}_m + \rho(2\ell+3) G^{(\ell+1)}_m.$$

Multiplication by e_m and summation give (28). At $\sigma = 1$, (27) becomes $\mathcal{Q}^{(1)}_c F = (Z - c)F - Z$; the two copies of $-Z$ cancel and (19) follows. □

At $\sigma = 1$ the coefficient of $\partial_\sigma F$ in (27) vanishes. Nevertheless all derivatives in σ at $\sigma = 1$ are determined recursively.

Theorem 6.4. *Put $F^{[k]}_\ell(Z;\rho) = \partial^k_\sigma F_\ell(Z;\sigma,\rho)\big|_{\sigma=1}$ for $k \geq 0$, with $F^{[-1]}_\ell = 0$. Then*

$$\begin{aligned} &(Z + k - 2\ell - 4) F^{[k]}_{\ell+2} + k(Z + k - 1) F^{[k-1]}_{\ell+2} \\ &\qquad = (Z + k + 2\ell + 2) F^{[k]}_\ell + k(Z + k - 1) F^{[k-1]}_\ell + \rho(2\ell+3) F^{[k]}_{\ell+1}. \end{aligned} \tag{29}$$

In particular, at the midpoint $\rho = 0$, the two parity ladders decouple and every $F^{[k]}_\ell$ is obtained recursively from the two base families F_0 and F_1.

Proof. For $k \geq 1$, Leibniz' rule applied to (27) gives

$$\partial^k_\sigma \mathcal{Q}^{(\sigma)}_c F\big|_{\sigma=1} = (Z + k - c) F^{[k]} + k(Z + k - 1) F^{[k-1]} - \delta_{k,1} Z : \tag{30}$$

only the first two derivatives of $\sigma(1 - \sigma)$ are nonzero, and their contributions combine with the derivative of $\sigma Z(F - 1)$ exactly as displayed. For $k = 0$ the formula is $\mathcal{Q}^{(1)}_c F = (Z - c)F^{[0]} - Z$. For $k \leq 1$ the terms $-Z$, which do not depend on c, cancel between the two sides of (28). Differentiating (28) k times and using (30) proves (29). □

The base data are explicit: from $\Pi_0 = 1$, $\Pi_1 = (1 + p\xi)(1 + q\xi)$ and (25),

$$F_0 = 1, \qquad F_1 = 1 + \frac{\rho + 2}{\sigma} e_1 + \frac{\rho + 2}{\sigma^2} e_2, \tag{31}$$

hence $F^{[k]}_0 = 0$ for $k \geq 1$ and

$$F^{[k]}_1 = (-1)^k k! \, (\rho + 2)\big(e_1 + (k+1)e_2\big) \qquad (k \geq 1). \tag{32}$$

Equations (29)–(32) therefore determine every $F^{[k]}_\ell$ recursively.

Remark 6.5. For each fixed order in Z^{-1} and each fixed k, the interpolation argument of Section 3 extends (29), written with $2\ell + 1 = 2\alpha$ and with $F_\ell, F_{\ell+1}, F_{\ell+2}$ replaced by the diagonal amplitudes at $\alpha, \alpha + 1, \alpha + 2$, from $\alpha = \ell + \frac{1}{2}$ to arbitrary α, each coefficient being a polynomial in α. Thus the diagonal amplitude is formally determined, to any prescribed order in $\sigma - 1$, in a neighbourhood of the physical curve.

7. Off the diagonal: the transverse variation and the dyadic law

This section contains the proofs of Theorems 1.6 and 1.5. We begin with two preliminary results for parameters off the diagonal: a recurrence for independent half-odd-integer parameters, and the identification of the amplitude (6) with a double hypergeometric series.

Take *independent* half-odd-integer parameters $\alpha = \ell + \frac{1}{2}$, $\beta = m + \frac{1}{2}$ at the midpoint $p = 1 + i$, $q = 1 - i$: both Hankel series terminate and $\Pi_{\ell,m}(\xi) = y_\ell(p\xi)y_m(q\xi)$ generates the coefficients of the rational function $F_{\ell,m}(Z) = \sum_n G_n^{(\ell,m)} e_n(Z)$. Set $\widetilde{C}_{\ell,m} = y_{\ell-1}(p\xi)y_m(q\xi)/p + y_\ell(p\xi)y_{m-1}(q\xi)/q$ and $\widetilde{D}_{\ell,m} = y_\ell(p\xi)y_{m-1}(q\xi)/p + y_{\ell-1}(p\xi)y_m(q\xi)/q$.

Theorem 7.1. *For all $\ell, m \geq 0$, at the midpoint,*

$$\xi^2\Pi'_{\ell,m} + \big(1 - (\ell+m)\xi\big)\Pi_{\ell,m} = \widetilde{C}_{\ell,m}, \tag{33}$$

$$\widetilde{C}_{\ell+1,m+1} = \widetilde{D}_{\ell,m} + 2i(\ell-m)\,\xi\,\Pi_{\ell,m}, \qquad \widetilde{D}_{\ell+1,m+1} = \widetilde{C}_{\ell,m} + 2(\ell+m+1)\,\xi\,\Pi_{\ell,m}, \tag{34}$$

$$\begin{aligned} &F_{\ell+2,m+2}(Z)\,\big(Z - (\ell+m) - 4\big) \\ &\qquad = F_{\ell,m}(Z)\,\big(Z + (\ell+m) + 2\big) + 2i(\ell-m)\,F_{\ell+1,m+1}(Z). \end{aligned} \tag{35}$$

Proof. (33) is the computation of Theorem 4.1(i) with the two orders kept distinct, using $\sigma = 1$. For (34), with $U = y_\ell(p\xi)$, $V = y_m(q\xi)$,

$$\widetilde{C}_{\ell+1,m+1} = \frac{UV_+}{p} + \frac{U_+V}{q} = \xi\Big[(2m+1)\frac{q}{p} + (2\ell+1)\frac{p}{q}\Big]\Pi_{\ell,m} + \widetilde{D}_{\ell,m},$$

and at the midpoint $q/p = -i$, $p/q = i$, so the bracket is $2i(\ell - m)$; the second identity has bracket $(2\ell+1) + (2m+1)$. Composing gives $\widetilde{C}_{\ell+2,m+2} = \widetilde{C}_{\ell,m} + 2(\ell+m+1)\xi\Pi_{\ell,m} + 2i(\ell-m)\xi\Pi_{\ell+1,m+1}$, and the translation of Theorem 5.2, with $\ell+m$ in place of 2ℓ, yields (35); the constants match because $G_1^{(\ell,m)} = \frac{\ell(\ell+1)}{2}p + \frac{m(m+1)}{2}q$ gives $G_1^{(\ell+2,m+2)} - G_1^{(\ell,m)} = \big[(\ell+m)+2\big] + \big[(\ell+m)+4\big] + 2i(\ell-m)$. □

On the diagonal $\ell = m$ the inhomogeneity vanishes and (35) is Proposition 2.8. Off the diagonal the transverse direction enters as the explicit term $2i(\ell-m)$: antisymmetric, purely imaginary, of constant coefficient along the diagonal step. Together with the two boundary strips $m = 0$ and $m = 1$, the recurrence determines every $F_{\ell,m}$ by induction on $\min(\ell, m)$, using the midpoint conjugation symmetry

$$F_{\ell,m}(Z) = \overline{F_{m,\ell}(\overline{Z})}$$

(and hence $F_{\ell,m}(Z) = \overline{F_{m,\ell}(Z)}$ for real Z) whenever the smaller of the two indices is ℓ rather than m. More explicitly, if $m \geq 2$, equation (35) expresses $F_{\ell,m}$ in terms of $F_{\ell-2,m-2}$ and $F_{\ell-1,m-1}$ after shifting the two indices down; repeating reaches $m = 0$ or $m = 1$. If $\ell < m$ one first interchanges the indices by the displayed conjugation symmetry and applies the same argument. The strip $m = 0$ is the terminating Gauss family of Lemma 7.2; the strip $m = 1$ is equally explicit from $y_1(q\xi) = 1 + q\xi$ and the finite defining sum. Thus the two-step recurrence, together with these two strips, determines every $F_{\ell,m}$.

Lemma 7.2. *For every $\ell \geq 0$, $F_{\ell,0}(Z) = {}_2\mathrm{F}_1\big(-\ell, \ell+1; 1-Z; x_0\big)$ with $x_0 = \frac{1+i}{2}$. More generally, for all α, β, as formal series,*

$$F(\alpha,\beta;Z) = \sum_{\mu,\nu\geq 0} \frac{(\frac{1}{2}-\alpha)_\mu(\frac{1}{2}+\alpha)_\mu(\frac{1}{2}-\beta)_\nu(\frac{1}{2}+\beta)_\nu}{(1-Z)_{\mu+\nu}\,\mu!\,\nu!}\, x_0^\mu y_0^\nu, \tag{36}$$

a Kampé de Fériet function $F^{0:2;2}_{1:0;0}$ *at arguments with* $x_0+y_0=1$, $x_0y_0=\frac{1}{2}$.

Proof. $e_n(Z)=(-1)^n/(1-Z)_n$ and $a_\mu(\alpha)(2x_0)^\mu(-1)^\mu=(\frac{1}{2}-\alpha)_\mu(\frac{1}{2}+\alpha)_\mu x_0^\mu/\mu!$, term by term in (6). At $\alpha=\ell+\frac{1}{2}$, $\beta=\frac{1}{2}$, only $\nu=0$ survives and the μ-series terminates. □

7.1. **The Appell F_3 specialization on the physical curve.**
Branch convention. All multivalued functions in this and the following subsections are continued from a common germ at the origin of the path described below: the Appell/Kampé de Fériet series at (x,y) small, the reduced Gauss function through the argument path $4x(\lambda)(1-x(\lambda))=2\lambda+2i\lambda(1-\lambda)$, which stays in the *upper* half-plane and reaches $2+i0$; every boundary value written ${}_2F_1^{\uparrow}(\cdots;2)$, $\log(1-z)\to-i\pi$, 2^+ or $-1-i0$ is the one obtained along this path, and the lower determinations (denoted $\downarrow$) are their Schwarz reflections. Replacing the upper boundary value by the lower one changes the functions; no identity below mixes the two determinations except where both appear explicitly, as in Appendix A.4.

The Kampé de Fériet notation in Lemma 7.2 hides an additional one-variable structure. We make it explicit because it removes one of the two Horn variables before any asymptotic or 2-adic argument is made. For independent variables x,y write

$$\mathcal{F}(\alpha,\beta;Z;x,y):=F_3\left(\begin{matrix}\frac{1}{2}-\alpha,\frac{1}{2}-\beta;\frac{1}{2}+\alpha,\frac{1}{2}+\beta\\ 1-Z\end{matrix}\middle|x,y\right),$$

where the convention is that the first and third upper parameters belong to x, and the second and fourth to y. Thus (36) is precisely $F(\alpha,\beta;Z)=\mathcal{F}(\alpha,\beta;Z;x_0,y_0)$.

Theorem 7.3. *Put* $c=1-Z$, $b_1=\frac{1}{2}+\alpha$, $b_2=\frac{1}{2}+\beta$. *On*

$$y=\frac{x}{2x-1},\qquad \textit{equivalently}\qquad x+y=2xy,$$

one has, for generic parameters and then by meromorphic continuation,

$$\mathcal{F}\left(\alpha,\beta;Z;x,\frac{x}{2x-1}\right)=(1-x)^{c-1}(1-2x)^{b_2}$$
$$\times\,{}_2F_1\left(\frac{b_2-b_1+c}{2},\frac{b_1+b_2+c-1}{2};c;4x(1-x)\right). \tag{37}$$

The branches are those obtained by analytic continuation from a neighbourhood of $x=0$ *along the chosen path.*

Proof. The four upper parameters of our F_3 are $1-b_1,1-b_2;b_1,b_2$. The fourth pair in Theorem 3.1 of Vidūnas [19] is therefore exactly the pair displayed in (37). Immediately after that theorem, Vidūnas states that in the first and fourth pairs the two functions are directly identical in a neighbourhood of $x=0$ (not merely solutions of the same second-order ODE). Hence (37) holds on the common germ at $x=0$. The relation $y=x/(2x-1)$ is the physical-curve identity already used in Section 5. Therefore the two sides represent the same analytic germ near $x=0$. Analytic continuation of this common germ along any path avoiding the singular locus gives the stated identity. Notice that this is an identity for fixed generic Z (and then meromorphically in the parameters); no passage to the large-Z limit is used in establishing it. □

At the midpoint,

$$x_0=\frac{1+i}{2},\qquad y_0=\frac{1-i}{2},\qquad y_0=\frac{x_0}{2x_0-1},\qquad 4x_0(1-x_0)=2.$$

For the straight path $x(\lambda) = \lambda x_0$, $0 < \lambda < 1$, one has

$$y(\lambda) = \frac{x(\lambda)}{2x(\lambda) - 1}, \qquad 4x(\lambda)(1 - x(\lambda)) = 2\lambda + 2i\lambda(1 - \lambda).$$

Thus the transformed Gauss argument stays strictly in the upper half-plane for $0 < \lambda < 1$ and tends to $2 + i0$ as $\lambda \uparrow 1$. In addition, $|x(\lambda)| = \lambda/\sqrt{2} < 1$ and

$$|y(\lambda)|^2 = \frac{\lambda^2}{2(2\lambda^2 - 2\lambda + 1)} \le \frac{1}{2},$$

so the defining Appell double series itself stays inside its bidisc of absolute convergence along the whole path. The path has $x, y \neq 0$, meets neither $x = 1$ nor $y = 1$, and the additional finite Appell singular relation $xy = x + y$, together with $x + y = 2xy$, would force $xy = 0$. Hence continuation may start at any sufficiently small $\lambda > 0$ inside the common germ and reaches the midpoint without meeting the finite singular locus. The only nontrivial continuation issue in the reduced expression is therefore the Gauss boundary value at its standard cut. If ${}_2F_1^{\uparrow}(a, b; c; 2)$ denotes this upper boundary value, then analytic continuation of the common germ gives

$$F(\alpha, \beta; Z) = y_0^{-Z}(-i)^{\frac{1}{2}+\beta}\, {}_2F_1^{\uparrow}\left(\frac{1 - Z + \beta - \alpha}{2}, \frac{1 - Z + \alpha + \beta}{2}; 1 - Z; 2\right). \tag{38}$$

This branch convention is essential: the two boundary values on the cut $[1, \infty)$ need not agree.

Corollary 7.4. *Set $s = (1 - Z)/2$ and*

$$\mathcal{H}(\alpha; Z) = {}_2F_1^{\uparrow}\left(s - \frac{\alpha}{2}, s + \frac{\alpha}{2}; 2s; 2\right).$$

Then the inverse-power expansion at $Z = \infty$ inherited coefficientwise from the defining inverse-factorial series for F satisfies

$$\sum_{n \ge 1} g_n Z^{-n} = \frac{1}{2}\,\frac{\partial^2}{\partial \alpha^2} \log \mathcal{H}(\alpha; Z)\Big|_{\alpha=0} = \frac{1}{2}\,\frac{\partial_\alpha^2 \mathcal{H}(0; Z)}{\mathcal{H}(0; Z)}. \tag{39}$$

The last expression is independent of a choice of logarithm. In particular the two numerator parameters in (39) sum to the denominator parameter: the series $\sum_n g_n Z^{-n}$ is obtained from a parameter variation of a single zero-balanced *Gauss function.*

Proof. Take $\beta = 0$ in (38). Its prefactor is independent of α. Since $A = \alpha^2$ and the left-hand side is even in α, $\partial_A|_{A=0} = \frac{1}{2}\partial_\alpha^2|_{\alpha=0}$. The function $\mathcal{H}$ is itself even in α, so $\partial_\alpha \mathcal{H}(0; Z) = 0$; therefore the second logarithmic derivative is $\partial_\alpha^2 \mathcal{H}(0; Z)/\mathcal{H}(0; Z)$. Finally, $F = 1 + O(Z^{-1})$ coefficientwise, so its formal logarithm at $Z = \infty$ is uniquely defined. Since (38) is an exact identity for generic fixed Z, the formal inverse-power expansion used in this paper may be taken on either side; the coefficients inherited from the left-hand inverse-factorial series are therefore exactly those in (39). No assertion about convergence of the resulting Z^{-1} series is needed. □

There is also a differential form of the same reduction. With $\Theta_x = x\partial_x$, $\Theta_y = y\partial_y$, the defining coefficients of F_3 give

$$\mathcal{L}_x^{(A)}\mathcal{F} = 0, \qquad \mathcal{L}_x^{(A)} = \Theta_x(\Theta_x + \Theta_y - Z) - x\big((\Theta_x + \tfrac{1}{2})^2 - A\big), \tag{40}$$

$$\mathcal{L}_y^{(B)}\mathcal{F} = 0, \qquad \mathcal{L}_y^{(B)} = \Theta_y(\Theta_x + \Theta_y - Z) - y\big((\Theta_y + \tfrac{1}{2})^2 - B\big). \tag{41}$$

Thus the parameters enter the Appell system *affinely*. Put

$$A_+ = \frac{A+B}{2}, \qquad A_- = \frac{A-B}{2},$$

and let $\mathcal{F}_0 = \mathcal{F}|_{A_-=0}$ and $\mathcal{F}_1 = \partial_{A_-}\mathcal{F}|_{A_-=0}$. Differentiating (40)–(41) yields the forced system

$$\mathcal{L}_x^{(A_+)}\mathcal{F}_1 = -x\mathcal{F}_0, \qquad \mathcal{L}_y^{(A_+)}\mathcal{F}_1 = +y\mathcal{F}_0, \qquad \mathcal{F}_1(y,x) = -\mathcal{F}_1(x,y). \tag{42}$$

At the midpoint $y_0 = \overline{x_0}$, coefficientwise conjugation and exchange of the two variables give $\overline{\mathcal{F}(A_+, A_-; x_0, y_0)} = \mathcal{F}(A_+, -A_-; x_0, y_0)$ for real Z and real $A_\pm$. Consequently, at $A_- = 0$, $\mathcal{F}_0$ and $\partial_{A_+}\mathcal{F}_0$ are real, whereas $\mathcal{F}_1$ is purely imaginary.

Remark 7.5. κ_r is a polynomial in (A, B); its values on the grid $A_\ell = (\ell + \frac{1}{2})^2$, $B_m = (m+\frac{1}{2})^2$ are determined by the rational functions $F_{\ell,m}$, and the node differences $A_\ell - A_m = (\ell - m)(\ell + m + 1)$ are integers. The recurrence (35) therefore expresses $[A^1 B^0]\kappa_r$ as a Lagrange interpolation functional with integer node differences.

7.2. **The boundary split and the transverse quadrature.** By Corollary 7.4, the series $\sum_n g_n Z^{-n}$ is obtained from the one-parameter family of boundary values $\mathcal{H}(\alpha; Z)$. We now split this boundary value into two classical constituents and compute its second variation in α by quadratures. Throughout, $s = (1 - Z)/2$, $a = s - \frac{\alpha}{2}$, $b = s + \frac{\alpha}{2}$, and $Z > 1$ is real, so $s < 0$; the degenerate lattice $2s \in \mathbb{Z}_{\le 0}$ is excluded.

Lemma 7.6.

$$\mathcal{H}(0; Z) = {}_2F_1^{\uparrow}(s, s; 2s; 2) = e^{i\pi s/2}\,\frac{\sqrt{\pi}\,\Gamma(s + \frac{1}{2})}{\Gamma(\frac{s+1}{2})^2}.$$

In particular $\operatorname{Im}\mathcal{H}(0; Z)/\operatorname{Re}\mathcal{H}(0; Z) = \tan\frac{\pi s}{2}$.

Proof. The quadratic transformation [3, Sec. 3.1]

$${}_2F_1(a, b; 2b; z) = (1 - z)^{-a/2}\,{}_2F_1\Big(\frac{a}{2},\, b - \frac{a}{2};\, b + \frac{1}{2};\, \frac{z^2}{4(z-1)}\Big)$$

at $a = b = s$ has transformed argument $z^2/(4(z-1)) \to 1$ as $z \to 2$, and the right-hand Gauss series is continuous up to argument 1 because there $\operatorname{Re}(c - a - b) = \frac{1}{2} > 0$; Gauss's summation gives ${}_2F_1(\frac{s}{2}, \frac{s}{2}; s + \frac{1}{2}; 1) = \Gamma(s + \frac{1}{2})\Gamma(\frac{1}{2})/\Gamma(\frac{s+1}{2})^2$. Approaching from $\operatorname{Im} z > 0$, $\log(1 - z) \to -i\pi$, so $(1 - z)^{-s/2} \to e^{i\pi s/2}$. □

Proposition 7.7. *For* $\operatorname{Re} s < \frac{1}{2}$ *and* $2s \notin \mathbb{Z}$,

$$\mathcal{H}(\alpha; Z) = \frac{\Gamma(2s)}{\Gamma(a)\Gamma(b)}\big[\mathsf{S}(\alpha; s) + i\pi\,\mathsf{K}(\alpha; s)\big], \tag{43}$$

where

$$\mathsf{K}(\alpha; s) = {}_2F_1(a, b; 1; -1), \qquad \mathsf{S}(\alpha; s) = \sum_{n\ge 0}(-1)^n \frac{(a)_n (b)_n}{(n!)^2}\big[2\psi(n+1) - \psi(a+n) - \psi(b+n)\big],$$

both series converging absolutely. In particular, for real α *the imaginary part of the boundary value is* $\pi\Gamma(2s)\,{}_2F_1(a, b; 1; -1)/(\Gamma(a)\Gamma(b))$. *At the Legendre point* $\alpha = 0$ *both blocks have closed forms:*

$$\mathsf{K}(0; s) = \frac{\Gamma(1 + \frac{s}{2})}{\Gamma(1 + s)\,\Gamma(1 - \frac{s}{2})} \quad \text{(Kummer)}, \qquad \mathsf{S}(0; s) = \cos\frac{\pi s}{2} \cdot \frac{\sqrt{\pi}\,\Gamma(s + \frac{1}{2})\,\Gamma(s)^2}{\Gamma(2s)\,\Gamma(\frac{s+1}{2})^2}. \tag{44}$$

Proof. Write $R_n(a,b) = 2\psi(n+1) - \psi(a+n) - \psi(b+n)$. The zero-balanced connection formula [3, Sec. 2.3] (see also [2, Eq. 15.8.10]) gives, for $|1-z|<1$, $|\arg(1-z)|<\pi$,

$$ {}_2F_1(a,b;a+b;z) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \sum_{n\ge 0} \frac{(a)_n (b)_n}{(n!)^2} \big[R_n(a,b) - \log(1-z)\big](1-z)^n. $$

On the upper approach $z \to 2+i0$ one has $1-z \to -1-i0$, so $\log(1-z) \to -i\pi$ and $(1-z)^n \to (-1)^n$; the two limiting series are absolutely convergent for $\mathrm{Re}(a+b) < 1$ since their terms are $O(n^{\mathrm{Re}(a+b)-2}\log n)$, and the boundary value is attained by Abel continuity along the path. The digamma-free part is the Taylor series of ${}_2F_1(a,b;1;-1)$ because $(1)_n = n!$. The first evaluation in (44) is Kummer's theorem ${}_2F_1(a,b;1+a-b;-1)$ at $a=b=s$; the second follows from Lemma 7.6 by taking real parts in (43). The $n=0$ kernel $2\psi(1)-\psi(a)-\psi(b)$ is precisely the constant $R(a,b) = -2\gamma-\psi(a)-\psi(b)$ governing the zero-balanced asymptotics studied in [23, 24]. □

Lemma 7.8. *For all α and generic s,*

$$ 4(s-1)(4s^2-\alpha^2)\,\mathsf{K}(\alpha;s+1) + 2\alpha^2(2s-1)\,\mathsf{K}(\alpha;s) + s\big(4(s-1)^2-\alpha^2\big)\,\mathsf{K}(\alpha;s-1) = 0. \tag{45} $$

At $\alpha = 0$ this degenerates to the two-term relation $4s\,\mathsf{K}(0;s+1) + (s-1)\,\mathsf{K}(0;s-1) = 0$, which is also satisfied by the closed form of $\mathsf{K}(0;s)$ in (44).

Proof. Assume first that $\mathrm{Re}\, s < -\frac{1}{2}$, so that the three series converge absolutely. With $t_n = (-1)^n (a)_n (b)_n/(n!)^2$, the shifted series have termwise ratios $t_n^+ = t_n (a+n)(b+n)/(ab)$ and $t_n^- = t_n (a-1)(b-1)/((a-1+n)(b-1+n))$, and the rational function

$$ r(n) = \frac{-n^2\,\mathsf{P}(n)}{(a-1+n)(b-1+n)}, $$
$$ \mathsf{P}(n) = 8(s-1)n^2 + 8(3s-2)(s-1)n + 4(5s-2)(s-1)^2 + (2-s)\alpha^2, $$

satisfies the identity

$$ c_+ \frac{(a+n)(b+n)}{ab} + c_0 + c_- \frac{(a-1)(b-1)}{(a-1+n)(b-1+n)} = -\frac{(a+n)(b+n)}{(n+1)^2}\, r(n+1) - r(n) $$

where c_+, c_0, c_- denote the three coefficients of (45), in this order (an identity of rational functions of (n,s,α)). Multiplying by t_n and summing over $n \ge 0$, the right side telescopes: $r(0) = 0$, and $t_n r(n) = O(n^{2\,\mathrm{Re}\, s}) \to 0$. This proves (45) for $\mathrm{Re}\, s < -\frac{1}{2}$, and the general case follows by analytic continuation in s, since ${}_2F_1(a,b;1;-1)$ is an entire function of a and b. The $\alpha = 0$ degeneration follows since the middle coefficient carries α^2; that the closed form of $\mathsf{K}(0;s)$ satisfies it is the elementary computation $\mathsf{K}(0;s+1)/\mathsf{K}(0;s-1) = -(s-1)/(4s)$. □

Theorem 7.9. *Let*

$$ \mathsf{K}_2(s) := \partial_\alpha^2 \mathsf{K}(\alpha;s)\big|_{\alpha=0} = -\frac{1}{2}\sum_{n\ge 1} (-1)^n \Big(\frac{(s)_n}{n!}\Big)^2 \sum_{j=0}^{n-1} \frac{1}{(s+j)^2}, $$

a series with rational terms. Then:

(i) K_2 satisfies the inhomogeneous two-term recurrence

$$ 8s^2(s-1)\,\mathsf{K}_2(s+1) + 2s(s-1)^2\,\mathsf{K}_2(s-1) = 4(s-1)\,\mathsf{K}_0(s+1) - 2(2s-1)\,\mathsf{K}_0(s) + s\,\mathsf{K}_0(s-1), \tag{46} $$

with $\mathsf{K}_0 = \mathsf{K}(0;\cdot)$. *Equivalently, the normalised quotient* $Q = \mathsf{K}_2/\mathsf{K}_0$ *obeys the exact difference equation*

$$Q(s+1) - Q(s-1) = \mathsf{f}(s), \qquad \mathsf{f}(s) = -\frac{2s-1}{2s^2(s-1)^2} - \frac{(2s-1)\,\mathsf{r}(s)}{4s^2(s-1)}, \tag{47}$$

where $\mathsf{r}(s) = \mathsf{K}_0(s)/\mathsf{K}_0(s+1) = 2\,\Gamma(1+\frac{s}{2})\Gamma(\frac{1-s}{2})\big/\big(\Gamma(1-\frac{s}{2})\Gamma(\frac{1+s}{2})\big)$.

(ii) On each lattice $\{s_0-2k\}_{k\ge 0}$, r *is bounded and* $\mathsf{f}(\varsigma) = O(\varsigma^{-2})$, *so* $\sum_{k\ge 0}\mathsf{f}(s-1-2k)$ *converges absolutely and*

$$\mathsf{K}_2(s) = \mathsf{K}_0(s)\Big[c_*(s_0) + \sum_{k\ge 0}\mathsf{f}(s-1-2k)\Big] \tag{48}$$

with a constant $c_*(s_0)$ *depending only on the lattice. This constant is in fact zero (Theorem A.12).*

Proof. Since $(a)_n(b)_n = \prod_{j=0}^{n-1}\big((s+j)^2 - \alpha^2/4\big)$, one has

$$\partial_\alpha^2\big[(a)_n(b)_n\big]_{\alpha=0} = -\frac{1}{2}(s)_n^2\sum_{j=0}^{n-1}\frac{1}{(s+j)^2},$$

which gives the series for K_2. (i) Differentiate (45) twice at $\alpha = 0$: the first α-derivatives vanish by evenness, and the α^2-parts of the three coefficients contribute $-8(s-1)$, $4(2s-1)$, $-2s$ against K_0, giving (46) after division by 2. Dividing (46) by $8s^2(s-1)\mathsf{K}_0(s+1)$ and eliminating $\mathsf{K}_0(s-1)$ through the two-term relation of Lemma 7.8 yields (47). (ii) By reflection and Stirling, $\Gamma(\frac{1-s}{2})/\Gamma(1-\frac{s}{2}) \sim (|s|/2)^{-1/2}$ while the quotient of the two left-hand Gammas is $O(|s|^{1/2})$ times a trigonometric factor that is periodic on the lattice, so r is bounded there; summing (47) in steps of two telescopes to (48), the general solution of the homogeneous equation on a fixed lattice being $\mathsf{K}_0 \times$ const. □

Corollary 7.10. *Fix real* $Z > 1$ *with* $s = (1-Z)/2$ *generic, and let*

$$\mathcal{A}(s) = \frac{1}{2}\,\frac{\partial_\alpha^2\mathcal{H}(0;Z)}{\mathcal{H}(0;Z)}$$

be the function on the right-hand side of (39). *Then*

$$\operatorname{Im}\mathcal{A} = \frac{\pi\,\mathsf{K}_2(s)}{2\,\mathsf{S}(0;s)} - \tan\frac{\pi s}{2}\Big(\operatorname{Re}\mathcal{A} + \frac{\psi'(s)}{4}\Big). \tag{49}$$

Every term on the right except K_2 *and* $\operatorname{Re}\mathcal{A}$ *is in closed form (*$\mathsf{S}(0;s)$ *by* (44)*).*

Proof. Write $\mathcal{H} = \mathsf{G}\,(\mathsf{S} + i\pi\mathsf{K})$ with $\mathsf{G} = \Gamma(2s)/(\Gamma(a)\Gamma(b))$, by Proposition 7.7. All three factors are even in α, so first α-derivatives vanish at 0 and $2\,\mathcal{A} = \partial_\alpha^2\log\mathcal{H}|_0 = \partial_\alpha^2\log\mathsf{G}|_0 + \big(\mathsf{S}_2 + i\pi\mathsf{K}_2\big)/\big(\mathsf{S}_0 + i\pi\mathsf{K}_0\big)$, where $\mathsf{S}_2, \mathsf{K}_2$ are the second α-derivatives and $\mathsf{S}_0, \mathsf{K}_0$ the values. Since $\partial_\alpha^2\log\mathsf{G}|_0 = -\frac{1}{2}\psi'(s)$ is real, taking real and imaginary parts and using $\pi\mathsf{K}_0/\mathsf{S}_0 = \operatorname{Im}\mathcal{H}(0)/\operatorname{Re}\mathcal{H}(0) = \tan\frac{\pi s}{2}$ (Lemma 7.6) to eliminate S_2 between the two resulting equations gives (49). □

7.3. **Formal quadratures and the isolation of the oscillation.** We now separate the quadrature (48) into a rational part, which is summed in closed form, and a part which is the product of a single oscillatory factor and a function with an asymptotic expansion in powers of s^{-1}.

Lemma 7.11. *The forcing in (47) splits as*

$$\mathsf{f} = \mathsf{f}_{\mathrm{rat}} + \mathsf{f}_{\Gamma}, \qquad \mathsf{f}_{\mathrm{rat}}(s) = \frac{1}{2s^2} - \frac{1}{2(s-1)^2}, \qquad \mathsf{f}_{\Gamma}(s) = -\frac{(2s-1)\,\mathsf{r}(s)}{4s^2(s-1)}.$$

Moreover the Gamma quotient obeys the multiplicative step-two identity

$$\mathsf{r}(s)\,\mathsf{r}(s-1) = -\frac{4s}{s-1}. \tag{50}$$

Proof. The partial-fraction identity $\frac{1}{2s^2} - \frac{1}{2(s-1)^2} = -\frac{2s-1}{2s^2(s-1)^2}$ is immediate. Since $\mathsf{r}(s) = \mathsf{K}_0(s)/\mathsf{K}_0(s+1)$, the two-term relation $4s\,\mathsf{K}_0(s+1) + (s-1)\mathsf{K}_0(s-1) = 0$ of Lemma 7.8 gives (50). □

Proposition 7.12. *Let $\mathbb{K}$ be a field of characteristic zero and let $f(s) \in s^{-2}\mathbb{K}[[s^{-1}]]$. There is a unique $Q(s) \in s^{-1}\mathbb{K}[[s^{-1}]]$ satisfying $Q(s+1) - Q(s-1) = f(s)$. If ∂_s^{-1} denotes the primitive with zero constant term in $s^{-1}\mathbb{K}[[s^{-1}]]$, then*

$$Q = \frac{1}{2\sinh \partial_s} f = \frac{1}{2}\partial_s^{-1} f + \sum_{j\geq 1} \frac{(1-2^{2j-1})\mathcal{B}_{2j}}{(2j)!}\,\partial_s^{2j-1} f. \tag{51}$$

In particular (51) applies to $\mathsf{f}_{\mathrm{rat}}$; to apply it to f_{Γ} one must first separate the lattice-periodic factor of r (Lemma 7.15 below).

Proof. On formal Laurent series, the central difference is $e^{\partial_s} - e^{-\partial_s} = 2\sinh \partial_s$. Since

$$\frac{x}{\sinh x} = \sum_{j\geq 0} \frac{2(1-2^{2j-1})\mathcal{B}_{2j}}{(2j)!} x^{2j},$$

its inverse on series with no constant term is (51). The leading operator is ∂_s, which is bijective from $s^{-1}\mathbb{K}[[s^{-1}]]$ to $s^{-2}\mathbb{K}[[s^{-1}]]$; this also proves existence and uniqueness. □

Corollary 7.13. *The unique formal solution of*

$$Q_{\mathrm{rat}}(s+1) - Q_{\mathrm{rat}}(s-1) = \mathsf{f}_{\mathrm{rat}}(s), \qquad Q_{\mathrm{rat}}(s) \in s^{-1}\mathbb{Q}[[s^{-1}]],$$

is also the distinguished analytic solution which tends to zero along a leftward step-two lattice. It has the closed form

$$Q_{\mathrm{rat}}(s) = \frac{1}{8}\left[\psi'\!\left(\frac{1-s}{2}\right) - \psi'\!\left(1-\frac{s}{2}\right)\right]. \tag{52}$$

Equivalently, $Q_{\mathrm{rat}}(s) = \sum_{k\geq 0} \mathsf{f}_{\mathrm{rat}}(s-1-2k)$, the series converging whenever the lattice avoids the poles.

Proof. From $\mathsf{f}_{\mathrm{rat}}(\varsigma) = \frac{1}{2}(\varsigma^{-2} - (\varsigma-1)^{-2})$ and $\sum_{k\geq 0}(a+2k)^{-2} = \frac{1}{4}\psi'(a/2)$, the lattice sum is $\frac{1}{2}\cdot\frac{1}{4}[\psi'(\frac{1-s}{2}) - \psi'(1-\frac{s}{2})]$, which is (52). Alternatively, the trigamma recurrence $\psi'(z+1) = \psi'(z) - z^{-2}$ shows directly that the central difference of (52) is $\mathsf{f}_{\mathrm{rat}}$. Its asymptotic expansion has no constant term, so uniqueness follows from Proposition 7.12. □

Remark 7.14. On functions, the kernel of the central-difference operator contains all 2-periodic functions; the constant $c_*(s_0)$ in (48) reflects this ambiguity. On formal series in $s^{-1}\mathbb{K}[[s^{-1}]]$ the kernel is zero.

Lemma 7.15. *Put*

$$\widehat{\mathsf{r}}(s) := s\left[\frac{\Gamma(s/2)}{\Gamma((1+s)/2)}\right]^2. \tag{53}$$

Then

$$\mathsf{r}(s) = \tan\frac{\pi s}{2}\,\widehat{\mathsf{r}}(s). \tag{54}$$

Moreover $\widehat{\mathsf{r}}$ *has a canonical Poincaré expansion*

$$\widehat{\mathsf{r}}(s) \sim 2\exp\left(2\sum_{n\ge1}\frac{(-1)^{n+1}\left[\mathcal{B}_{n+1}(0)-\mathcal{B}_{n+1}(\frac12)\right]}{n(n+1)}\left(\frac{2}{s}\right)^n\right) \in 2+s^{-1}\mathbb{Q}[[s^{-1}]]. \tag{55}$$

Thus every non-formal dependence of r *on the step-two lattice is carried by the single factor* $\tan(\pi s/2)$.

Proof. The reflection formulas

$$\Gamma\Big(\frac{1-s}{2}\Big)\Gamma\Big(\frac{1+s}{2}\Big) = \frac{\pi}{\cos(\pi s/2)}, \qquad \Gamma\Big(\frac{s}{2}\Big)\Gamma\Big(1-\frac{s}{2}\Big) = \frac{\pi}{\sin(\pi s/2)},$$

together with $\Gamma(1+s/2) = (s/2)\Gamma(s/2)$, give (54). Formula (55) is the standard formal logarithmic expansion of the Gamma quotient at $z = s/2$, *squared* (whence the factor 2 in the exponent), followed by exponentiation; the leading term is $s\cdot(s/2)^{-1} = 2$. □

Proposition 7.16. *Define the formal forcing*

$$\widehat{\mathsf{f}}_\Gamma(s) := -\frac{(2s-1)\,\widehat{\mathsf{r}}(s)}{4s^2(s-1)} \in s^{-2}\mathbb{Q}[[s^{-1}]], \tag{56}$$

and let

$$\widehat{Q}_\Gamma := \frac{1}{2\sinh\partial_s}\,\widehat{\mathsf{f}}_\Gamma \in s^{-1}\mathbb{Q}[[s^{-1}]]. \tag{57}$$

For a fixed step-two lattice avoiding the poles, the Gamma part of the convergent quadrature has the exact factorisation

$$\sum_{k\ge0}\mathsf{f}_\Gamma(s-1-2k) = -\cot\frac{\pi s}{2}\sum_{k\ge0}\widehat{\mathsf{f}}_\Gamma(s-1-2k). \tag{58}$$

In the upper half-plane the second sum is asymptotic to $\widehat{Q}_\Gamma(s)$ *up to an exponentially small error (Lemma 7.27(c)); this expansion holds in sectors off the real axis, not on real step-two lattices.*

Proof. By Lemma 7.15, $\mathsf{f}_\Gamma(\varsigma) = \tan(\pi\varsigma/2)\,\widehat{\mathsf{f}}_\Gamma(\varsigma)$. For $\varsigma = s-1-2k$,

$$\tan\frac{\pi(s-1-2k)}{2} = -\cot\frac{\pi s}{2},$$

independently of k, which proves (58). □

Proposition 7.17. *Let* $Q = \mathsf{K}_2/\mathsf{K}_0$ *and*

$$T(s) = \sum_{k\ge0}\widehat{\mathsf{f}}_\Gamma(s-1-2k). \tag{59}$$

On a fixed step-two lattice the quadrature (48) *reads*

$$Q(s) = c_*(s_0) + Q_{\mathrm{rat}}(s) - \cot\frac{\pi s}{2}\,T(s), \tag{60}$$

with Q_{rat} *given by* (52)*, and the identity* (49) *is equivalent to*

$$\operatorname{Im}\mathcal{A} = -\frac12\,T(s) + \tan\frac{\pi s}{2}\,\mathcal{E}(s), \tag{61}$$

where

$$\mathcal{E}(s) = \frac{c_*(s_0)}{2} + \frac{Q_{\mathrm{rat}}(s)}{2} - \operatorname{Re}\mathcal{A} - \frac{\psi'(s)}{4}. \tag{62}$$

Proof. Since $\pi\mathsf{K}_0/\mathsf{S}_0 = \tan(\pi s/2)$, formula (49) reads

$$\operatorname{Im}\mathcal{A} = \frac{1}{2}\tan\frac{\pi s}{2}\,Q(s) - \tan\frac{\pi s}{2}\left(\operatorname{Re}\mathcal{A} + \frac{\psi'(s)}{4}\right).$$

The decomposition (60) is (48) combined with Corollary 7.13 and (58). Inserting it, the product of tangent and cotangent is 1, and (61)–(62) follow. □

Remark 7.18. By Corollary 1.4, the series $\sum_n \operatorname{Re} g_n Z^{-n}$ is the asymptotic expansion of

$$\int_0^\infty \frac{t\,e^{2st}}{\sinh 2t}\,dt = \frac{1}{8}\,\psi'\!\left(\frac{1-s}{2}\right), \qquad \operatorname{Re} s < 0.$$

On the real axis the function $\operatorname{Re}\mathcal{A}$ does not coincide with this function: at $s = 1$ the value $\operatorname{Re}\mathcal{A} = -\pi^2/48$ is finite (Theorem 7.24 below), whereas $\psi'\big(\frac{1-s}{2}\big)$ has a double pole there. We denote the difference by

$$\Delta(s) := \operatorname{Re}\mathcal{A}(s) - \frac{1}{8}\,\psi'\!\left(\frac{1-s}{2}\right).$$

It is evaluated on the real axis in Subsections 7.4 and 7.5 and, under a periodicity hypothesis, in closed form in Proposition 7.28.

7.4. **Derivative-free contiguities and the boundary transfer.** For the boundary analysis it is convenient to pass to the symmetric normalisation $\eta = \alpha/2$: put

$$Y_s(\eta; z) := {}_2F_1(s-\eta, s+\eta; 2s; z), \qquad \mathcal{H}(\alpha; Z) = Y_s(\alpha/2; 2^+),$$

$$\mathcal{A} = \frac{1}{8}\,\partial_\eta^2 \log Y_s(\eta; 2^+)\big|_{\eta=0},$$

and write, at the upper boundary value $z = 2^+ = 2 + i0$,

$$Y_s^{[0]} = Y_s(0; 2^+), \qquad \dot Y_s^{[0]} = \partial_z Y_s(0; 2^+),$$

$$Y_s^{[2]} = \partial_\eta^2 Y_s(\eta; 2^+)\big|_{\eta=0}, \qquad \dot Y_s^{[2]} = \partial_z\partial_\eta^2 Y_s(\eta; 2^+)\big|_{\eta=0},$$

$$R_s = \frac{Y_s^{[2]}}{Y_s^{[0]}}, \qquad \mathfrak{p}_s = \frac{\dot Y_s^{[0]}}{Y_s^{[0]}}, \qquad J_s = \frac{\dot Y_s^{[2]}}{Y_s^{[0]}} - \frac{Y_s^{[2]}\dot Y_s^{[0]}}{(Y_s^{[0]})^2} = \partial_z R_s(2^+),$$

so that $\mathcal{A} = R_s/8$ and $\Delta(s)$ of Remark 7.18 is $\frac{1}{8}\operatorname{Re} R_s - \frac{1}{8}\psi'(\frac{1-s}{2})$. All identities below hold first away from parameter poles and then by meromorphic continuation; the branch is unchanged throughout because every function is continued to the same upper boundary value.

Lemma 7.19. *Write* $F^{\uparrow}(a,b) = {}_2F_1^{\uparrow}(a,b;a+b;2)$. *Then*

$$a\,{}_2F_1^{\uparrow}(a+1,b;a+b;2) + (b-a)\,F^{\uparrow}(a,b) + b\,{}_2F_1^{\uparrow}(a-1,b;a+b;2) = 0, \tag{63}$$

and symmetrically in (a,b). *Consequently the argument derivative is eliminated:*

$$\begin{aligned} 2\,\partial_z {}_2F_1^{\uparrow}(a,b;a+b;z)\big|_{z=2} &= -b\big(F^{\uparrow}(a,b) + {}_2F_1^{\uparrow}(a-1,b;a+b;2)\big) \\ &= -a\big(F^{\uparrow}(a,b) + {}_2F_1^{\uparrow}(a,b-1;a+b;2)\big). \end{aligned} \tag{64}$$

Proof. Gauss's contiguous relation $a(z-1)F(a+1,b;c;z)+(2a-c-az+bz)F(a,b;c;z)+(c-a)F(a-1,b;c;z)=0$ [18] at $c=a+b$, $z=2$ gives (63). The derivative contiguities $zF' = a(F(a+1,b;c;z)-F) = b(F(a,b+1;c;z)-F)$ are standard [2, Sec. 15.5]; substituting (63) and its (a,b)-exchange gives (64). □

The lower-parameter contiguity $(\vartheta+c)F(a,b;c+1;z)=cF(a,b;c;z)$, $\vartheta=z\partial_z$, combined with $\partial_z F(a,b;c;z)=\frac{ab}{c}F(a+1,b+1;c+1;z)$, gives the first s-shift of the zero-balanced family,

$$(\vartheta+2s+1)\,Y_{s+1}(\eta;z)=\frac{2s(2s+1)}{s^2-\eta^2}\,\partial_z Y_s(\eta;z). \tag{65}$$

Proposition 7.20. *With $d=d_s(\eta)=s^2-\eta^2$,*

$$d\,(\vartheta+2s)(\vartheta+2s+1)\,Y_{s+1}(\eta;z)=2s(2s+1)\big((\vartheta+s)^2-\eta^2\big)Y_s(\eta;z), \tag{66}$$

an identity of power series at $z=0$, hence of upper boundary values. At $z=2$ it is functionally independent of (65): after use of the two hypergeometric equations, the determinant of the coefficients of $Y_{s+1}(\eta;2)$ and $\partial_z Y_{s+1}(\eta;2)$ is $4d^2$.

Proof. The quotient of the nth Taylor coefficients of Y_{s+1} and Y_s is

$$\frac{(n+s)^2-\eta^2}{s^2-\eta^2}\cdot\frac{2s(2s+1)}{(n+2s)(n+2s+1)},$$

which is (66) coefficientwise, since $\vartheta z^n=nz^n$. □

Theorem 7.21. *At the upper boundary value, the pair (65)–(66) reduces to the two boundary rows*

$$2\,\partial_z Y_{s+1}(\eta;2)+(2s+1)Y_{s+1}(\eta;2)=\frac{2s(2s+1)}{d}\,\partial_z Y_s(\eta;2), \tag{67}$$

$$\begin{aligned} &d\left\{4(s-1)\partial_z Y_{s+1}(\eta;2)+2(s^2-s-1+\eta^2)Y_{s+1}(\eta;2)\right\}\\ &\qquad=2s(2s+1)\left\{-2\partial_z Y_s(\eta;2)-d\,Y_s(\eta;2)\right\}, \end{aligned} \tag{68}$$

an invertible two-by-two system. Eliminating $\partial_z Y_{s+1}$ gives the scalar transfer

$$Y_{s+1}(\eta;2^+)=\frac{s(2s+1)}{d^2}\left[2s\,\partial_z Y_s(\eta;2^+)+d\,Y_s(\eta;2^+)\right], \tag{69}$$

and, twice differentiated in η at zero,

$$Y_{s+1}^{[2]}=\frac{2s+1}{s^3}\big(2s\,\dot Y_s^{[2]}+s^2Y_s^{[2]}-2Y_s^{[0]}\big)+\frac{4(2s+1)}{s^5}\big(2s\,\dot Y_s^{[0]}+s^2Y_s^{[0]}\big). \tag{70}$$

Equivalently, with $U_s(\eta)=\big(Y_s(\eta;2^+),\,\partial_z Y_s(\eta;2^+)\big)^T$,

$$U_{s+1}(\eta)=M_s(\eta)\,U_s(\eta),\qquad M_s(\eta)=\begin{pmatrix}\dfrac{s(2s+1)}{d} & \dfrac{2s^2(2s+1)}{d^2}\\[2mm] -\dfrac{s(2s+1)^2}{2d} & \dfrac{s(2s+1)\{d-s(2s+1)\}}{d^2}\end{pmatrix}, \tag{71}$$

and, since M_s is even in η, the second derivatives with respect to η satisfy

$$U_{s+1}^{[2]}=M_s(0)\,U_s^{[2]}+M_s''(0)\,U_s(0),\qquad U_s^{[2]}=\partial_\eta^2 U_s(\eta)\big|_{\eta=0}=(Y_s^{[2]},\dot Y_s^{[2]})^T. \tag{72}$$

Proof. The hypergeometric equation for Y_{s+1}, respectively Y_s, evaluated at $z=2$, converts the second-order operators in (66) into the displayed first-order boundary rows; (67) is (65) at $z=2$. The determinant of the system is proportional to $d^2\neq 0$; elimination gives (69), and the first row returns $\partial_z Y_{s+1}$, i.e. (71). Formula (70) is ∂_η^2 of (69) at $\eta=0$ using $d=s^2-\eta^2$ and evenness in η; (72) is the same statement for the pair. □

Proposition 7.22. *For the upper boundary value,*

$$\mathfrak{p}_s=-\frac{s}{2}+\frac{i\,s^2(s+\frac12)}{2(2s+1)}\left[\frac{\Gamma(\frac{s+1}{2})}{\Gamma(\frac{s+2}{2})}\right]^2. \tag{73}$$

Consequently the coefficients in (72) are all explicit, and

$$J_s=\frac{1}{s}+\frac{2s\mathfrak{p}_s+s^2}{2s}\left(R_{s+1}-R_s-\frac{4}{s^2}\right): \tag{74}$$

thus J_s is expressed through the first difference $R_{s+1}-R_s$.

Proof. At $\eta=0$, (69) reads $Y^{[0]}_{s+1}=\frac{2s+1}{s^3}(2s\,\dot Y^{[0]}_s+s^2Y^{[0]}_s)$, so $\mathfrak{p}_s=\frac{s^2}{2(2s+1)}\frac{Y^{[0]}_{s+1}}{Y^{[0]}_s}-\frac{s}{2}$. By Lemma 7.6, $Y^{[0]}_{s+1}/Y^{[0]}_s=e^{i\pi/2}\,(s+\frac12)\,[\Gamma(\frac{s+1}{2})/\Gamma(\frac{s+2}{2})]^2$, which gives (73). Formula (74) follows by taking two η-derivatives at $\eta=0$ of the logarithm of

$$\frac{Y_{s+1}(\eta;2^+)}{Y_s(\eta;2^+)}=\frac{s(2s+1)}{d^2}\left\{2s\,\frac{\partial_z Y_s(\eta;2^+)}{Y_s(\eta;2^+)}+d\right\}.$$

□

Remark 7.23. The transfer (71) is invertible away from $s^2=\eta^2$, and it is two-dimensional: formula (74) exchanges the derivative J_s for the adjacent value R_{s+1} but does not eliminate it. The recurrence therefore determines (R_s,J_s) on a class $s_*+\mathbb{Z}$ once the pair (R_{s_*},J_{s_*}) is known at one point of the class. Terminating points $s=-N$ cannot be used for this purpose, because $2s=-2N$ is then a pole of the denominator parameter away from $\eta=0$. The next subsection supplies these initial values, which we call *anchors*.

7.5. Exact anchors: polylogarithms at $s=1$, Euler moments for every class.

Theorem 7.24. *At $s=1$, $Y_1(0;z)={}_2F_1(1,1;2;z)=-\log(1-z)/z$, so for the upper boundary convention*

$$Y_1^{[0]}=\frac{i\pi}{2},\qquad \dot Y_1^{[0]}=-\frac12-\frac{i\pi}{4}. \tag{75}$$

Furthermore

$$R_1=-\frac{\pi^2}{6}-\frac{7i}{\pi}\,\zeta(3),\qquad J_1=2\log 2-\frac{7}{\pi^2}\,\zeta(3)-\frac{i\pi}{3}, \tag{76}$$

equivalently

$$Y_1^{[2]}=\frac72\zeta(3)-\frac{i\pi^3}{12},\qquad \dot Y_1^{[2]}=\frac{\pi^2}{4}-\frac74\zeta(3)+i\Big(\pi\log 2+\frac{\pi^3}{24}\Big).$$

Proof. Formulae (75) are immediate from $\log(1-z)\to-i\pi$. Define $I(u)=\int_0^u\log^2(1-t)\,t^{-1}\,dt$; an antiderivative calculation gives

$$I(u)=\log^2(1-u)\log u+2\log(1-u)\,\mathrm{Li}_2(1-u)-2\,\mathrm{Li}_3(1-u)+2\zeta(3),$$

whence, at the upper boundary, $I(2^+) = \frac{7}{2}\zeta(3) - \pi^2\log 2 + \frac{i\pi^3}{6}$. The transverse variational equation at $s = 1$ and reduction of order over the explicit solution $Y_1(0; z)$ yield

$$R_1 = -2\int_0^{2^+} \frac{I(u)}{(1-u)\log^2(1-u)}\,du, \qquad J_1 = -\frac{2}{\pi^2}\,I(2^+),$$

which proves the J_1 formula. For R_1, substitute $x = \log(1-u)$; since $dI/dx = -x^2e^x/(1-e^x)$, integration by parts gives

$$R_1 = 2\left[-\frac{I}{x} + x\log(1-e^x) + \mathrm{Li}_2(e^x)\right]_{x=0}^{x=-i\pi},$$

and the endpoint values $\mathrm{Li}_2(1) = \pi^2/6$, $\mathrm{Li}_2(-1) = -\pi^2/12$, $\mathrm{Li}_3(-1) = -\frac{3}{4}\zeta(3)$ with $\log(-1-i0) = -i\pi$ give (76); the $\log 2$ terms cancel exactly. □

Combining Theorem 7.24 with the recurrence (72) determines, by finite algebra, the pair (R_s, J_s) at every positive integer s; the first propagated value is

$$R_2 = -\frac{\pi^2}{2} + 4 - \frac{7i}{\pi}\,\zeta(3) - \frac{i\pi}{2}\Big(4\log 2 - \frac{14}{\pi^2}\,\zeta(3) - 2\Big). \tag{77}$$

Theorem 7.25. *Let $0 < \mathrm{Re}\, s_* < 1$ with $Y_{s_*}^{[0]} \neq 0$. For $\varepsilon > 0$ put*

$$\mathsf{w}_{s_*,\varepsilon}(t) = t^{s_*-1}(1-t)^{s_*-1}\big(1-(2+i\varepsilon)t\big)^{-s_*}, \qquad \mathsf{T}_\varepsilon(t) = \frac{t}{1-(2+i\varepsilon)t},$$
$$\mathsf{L}_\varepsilon(t) = \log t - \log(1-t) + \log\big(1-(2+i\varepsilon)t\big),$$

and the normalised complex expectation $\langle f\rangle_{s_*,\varepsilon} = \int_0^1 \mathsf{w}_{s_*,\varepsilon} f\,dt \big/ \int_0^1 \mathsf{w}_{s_*,\varepsilon}\,dt$; all integrals converge absolutely for every $\varepsilon > 0$. Write $\langle f\rangle_{s_*} = \lim_{\varepsilon\downarrow 0}\langle f\rangle_{s_*,\varepsilon}$ whenever the limit exists. Then $\langle \mathsf{L}^2\rangle_{s_*}$ exists as an absolutely convergent integral at $\varepsilon = 0$ (boundary values $1-2t-i0$), the T-moments exist as $\varepsilon \downarrow 0$ limits (equivalently, as absolutely convergent integrals over any contour from 0 to 1 that leaves the segment near $t = \frac{1}{2}$ into the upper half-plane), and

$$R_{s_*} = -2\psi'(s_*) + \langle \mathsf{L}^2\rangle_{s_*}, \qquad J_{s_*} = \lim_{\varepsilon\downarrow 0}\Big[s_*\big\{\langle \mathsf{L}_\varepsilon^2\mathsf{T}_\varepsilon\rangle_{s_*,\varepsilon} - \langle \mathsf{L}_\varepsilon^2\rangle_{s_*,\varepsilon}\langle \mathsf{T}_\varepsilon\rangle_{s_*,\varepsilon}\big\} - 2\langle \mathsf{L}_\varepsilon\mathsf{T}_\varepsilon\rangle_{s_*,\varepsilon}\Big]. \tag{78}$$

Consequently every real *residual class $s_0 + \mathbb{Z}$ avoiding the parameter poles and the zeros of the boundary solution carries a canonical anchor: choose the representative $s_* \in s_0 + \mathbb{Z}$ with $0 < s_* < 1$, the integer class being covered by Theorem 7.24 at $s_* = 1$; the invertible transfer (71)–(72) and its inverse then determine (R_s, J_s) uniquely on the whole class, independently of the representative used.*

Proof. Convergence and existence of the limits. For $\varepsilon > 0$ the factor $1-(2+i\varepsilon)t$ is bounded away from zero on $[0,1]$, so all moments converge absolutely; at $\varepsilon = 0$ the integrand of $\langle \mathsf{L}^2\rangle_{s_*}$ is $O\big(|1-2t|^{-\mathrm{Re}\, s_*}\log^2|1-2t|\big)$ near $t = \frac{1}{2}$, absolutely integrable for $\mathrm{Re}\, s_* < 1$, while the T-moments carry an extra factor $(1-2t-i0)^{-1}$ and are *not* absolutely convergent at $\varepsilon = 0$: they must be taken as the stated limits. The limits exist by contour deformation: the singularity of the integrand sits at $t_\varepsilon = 1/(2+i\varepsilon)$, which lies *below* the real segment and tends to $\frac{1}{2}$ as $\varepsilon \downarrow 0$; the integrand is analytic in t in the upper half-disc around $t = \frac{1}{2}$, so $\int_0^1$ may be replaced, for every $\varepsilon \geq 0$, by the integral over the contour $\mathcal{C}_{r_0}$ which follows the segment but crosses a semicircle of fixed radius $r_0 \in (0, \frac{1}{4})$ *above* $t = \frac{1}{2}$. On $\mathcal{C}_{r_0}$ the integrands converge uniformly as $\varepsilon \downarrow 0$ (the

denominator is bounded away from zero), so each moment tends to the corresponding absolutely convergent $\mathcal{C}_{r_0}$-integral, which is independent of r_0; this realises the boundary prescription $1-2t-i0$.

The formulae. Euler's integral representation gives, for η near zero and $\varepsilon \geq 0$ interpreted as above,

$$Y_{s_*}(\eta; 2^+) = \frac{\Gamma(2s_*)}{\Gamma(s_*+\eta)\Gamma(s_*-\eta)} \int_{\mathcal{C}_{r_0}} \mathsf{w}_{s_*}(t)\, e^{\eta \mathsf{L}(t)}\, dt.$$

The family is even in η, hence $\langle \mathsf{L} \rangle_{s_*} = 0$. Two η-derivatives of the logarithm give the R_{s_*} formula, the $-2\psi'(s_*)$ coming from the Gamma prefactor. For fixed η, logarithmic z-differentiation of the integral has score $s_*\mathsf{T}$ at $\eta = 0$, while $\partial_z \mathsf{L} = -\mathsf{T}$; differentiating the normalised second moment gives the J_{s_*} formula, first at $\varepsilon > 0$ and then in the limit, which is (78). The transfer matrix is invertible away from its explicit divisors, so the anchor propagates uniquely in both directions; if two admissible representatives are chosen, Euler's formula identifies both with the same analytic boundary solution, and uniqueness of the propagation proves independence of the choice. □

Theorems 7.24 and 7.25, together with the transfer (71)–(72), give R_s, and hence $\mathcal{A} = R_s/8$ and the difference $\Delta(s)$ of Remark 7.18, as explicit convergent expressions on every real class.

7.6. **The asymptotic matching: proof of Theorem 1.6.** This subsection proves Theorem 1.6. The real part is obtained first, directly from Corollary 1.4. The imaginary part is obtained by an asymptotic matching in the lower half-plane; the estimates on which it relies are stated in Lemma 7.27 and proved in Appendix A.

Theorem 7.26. *As formal series in Z^{-1},*

$$\sum_{n\geq 1} \operatorname{Re} g_n\, Z^{-n} = \frac{1}{16}\,\psi'\Big(\frac{Z+1}{4}\Big) - \frac{1}{16}\,\psi'\Big(\frac{Z+3}{4}\Big) + \frac{1}{4}\,\psi'\Big(\frac{Z+1}{2}\Big), \tag{79}$$

where each ψ' stands for its Bernoulli asymptotic expansion.

Proof. From $\psi'(x) = \int_0^\infty u e^{-xu}(1-e^{-u})^{-1}\,du$ with $u = 4t$, $4t$, $2t$ respectively, the right side of (79) is the Watson expansion of $\int_0^\infty t e^{-Zt}\varphi_{\mathrm{R}}(t)\,dt$ with

$$\varphi_{\mathrm{R}}(t) = \frac{e^{-t}-e^{-3t}}{1-e^{-4t}} + \frac{e^{-t}}{1-e^{-2t}} = \frac{1}{2\cosh t} + \frac{1}{2\sinh t}.$$

Watson's correspondence $\int_0^\infty t^n e^{-Zt} dt = n!\, Z^{-n-1}$ is coefficientwise exact, so the n-th coefficient of the right side of (79) is the $(n-1)$-st Taylor coefficient of $t\varphi_{\mathrm{R}}(t)$. By Corollary 1.4, $\sum_{n\geq 1} \operatorname{Re} g_n\, t^{n-1}/(n-1)! = \operatorname{Re}\Phi(t)/t = te^t/\sinh 2t = t(\operatorname{sech} t + \operatorname{csch} t)/2 = t\varphi_{\mathrm{R}}(t)$, which proves (79). □

By the duplication formula $\psi'(2x) = \frac{1}{4}\big[\psi'(x)+\psi'(x+\frac{1}{2})\big]$, the right-hand side of (79) is also equal to $\frac{1}{8}\psi'\big(\frac{Z+1}{4}\big)$; with $Z = 1-2s$ this is the expression $\frac{1}{8}\psi'\big(\frac{1-s}{2}\big)$ of Remark 7.18.

Lemma 7.27.

(a) *Fix $0 < \theta_0 < 1$. For every N there is C_N such that, uniformly on the bilateral cone $|\operatorname{Re} Z| \leq \theta_0 |\operatorname{Im} Z|$, $|\operatorname{Im} Z| \geq y_{\min}(\theta_0, N)$ (*both *signs of* $\operatorname{Im} Z$*, in particular*

on every vertical strip)

$$\Big|\mathcal{A} - \sum_{n=1}^{N} g_n Z^{-n}\Big| \le C_N\,|Z|^{-N-1}.$$

(b) *On every closed subsector of* $|\arg(\pm is)| < \pi/2$ *the closed blocks* K_0, S_0, $\widehat{\mathsf{r}}$, Q_{rat} *and the trigamma functions are asymptotic to their Bernoulli–Stirling series, with uniform remainders after each truncation.*

(c) *For every* $\delta > 0$ *and every* N, *uniformly on* $\operatorname{Im} s \ge \delta|s|$,

$$\Big|T(s) - \sum_{j=1}^{N} \omega_j\, s^{-j}\Big| \le C_{\delta,N}\,|s|^{-N-1} + C_\delta\,\frac{e^{-\pi \operatorname{Im} s}}{\operatorname{Im} s}, \qquad T(s) = \sum_{k\ge 0} \widehat{\mathsf{f}}_\Gamma(s-1-2k),$$

where $\sum_j \omega_j s^{-j}$ *is the formal series* $\widehat{Q}_\Gamma = (2\sinh\partial_s)^{-1}\widehat{\mathsf{f}}_\Gamma$ *of Proposition 7.12.*

The mirror statements hold in the lower half-plane by Schwarz reflection, with the same (rational) coefficients.

Proof. Appendix A: (a) is Proposition A.3, proved from the coefficient bound $|G_n| \le C(n+1)!\,2^{-n/2}$ (Lemma A.1), the elementary product inequality $(j+1) \le \sqrt{2}\,|Z-j|$ valid on the cone (Lemma A.2), and the exact function-level identity (38) between the convergent inverse-factorial series (6) and the boundary family $\mathcal{H}$; (b) is Lemma A.5; (c) is Proposition A.8, proved from a two-regime uniform expansion of $\widehat{\mathsf{f}}_\Gamma$ on horizontal lines (Stirling on $|\arg\varsigma| \le \frac{3}{4}\pi$; reflection, a Beta-integral bound and $\cot(\pi\varsigma/2) = -i + O(e^{-\pi\operatorname{Im}\varsigma})$ on the left regime), the $k \le |s|$ / $k > |s|$ splitting of the lattice sum, the uniform Hurwitz–Bernoulli expansions of the moment sums $\sum_k (s-1-2k)^{-j}$, and the formal uniqueness of Proposition 7.12. □

The following proposition gives a closed form on the real axis for the difference Δ of Remark 7.18, under a periodicity hypothesis.

Proposition 7.28. *Define the analytic extension of the difference* Δ *of Remark 7.18,*

$$\widetilde{\Delta}(s) := \frac{\mathcal{A}^{\uparrow}(s) + \mathcal{A}^{\downarrow}(s)}{2} - \frac{1}{16}\,\psi'\Big(\frac{1-s}{2}\Big) + \frac{1}{16}\,\psi'\Big(1-\frac{s}{2}\Big) - \frac{1}{4}\,\psi'(1-s),$$

with $\mathcal{A}^{\uparrow,\downarrow}$ *the two boundary components of Appendix A.4, so that* $\widetilde{\Delta} = \operatorname{Re}\mathcal{A} -$ *(trigamma part) on the real axis. Assume*

(P) $\widetilde{\Delta}$ *is 2-periodic, with poles confined to the odd class, of order at most two.*

Then, on the real axis,

$$\operatorname{Re}\mathcal{A}(s) = \frac{1}{16}\,\psi'\Big(\frac{1-s}{2}\Big) - \frac{1}{16}\,\psi'\Big(1-\frac{s}{2}\Big) + \frac{1}{4}\,\psi'(1-s) - \frac{\pi^2}{8}\,\sec^2\frac{\pi s}{2}, \tag{80}$$

with convergent trigamma functions; that is, $\widetilde{\Delta} = -\frac{\pi^2}{8}\sec^2\frac{\pi s}{2}$. *Consequently*

$$\mathcal{E}(s) = \frac{\pi^2}{8}\,\sec^2\frac{\pi s}{2} - \frac{\pi^2}{4}\,\csc^2(\pi s), \tag{81}$$

and the identity (61) becomes explicit:

$$\operatorname{Im}\mathcal{A}(s) = -\frac{1}{2}\,T(s) + \tan\frac{\pi s}{2}\,\mathcal{E}(s), \qquad T(s) = \sum_{k\ge 0}\widehat{\mathsf{f}}_\Gamma(s-1-2k). \tag{82}$$

Proof. The function $\widetilde{\Delta}$ decays at both vertical ends of the strip $\operatorname{Re} s \in [0,2]$: by the two-component relation (106) and Lemma 7.27(a), in the upper half-plane $\mathcal{A}^{\downarrow} = -\frac{i}{2}X + O(e^{-\pi \operatorname{Im} s})$ while $\mathcal{A}^{\uparrow}$ is asymptotic to the series $\sum_n g_n Z^{-n}$, so $(\mathcal{A}^{\uparrow} + \mathcal{A}^{\downarrow})/2$ is asymptotic to the *real* trigamma series of Theorem 7.26, which the convergent trigamma part reproduces (Lemma 7.27(b)); the lower end is the mirror statement. Under hypothesis (P) the classification lemma (Lemma A.11, with $k_1 = 2$, $k_0 = 0$) gives $\widetilde{\Delta} = c \sec^2 \frac{\pi s}{2}$ for some constant c. The principal part at $s = 1$ determines c: the trigamma part of $\widetilde{\Delta}$ has the double pole $-\left[\frac{1}{16} \cdot 4 + \frac{1}{4}\right](s-1)^{-2} = -\frac{1}{2}(s-1)^{-2}$, while $\mathcal{A}^{\uparrow,\downarrow}$ are finite at $s = 1$ (Theorem 7.24), and $\sec^2 \frac{\pi s}{2} \sim 4/(\pi^2(s-1)^2)$; hence $c = -\pi^2/8$. The finite part at $s = 1$ is consistent with the value given by the polylogarithmic evaluation: $\operatorname{Re} R_1/8 = -\pi^2/48$ (Theorem 7.24). Formula (81) follows from (62) with $c_*(s_0) = 0$ (Theorem A.12) and the reflection $\psi'(s) = \pi^2 \csc^2(\pi s) - \psi'(1-s)$; and (82) is (61). □

We can now prove Theorem 1.6.

Proof of Theorem 1.6. We describe the structure of the argument; the details are given in Appendix A.6. The two boundary components obey the exact two-component relation (Lemma A.9, Corollary A.13)

$$\Big(\tan\frac{\pi s}{2} - i\Big)\mathcal{A}^{\uparrow} + \Big(\tan\frac{\pi s}{2} + i\Big)\mathcal{A}^{\downarrow} = -T(s) + \tan\frac{\pi s}{2}\Big(Q_{\mathrm{rat}}(s) - \frac{\psi'(s)}{2}\Big) =: X(s),$$

which uses the vanishing of the lattice constant (Theorem A.12). Along the vertical line $s = -\frac{1}{2} + iY$, $Y \to -\infty$, one has $\tan\frac{\pi s}{2} + i = O(e^{-\pi|Y|})$, so that the term in $\mathcal{A}^{\downarrow}$ is exponentially small, $\tan\frac{\pi s}{2} - i \to -2i$, and $\mathcal{A}^{\uparrow}$ is asymptotic to $\sum_n g_n Z^{-n}$ by the *bilateral* expansion of Lemma 7.27(a); hence $\mathcal{A}^{\uparrow} = \frac{i}{2}X + O(e^{-\pi|Y|})$. Expanding X by Lemma 7.27(b),(c), with $\tan\frac{\pi s}{2} = -i + O(e^{-\pi|Y|})$, and using the formal reflection parity $S_\psi(1-s) = -S_\psi(s)$ of the Stirling series of ψ' (Lemma A.14) to put the real part into trigamma-kernel form, one obtains exactly (13): the real part is Theorem 7.26, and the imaginary part is (14). □

Lemma 7.29. *Write the series (56) in powers of Z^{-1}, $\widehat{\mathsf{f}}_\Gamma = \sum_{n\ge 2} f_n Z^{-n}$, and let $\varphi(t) = \sum_{n\ge 2} f_n t^{n-1}/(n-1)!$ be its Borel transform. Then the Borel transform of $\widehat{Q}_\Gamma$ is $\varphi(t)/(2\sinh 2t)$, and*

$$W(t) = -\frac{\varphi(t)}{4t}.$$

Proof. Since $s \pm 1$ corresponds to $Z \mp 2$, the equation $\widehat{Q}_\Gamma(s+1) - \widehat{Q}_\Gamma(s-1) = \widehat{\mathsf{f}}_\Gamma(s)$ reads $\widehat{Q}_\Gamma(Z-2) - \widehat{Q}_\Gamma(Z+2) = \widehat{\mathsf{f}}_\Gamma(Z)$. A shift $Z \mapsto Z + c$ multiplies Borel transforms by e^{-ct} (Remark 3.3), so the Borel transform $\widehat{q}(t)$ of $\widehat{Q}_\Gamma$ satisfies $(e^{2t} - e^{-2t})\,\widehat{q}(t) = \varphi(t)$. By (14), $\operatorname{Im}\Phi(t)/t = -\frac{1}{2}\widehat{q}(t) = -\varphi(t)/(4\sinh 2t)$, and the formula for W follows from (12). □

7.7. **The arithmetic step: proof of Theorem 1.5.** By Theorem 1.6 and Lemma 7.29, Theorem 1.5 is a statement about the 2-adic valuations of the coefficients of the explicit series $\widehat{\mathsf{f}}_\Gamma$. Throughout, $\mathfrak{G}_{2j} = 2(1-2^{2j})\mathcal{B}_{2j}$ denotes the Genocchi numbers.

Lemma 7.30. *The formal exponent of Lemma 7.15, written in the variable Z, is*

$$\log\frac{\widehat{\mathsf{r}}}{2} = \mathsf{b}(Z) = \sum_{j\ge 1} b_j\,(Z-1)^{1-2j}, \qquad b_j = \frac{2^{2j-1}\mathfrak{G}_{2j}}{(2j)(2j-1)}, \tag{83}$$

and coefficientwise b *is the Watson expansion of* $-\int_0^\infty e^{-(Z-1)t}\,t^{-1}\tanh t\,dt$. *Moreover*

$$\nu_2(b_j) = 2j - 2 - \nu_2(j) \;\geq\; 0, \tag{84}$$

with equality only for $j = 1$, *where* $b_1 = \mathfrak{G}_2 = -1$.

Proof. In (55) the bracket is $\mathcal{B}_{n+1}(0) - \mathcal{B}_{n+1}(\frac{1}{2}) = (2 - 2^{-n})\mathcal{B}_{n+1}$ by $\mathcal{B}_m(\frac{1}{2}) = (2^{1-m} - 1)\mathcal{B}_m$; it vanishes for even $n \geq 2$ (odd-index Bernoulli numbers vanish), and for $n = 2j - 1$ the coefficient of s^{1-2j} is $(2^{2j+1} - 2)\mathcal{B}_{2j}/((2j)(2j-1)) = -\mathfrak{G}_{2j}/((2j)(2j-1))$. Substituting $s^{-1} = -2/(Z-1)$ gives (83). For the kernel form, $\tanh t = \sum_{j\geq 1} 2^{2j}(2^{2j} - 1)\mathcal{B}_{2j}\,t^{2j-1}/(2j)!$, so $-(2j-2)!\,[t^{2j-1}]\tanh t = 2^{2j-1}\mathfrak{G}_{2j}/((2j)(2j-1)) = b_j$, and Watson's correspondence $(Z-1)^{1-2j} \leftrightarrow t^{2j-2}/(2j-2)!$ is coefficientwise exact. For (84): by von Staudt–Clausen $\nu_2(\mathcal{B}_{2j}) = -1$, hence $\mathfrak{G}_{2j} = 2(1 - 2^{2j})\mathcal{B}_{2j}$ is an odd integer, and $\nu_2(b_j) = (2j-1) - \nu_2\big((2j)(2j-1)\big) = 2j - 2 - \nu_2(j)$; finally $\nu_2(j) \leq \log_2 j \leq 2j - 2$, with equality throughout only at $j = 1$. □

Lemma 7.31. *As formal series in* Z^{-1},

$$\mathsf{b}(Z) + \sum_{j\geq 1} b_j (Z+1)^{1-2j} = \log\frac{Z-1}{Z+1}, \tag{85}$$

and consequently the forcing of (56), *written in* Z^{-1},

$$\widehat{\mathsf{f}}_\Gamma = \mathsf{F}(Z) := -\frac{4Z\,e^{\mathsf{b}(Z)}}{(Z-1)^2(Z+1)}, \tag{86}$$

is an even *series:* $[Z^{-n}]\mathsf{F} = 0$ *for odd* n.

Proof. By Lemma 7.30 the left side of (85) is the Watson expansion of

$$-\int_0^\infty e^{-Zt}\,(e^t + e^{-t})\,\frac{\tanh t}{t}\,dt = -\int_0^\infty e^{-Zt}\,\frac{2\sinh t}{t}\,dt,$$

whose coefficient series is $-2\sum_{k\geq 0} Z^{-2k-1}/(2k+1) = \log\frac{Z-1}{Z+1}$. Formula (86) is the substitution $s = (1-Z)/2$ in (56): $2s - 1 = -Z$, $4s^2(s-1) = -(Z-1)^2(Z+1)/2$, and $\widehat{\mathsf{r}} = 2e^{\mathsf{b}}$. Then

$$\frac{\mathsf{F}(-Z)}{\mathsf{F}(Z)} = e^{-\left(\mathsf{b}(Z) + \sum_j b_j (Z+1)^{1-2j}\right)} \cdot \frac{Z-1}{Z+1} = 1$$

by (85), which is evenness. □

Lemma 7.32. *Write* $e^{\mathsf{b}(Z)} = \sum_{N\geq 0} u_N (Z-1)^{-N}$. *Then for every* $N \geq 0$,

$$\nu_2(u_N) = -\nu_2(N!),$$

and $N!\,u_N$ *is a 2-adic unit.*

Proof. Expanding the exponential, $u_N = \sum_\lambda \prod_j b_j^{m_j}/m_j!$, the sum over partitions λ of N into odd parts $2j-1$ with multiplicities m_j. For any λ, since every $\nu_2(b_j) \geq 0$ and $\prod_j m_j! \mid \ell!$ with $\ell = \sum_j m_j \leq N$ (multinomial integrality),

$$\nu_2\Big(\prod_j b_j^{m_j}/\prod_j m_j!\Big) \;\geq\; -\nu_2\Big(\prod_j m_j!\Big) \;\geq\; -\nu_2(\ell!) \;\geq\; -\nu_2(N!).$$

The all-ones partition contributes $b_1^N/N! = (-1)^N/N!$, of valuation exactly $-\nu_2(N!)$. Every other partition has some part $2j-1 \geq 3$, hence $\sum_j m_j \nu_2(b_j) \geq \nu_2(b_j) \geq 1$ by (84), so its valuation is at least $1 - \nu_2(N!)$: strictly larger. By the ultrametric inequality the valuation of u_N is exactly $-\nu_2(N!)$ and $N!\,u_N \equiv (-1)^N \pmod 2$. □

Theorem 7.33. *For every $m \geq 0$,*

$$\nu_2\big([Z^{-(2m+2)}]\,\mathsf{F}\big) = 2 - 2m + s_2(m). \tag{87}$$

Proof. Expand the rational prefactor of (86) in Z^{-1}:

$$\frac{-4Z}{(Z-1)^2(Z+1)} = -4\sum_{k\geq 0} c_k\, Z^{-k-2}, \qquad c_k = [x^k]\,\frac{1}{(1-x)^2(1+x)} = \Big\lfloor \frac{k}{2} \Big\rfloor + 1 \in \mathbb{Z},$$

and $(Z-1)^{-N} = \sum_{m\geq N} \binom{m-1}{N-1} Z^{-m}$ for $N \geq 1$, with integer coefficients. Hence, for $n \geq 2$,

$$[Z^{-n}]\mathsf{F} = -4\sum_{N=0}^{n-2} u_N\, M_{n,N}, \qquad M_{n,N} = \sum_{m=N}^{n-2} c_{n-2-m}\binom{m-1}{N-1} \in \mathbb{Z} \quad (N \geq 1),$$

with $M_{n,0} = c_{n-2}$; in particular $M_{n,n-2} = 1$. The term $N = n-2$ contributes $-4u_{n-2}$, of valuation exactly $2 - \nu_2((n-2)!)$ by Lemma 7.32. For $N \leq n-3$ and n *even*,

$$\nu_2(u_N M_{n,N}) \;\geq\; -\nu_2(N!) \;\geq\; -\nu_2((n-3)!) = -\nu_2((n-2)!) + \nu_2(n-2) \;\geq\; -\nu_2((n-2)!) + 1,$$

so every other term has strictly larger valuation. Hence $\nu_2([Z^{-n}]\mathsf{F}) = 2 - \nu_2((n-2)!)$ exactly; with $n = 2m+2$ this is $2 - (2m - s_2(2m)) = 2 - 2m + s_2(m)$. □

Theorem 7.34. *For every $m \geq 0$,*

$$w_m = -\,\frac{[Z^{-(2m+2)}]\,\mathsf{F}}{4\,(2m+1)!}\,, \tag{88}$$

and consequently $\nu_2(w_m) = -2\,\nu_2\big((2m)!\big)$.

Proof. By Lemma 7.29, $W(t) = -\varphi(t)/(4t)$ with $\varphi(t) = \sum_n f_n t^{n-1}/(n-1)!$ the Borel transform of $\widehat{\mathsf{f}_\Gamma} = \mathsf{F}$; the coefficient of t^{2m} in W is therefore $-f_{2m+2}/(4(2m+1)!)$, which is (88). Then Theorem 7.33 and $\nu_2((2m+1)!) = \nu_2((2m)!) = 2m - s_2(m)$ give $\nu_2(w_m) = \big(2 - 2m + s_2(m)\big) - 2 - \big(2m - s_2(m)\big) = -2\big(2m - s_2(m)\big) = -2\nu_2((2m)!)$. □

The passage from the w_m to the numbers $\operatorname{Im} g_{2r-1}$ is a convolution which preserves valuations. It was given in [17]; we repeat the argument for completeness.

Lemma 7.35. *Put*

$$\mathsf{d}(t) = \frac{2t}{\sinh 2t} = \sum_{k\geq 0} \mathsf{d}_k t^{2k}, \qquad \mathsf{e}(t) = \frac{\sinh 2t}{2t} = \sum_{k\geq 0} \mathsf{e}_k t^{2k},$$

so that $\mathsf{d}(t)\,\mathsf{e}(t) = 1$. Then $\mathsf{d}_0 = \mathsf{e}_0 = 1$ and $\nu_2(\mathsf{d}_k) = \nu_2(\mathsf{e}_k) = s_2(k)$ for every $k \geq 1$.

Proof. $\mathsf{e}_k = 2^{2k}/(2k+1)!$ gives $\nu_2(\mathsf{e}_k) = 2k - \big(2k + 1 - s_2(2k+1)\big) = s_2(k)$, using $s_2(2k+1) = s_2(k) + 1$. For d, the expansion $x/\sinh x = \sum_k (2 - 2^{2k})\mathcal{B}_{2k} x^{2k}/(2k)!$ at $x = 2t$ gives $\mathsf{d}_k = (2 - 2^{2k})\mathcal{B}_{2k}\, 2^{2k}/(2k)!$; for $k \geq 1$, $\nu_2(2 - 2^{2k}) = 1$ and $\nu_2(\mathcal{B}_{2k}) = -1$ by von Staudt–Clausen, so $\nu_2(\mathsf{d}_k) = 1 - 1 + 2k - (2k - s_2(k)) = s_2(k)$. □

Theorem 7.36. *For every $k \geq 1$, $\operatorname{Im} g_{2k} = 0$; and for every $r \geq 1$,*

$$\nu_2\big(\operatorname{Im} g_{2r-1}\big) = -H_r, \qquad H_r = r + \nu_2\big((r-1)!\big).$$

Proof. Even indices. By Lemma 7.31 the coefficients of the odd powers of Z^{-1} in F vanish, so its Borel transform φ is an odd function of t. By Lemma 7.29 the Borel transform of $\widehat{Q}_\Gamma$ is $\varphi(t)/(2\sinh 2t)$, an even function, so $\widehat{q}_n = 0$ for even n, and $\operatorname{Im} g_{2k} = -\widehat{q}_{2k}/2 = 0$ by (14).

Odd indices. Put $\iota_m = [t^{2m}]\big(2\operatorname{Im}\Phi(t)/t\big) = 2\operatorname{Im} g_{2m+1}/(2m)!$ and $\tau(m) = -2\nu_2((2m)!) = -4m + 2s_2(m)$. By the first part, $2\operatorname{Im}\Phi(t)/t$ is an even series, and by (12)

$$W(t) = \frac{2\operatorname{Im}\Phi(t)}{t}\,\mathsf{e}(t), \qquad \frac{2\operatorname{Im}\Phi(t)}{t} = W(t)\,\mathsf{d}(t),$$

that is, $w_m = \sum_{k=0}^{m} \mathsf{e}_k\,\iota_{m-k}$ and $\iota_m = \sum_{k=0}^{m} \mathsf{d}_k\,w_{m-k}$. For $1 \le k \le m$, Lemma 7.35 and the subadditivity $s_2(m) \le s_2(m-k) + s_2(k)$ give

$$\big[s_2(k) + \tau(m-k)\big] - \tau(m) = s_2(k) + 4k + 2s_2(m-k) - 2s_2(m) \;\ge\; 4k - s_2(k) \;>\; 0.$$

Since $\nu_2(w_j) = \tau(j)$ for every j by Theorem 7.34, every term with $k \ge 1$ in $\iota_m = \sum_k \mathsf{d}_k w_{m-k}$ has valuation strictly larger than $\tau(m)$, while the term $k = 0$ is w_m. Hence $\nu_2(\iota_m) = \tau(m)$, that is,

$$\nu_2\big(\operatorname{Im} g_{2m+1}\big) = \tau(m) - 1 + \nu_2\big((2m)!\big) = -\nu_2\big((2m)!\big) - 1 = -\big(2m + 1 - s_2(m)\big) = -H_{m+1}.$$

□

Proof of Theorem 1.5. The statements on $\operatorname{Im} g_{2k}$ and on $\operatorname{Im} g_{2r-1}$ are Theorem 7.36, and the statement on w_m is Theorem 7.34. By (11),

$$\nu_2\big([A^1B^0]\kappa_r\big) = -(2r-1) - H_r = -E_r.$$

□

Proposition 7.37. *For every $r \le 8$, the polynomial $\kappa_r \in \mathbb{Q}[A,B]$ defined by (7) coincides with the constant $\Psi_r(\pi/2)$ of the phase expansion (3).*

Proof. Both sides are polynomials in (A,B) of degree at most $2r-1 \le 15$ in each variable: for (7) this is Lemma 3.1, and $\Psi_r(\pi/2)$ is obtained from the phase recursion as an explicit polynomial, whose degrees are read off. The two polynomials have been evaluated in exact rational arithmetic on the 16×16 product grid $A = i^2/4$, $B = j^2/9$, $1 \le i, j \le 16$, on which they agree for every $r \le 8$; since a polynomial of degree at most 15 in each variable is determined by its values on such a grid, they coincide. □

8. The conjugate-layer factorisation

The double series (36) admits an expansion of Burchnall–Chaundy type [20, 21]. We give a self-contained proof by the Wilf–Zeilberger method [22].

Theorem 8.1. *For all parameters a, b, a', b' and generic c, as an identity of formal double series in x, y,*

$$\sum_{\mu,\nu\ge 0} \frac{(a)_\mu (b)_\mu (a')_\nu (b')_\nu}{(c)_{\mu+\nu}\,\mu!\,\nu!} x^\mu y^\nu = \sum_{r\ge 0} \frac{(-1)^r (a)_r (b)_r (a')_r (b')_r}{r!\,(c)_{2r}\,(c+r-1)_r}\,(xy)^r \times {}_2\mathrm{F}_1(a+r, b+r; c+2r; x)\;{}_2\mathrm{F}_1(a'+r, b'+r; c+2r; y). \tag{89}$$

Proof. Comparing coefficients of $x^\mu y^\nu$ and cancelling $(a)_r(a+r)_{\mu-r} = (a)_\mu$ and its three companions (the parameter dependence cancels termwise), the claim is equivalent to the pure identity

$$\sum_{r=0}^{\min(\mu,\nu)} t_r = \frac{1}{(c)_{\mu+\nu}\,\mu!\,\nu!}, \tag{90}$$

$$t_r = \frac{(-1)^r}{r!\,(c)_{2r}(c+r-1)_r\,(c+2r)_{\mu-r}(c+2r)_{\nu-r}\,(\mu-r)!\,(\nu-r)!}.$$

Set $f(\mu,r) = t_r\,(c)_{\mu+\nu}\,\mu!\,\nu!$, extended to real r by $1/(\mu-r)! = 1/\Gamma(\mu-r+1)$ and $1/(\nu-r)! = 1/\Gamma(\nu-r+1)$; the claim is $\Sigma(\mu) := \sum_r f(\mu,r) = 1$. The certificate

$$R(r) = -\frac{r\,(c+\nu+r-1)}{(c+2r-1)(\mu+1-r)}, \qquad h(r) = R(r)f(\mu,r),$$

satisfies the rational-function identity $f(\mu+1,r)-f(\mu,r) = h(r+1)-h(r)$ (after division by $f(\mu,r)$ both sides are rational functions of r, and the identity is checked by clearing denominators). Both sides are finite at every integer $r \geq 0$: the only delicate points are the simple zeros of $1/\Gamma$ against the simple pole of R at $r = \mu+1$, where the limits exist and the identity holds by continuity. Summing over $0 \leq r \leq K$, $K = \min(\mu+1,\nu)$, telescopes to $\Sigma(\mu+1) - \Sigma(\mu) = h(K+1) - h(0)$. Now $h(0) = 0$ since $R(0) = 0$; and $h(K+1) = 0$: if $\nu \leq \mu$ then $K+1 = \nu+1$, where $1/\Gamma(\nu-r+1)$ vanishes and R is regular unless $\mu = \nu$, in which case the double zero of $1/\big(\Gamma(\mu-r+1)\Gamma(\nu-r+1)\big)$ beats the simple pole; if $\nu \geq \mu+1$ then $K+1 = \mu+2$, where $1/\Gamma(\mu-r+1)$ vanishes and R is regular. Hence $\Sigma(\mu+1) = \Sigma(\mu)$, and $\Sigma(0) = 1$. Symmetry in (μ,ν) completes the induction. □

Corollary 8.2. *For all α,β, as formal series in Z^{-1}, with $\gamma_r(A) = (\frac12-\alpha)_r(\frac12+\alpha)_r = \prod_{j=0}^{r-1}\big((j+\frac12)^2 - A\big)$,*

$$F(\alpha,\beta;Z) = \sum_{r\geq 0} \frac{(-1)^r\,\gamma_r(A)\,\gamma_r(B)}{2^r\,r!\,(1-Z)_{2r}\,(-Z+r)_r}\,\mathcal{P}_r(A;Z)\,\overline{\mathcal{P}}_r(B;Z), \tag{91}$$

where $\mathcal{P}_r(A;Z) = {}_2\mathrm{F}_1(\frac12-\alpha+r, \frac12+\alpha+r; 1-Z+2r; x_0)$ and $\overline{\mathcal{P}}_r$ is the same series at $y_0 = \bar{x}_0$. Each layer is $O(Z^{-3r})$: with $v = 1/Z$,

$$\frac{(-1)^r\,(x_0y_0)^r}{r!\,(1-Z)_{2r}\,(-Z+r)_r} = \frac{v^{3r}}{2^r\,r!}\prod_{j=1}^{2r}\frac{1}{1-jv}\prod_{j=r}^{2r-1}\frac{1}{1-jv}. \tag{92}$$

Proof. Apply Theorem 8.1 to (36) with $a = \frac12-\alpha$, $b = \frac12+\alpha$, $a' = \frac12-\beta$, $b' = \frac12+\beta$, $c = 1-Z$, $x_0y_0 = \frac12$, $(c+r-1)_r = (-Z+r)_r$. Per coefficient of v^n every sum is finite (the monomial (μ,ν) contributes from order $\mu+\nu$ on, the layer r from order $3r$ on), so the rearrangement is legitimate v-adically. For (92) write $(1-Z+j) = -Z(1-(1+j)v)$ and $(-Z+r+j) = -Z(1-(r+j)v)$. □

Remark 8.3. At the Legendre point $A = B = 0$, $\gamma_r(0) = (\frac12)_r^2$ with $(\frac12)_r = (2r)!/(4^r r!)$, so the layer-$r$ prefactor in (91) carries $\gamma_r(0)^2\,2^{-r}/r! = \big((2r)!\big)^4/\big(2^{9r}(r!)^5\big)$ against three inverse powers of Z per layer.

Remark 8.4. ${}_2\mathrm{F}_1(a,1-a;c;x)$ is an associated Legendre function, so $\mathcal{P}_0(A;Z)$ is, up to a Gamma-and-power prefactor independent of α, the Legendre function $P^Z_{\alpha-1/2}(-i)$ of *order* Z and degree $\alpha-\frac12$, and its partner is the conjugate function at $+i$.

9. The transverse linearisation

9.1. **Diagonal and transverse coordinates.** In this section F denotes the Appell function $\mathcal{F}(\alpha,\beta;Z;x,y)$ of Subsection 7.1, regarded as a formal series in x, y and Z^{-1}. Put $A_+ = \frac{A+B}{2}$, $A_- = \frac{A-B}{2}$, so $\partial_A = \frac12\partial_{A_+} + \frac12\partial_{A_-}$, and define

$$\mu_\alpha = A - \frac14, \qquad \mu_\beta = B - \frac14.$$

Let $\mathcal{V}_0 = \log F|_{A_-=0}$, $\mathcal{V}_1 = \partial_{A_-}\log F|_{A_-=0}$, and

$$\mathsf{u} = \Theta_x\mathcal{V}_0, \quad \mathsf{v} = \Theta_y\mathcal{V}_0, \quad \mathsf{t} = \mathsf{u}+\mathsf{v}, \qquad \mathsf{p} = \Theta_x\mathcal{V}_1, \quad \mathsf{q} = \Theta_y\mathcal{V}_1, \quad \mathsf{h} = \mathsf{p}+\mathsf{q},$$

with $\Theta_x = x\partial_x$ and $\Theta_y = y\partial_y$; the sans-serif letters distinguish these fields from the midpoint constants $p = 1+i$, $q = 1-i$ of Section 2. Dividing the Appell equations (40)–(41) by F gives, before linearisation, the logarithmic Horn system

$$Z\mathsf{u} = \Theta_x\mathsf{t} + \mathsf{u}\mathsf{t} - x\big(\Theta_x\mathsf{u} + \mathsf{u}^2 + \mathsf{u} - \mu_\alpha\big), \tag{93}$$

$$Z\mathsf{v} = \Theta_y\mathsf{t} + \mathsf{v}\mathsf{t} - y\big(\Theta_y\mathsf{v} + \mathsf{v}^2 + \mathsf{v} - \mu_\beta\big), \tag{94}$$

together with $\Theta_y\mathsf{u} = \Theta_x\mathsf{v}$. Indeed, $(\Theta_x+\frac12)^2F/F = \Theta_x\mathsf{u} + \mathsf{u}^2 + \mathsf{u} + \frac14$, and similarly in y, so these equations are exactly equivalent to the Appell PDEs wherever F is invertible as a formal series (its constant term is 1). At the midpoint, the symmetry $\overline{L_n(A,B)} = L_n(B,A)$ makes $\partial_{A_+}\log F$ real and $\partial_{A_-}\log F$ purely imaginary on the diagonal $A_- = 0$; the diagonal part is given by Theorem 1.1.

Theorem 9.1. *Differentiating* (93)–(94) *at* $A_- = 0$ *gives the exact* linear *forced system*

$$Z\mathsf{p} = \Theta_x\mathsf{h} + \mathsf{p}\mathsf{t} + \mathsf{u}\mathsf{h} - x\big(\Theta_x\mathsf{p} + 2\mathsf{u}\mathsf{p} + \mathsf{p} - 1\big), \tag{95}$$

$$Z\mathsf{q} = \Theta_y\mathsf{h} + \mathsf{q}\mathsf{t} + \mathsf{v}\mathsf{h} - y\big(\Theta_y\mathsf{q} + 2\mathsf{v}\mathsf{q} + \mathsf{q} + 1\big), \tag{96}$$

with the compatibility $\Theta_y\mathsf{p} = \Theta_x\mathsf{q}$ *and forcing* $(x,-y)$.

Proof. At fixed A_+ one has $A = A_+ + A_-$, $B = A_+ - A_-$, hence $\partial_{A_-}\mu_\alpha = 1$ and $\partial_{A_-}\mu_\beta = -1$. Also

$$\partial_{A_-}\mathsf{u} = \Theta_x\mathcal{V}_1 = \mathsf{p}, \qquad \partial_{A_-}\mathsf{v} = \Theta_y\mathcal{V}_1 = \mathsf{q}, \qquad \partial_{A_-}\mathsf{t} = \mathsf{h}.$$

Differentiating (93) term by term therefore gives

$$Z\mathsf{p} = \Theta_x\mathsf{h} + \mathsf{p}\mathsf{t} + \mathsf{u}\mathsf{h} - x\big(\Theta_x\mathsf{p} + 2\mathsf{u}\mathsf{p} + \mathsf{p} - 1\big),$$

because the derivative of $-\mu_\alpha$ is -1. The y equation is identical except that $\partial_{A_-}\mu_\beta = -1$, so the derivative of $-\mu_\beta$ is $+1$, which yields

$$Z\mathsf{q} = \Theta_y\mathsf{h} + \mathsf{q}\mathsf{t} + \mathsf{v}\mathsf{h} - y\big(\Theta_y\mathsf{q} + 2\mathsf{v}\mathsf{q} + \mathsf{q} + 1\big).$$

Finally, $\Theta_y\mathsf{u} = \Theta_x\mathsf{v}$ follows from commutation of Θ_x, Θ_y on $\log F$; differentiating this identity in A_- gives $\Theta_y\mathsf{p} = \Theta_x\mathsf{q}$. Every occurrence of the unknown transverse quantities is linear, while $\mathsf{u},\mathsf{v},\mathsf{t}$ belong to the fixed diagonal background. □

Corollary 9.2. *In the class of pairs*

$$(\mathsf{p},\mathsf{q}) \in Z^{-1}\,\mathbb{C}[[x,y]][[Z^{-1}]]^2$$

satisfying the compatibility $\Theta_y\mathsf{p} = \Theta_x\mathsf{q}$, *system* (95)–(96) *has at most one solution for the fixed diagonal background* $\mathsf{u},\mathsf{v},\mathsf{t}$; *hence the solution obtained by differentiating* $\mathcal{V}_1$ *is the unique formal solution with zero constant term in* Z^{-1}.

Proof. Existence is already supplied by $\mathsf{p} = \Theta_x \mathcal{V}_1$ and $\mathsf{q} = \Theta_y \mathcal{V}_1$. For uniqueness, let two such solutions be given and denote their difference by $(\mathsf{p}^\circ, \mathsf{q}^\circ)$. The forcing terms cancel. Because $F = 1 + O(Z^{-1})$, one has $\mathsf{u}, \mathsf{v}, \mathsf{t} = O(Z^{-1})$. Suppose, for contradiction, that the difference is nonzero and let $n \geq 1$ be the least index for which the coefficient of Z^{-n} in either p° or q° is nonzero. Multiplication by Z makes the left-hand side of the corresponding difference equation have a nonzero term of order $Z^{-(n-1)}$. Every term on its right-hand side has order at least Z^{-n}: the Euler operators do not change the order in Z^{-1}, multiplication by x or y does not change it, and multiplication by $\mathsf{u}, \mathsf{v}, \mathsf{t}$ raises it by at least one. Thus the coefficient of $Z^{-(n-1)}$ on the right is zero, a contradiction. Hence $\mathsf{p}^\circ = \mathsf{q}^\circ = 0$. □

The system (93)–(94) is nonlinear; its first transverse variation is linear, with coefficients given by the diagonal fields $\mathsf{u}, \mathsf{v}, \mathsf{t}$.

Corollary 9.3. *At the midpoint and at $A_+ = 0$, write $\mathcal{V}_1 = \sum_{n \geq 1} \mathcal{V}_{1,n} Z^{-n}$ and $\widehat{\mathcal{V}}_1(t) = \sum_{n \geq 1} \mathcal{V}_{1,n}\, t^n/(n-1)!$. Then, with W as in (12),*

$$W(t) = \frac{\sinh 2t}{2t^2}\ \operatorname{Im} \widehat{\mathcal{V}}_1(t).$$

Proof. At $A = B = 0$,

$$\partial_A \log F = \frac{1}{2}\partial_{A_+} \log F + \frac{1}{2}\partial_{A_-} \log F.$$

At the midpoint the first term is real and the second is purely imaginary. Passing coefficientwise to the Borel transforms gives $\operatorname{Im} \Phi(t) = \frac{1}{2} \operatorname{Im} \widehat{\mathcal{V}}_1(t)$, and the formula follows from (12). □

9.2. **A Green kernel at the Legendre background.** The first variation of the Jacobi polynomial itself can be written by reduction of order. Put $\alpha = a + \delta$, $\beta = a - \delta$, and $\widetilde{P}_a(x) = P_d^{(a,a)}(x)/P_d^{(a,a)}(1)$,

$$v_a = \partial_\delta \left[\frac{P_d^{(a+\delta, a-\delta)}}{P_d^{(a+\delta, a-\delta)}(1)} \right]_{\delta = 0}.$$

Only $\beta - \alpha = -2\delta$ varies in the Jacobi equation, so $\mathcal{J}_{a,d} v_a = 2\widetilde{P}_a'$ with $\mathcal{J}_{a,d} y = (1 - x^2)y'' - 2(a+1)xy' + d(d + 2a + 1)y$ and $v_a(1) = 0$. Writing $v_a = \widetilde{P}_a h_a$ and using $\mathcal{J}_{a,d}\widetilde{P}_a = 0$, $\big((1-x^2)^{a+1}\widetilde{P}_a^2 h_a'\big)' = 2(1-x^2)^a \widetilde{P}_a \widetilde{P}_a'$, so that

$$v_a(x) = -2 \int_x^1 G_{a,d}(x, x_1)\, \widetilde{P}_a'(x_1)\, dx_1,$$

$$G_{a,d}(x, x_1) = \widetilde{P}_a(x)\, (1 - x_1^2)^a \widetilde{P}_a(x_1) \int_{x_1}^x \frac{dw}{(1-w^2)^{a+1} \widetilde{P}_a(w)^2}.$$

Proposition 9.4. *At $a = 0$, with P_d, Q_d the Legendre functions of the first and second kind and $\mathrm{h}_d = \sum_{j=1}^d 1/j$,*

$$\frac{v_0(x)}{P_d(x)} = \operatorname{artanh} x - \frac{Q_d(x)}{P_d(x)} - \mathrm{h}_d.$$

Proof. $P_dQ_d' - Q_dP_d' = (1-x^2)^{-1}$ gives $(v_0/P_d)' = \frac{P_d^2-1}{(1-x^2)P_d^2} = \frac{1}{1-x^2} - \big(Q_d/P_d\big)'$, so $v_0/P_d = \operatorname{artanh} x - Q_d/P_d + c$. As $x \to 1^-$, $Q_d(x) = \frac{1}{2}\log\frac{2}{1-x} - \mathrm{h}_d + o(1)$ and $\operatorname{artanh} x = \frac{1}{2}\log\frac{2}{1-x} + o(1)$, while $P_d(1) = 1$; hence $v_0/P_d \to \mathrm{h}_d + c$, and $v_0(1) = 0$ forces $c = -\mathrm{h}_d$. □

Remark 9.5. Proposition 9.4 does not give the numbers g_n. With $\alpha = a+\delta$, $\beta = a-\delta$ one has $A - B = 4a\delta$, so $\partial_\delta(A-B)|_{a=0} = 0$: the first δ-variation at $a = 0$ does not move $A - B$ at all. Since $\kappa_r = (A-B)\,\widetilde{\kappa}_r(A,B)$ by (5),

$$[A^1B^0]\kappa_r = \widetilde{\kappa}_r(0,0) = \tfrac{1}{4}\,\partial_a\partial_\delta\,\kappa_r(a+\delta, a-\delta)\big|_{a=\delta=0},$$

a *mixed* derivative. Moreover the expansion must be taken with $\mathcal{N} = d + a + \frac{1}{2}$ held fixed, not with the degree d held fixed; the two differ by the variation of the eigenvalue.

10. The tangent from the layers, and four valuation formulas

The layer expansion of Corollary 8.2 expresses the series $\sum_n g_n Z^{-n}$ as a quotient of two explicit series. In this section we determine the 2-adic valuations of the coefficients of the numerator and of the denominator, using Theorem 7.36.

Write $\mathcal{P}_r' = \partial_A \mathcal{P}_r|_{A=0}$, denote by $\varrho_r(Z)$ the coefficient of $\gamma_r(A)\gamma_r(B)\mathcal{P}_r(A;Z)\overline{\mathcal{P}}_r(B;Z)$ in (91), and set, at the Legendre point $A = B = 0$,

$$\mathrm{Num} = \sum_{r\ge 0} \varrho_r \left[\gamma_r\mathcal{P}_r\right]' \overline{\gamma_r\mathcal{P}_r}, \qquad \mathrm{Den} = \sum_{r\ge 0} \varrho_r \left(\gamma_r\mathcal{P}_r\right)\overline{\left(\gamma_r\mathcal{P}_r\right)}, \tag{97}$$

where the bar denotes the series at y_0 (coefficientwise conjugation). Both γ_r and $\mathcal{P}_r$ are polynomials in A coefficientwise, so the derivative is taken term by term. Then, coefficientwise in $v = 1/Z$,

$$\sum_{n\ge 1} g_n v^n = \partial_A \log F\big|_{A=B=0} = \frac{\mathrm{Num}}{\mathrm{Den}}, \qquad \operatorname{Im} g_n = \left[v^n\right]\frac{\operatorname{Im}\mathrm{Num}}{\mathrm{Den}}, \tag{98}$$

the last equality because Den has real coefficients.

Corollary 10.1. $\mathrm{Den} = F(0,0;Z) = \exp\Lambda$, *where, by Theorem 1.1 at* $\alpha = 0$,

$$\Lambda(v) = \sum_{n\ge 1}\lambda_n\, v^n, \qquad \lambda_n = L_n^{\mathrm{diag}}\big|_{\alpha=0} = \frac{4^n}{n(n+1)}\Big[2\mathcal{B}_{n+1}\big(\tfrac{3}{4}\big) - \mathcal{B}_{n+1}(1) - \mathcal{B}_{n+1}\big(\tfrac{1}{2}\big)\Big].$$

We write

$$\mathrm{Den}^{\pm 1} = \exp(\pm\Lambda) = \sum_{n\ge 0} d_n^{\pm} v^n.$$

Lemma 10.2. *For* $k \ge 1$,

$$\lambda_{2k} = \frac{\mathsf{E}_{2k}}{4k}, \qquad \lambda_{2k-1} = -\frac{(2^{2k}-1)\mathcal{B}_{2k}}{(2k)(2k-1)}.$$

Hence

$$\nu_2(\lambda_{2k}) = \nu_2(\lambda_{2k-1}) = -2 - \nu_2(k);$$

in particular $\lambda_1 = -1/4$, *and* $\nu_2(\lambda_n) \ge -n$ *for every* $n \ge 2$.

Proof. By Corollary 1.3(iii),

$$\Upsilon_0(t) = \frac{1}{2}\left(\operatorname{sech} t - \tanh\frac{t}{2} - 1\right) = \sum_{n\ge 1}\frac{\lambda_n}{(n-1)!}t^n.$$

Using $\operatorname{sech} t = \sum_{k\ge 0} \mathsf{E}_{2k} t^{2k}/(2k)!$ and $\tanh \frac{t}{2} = \sum_{k\ge 1} 2(2^{2k}-1)\mathcal{B}_{2k} t^{2k-1}/(2k)!$ gives the displayed formulas. Euler numbers are odd, von Staudt–Clausen gives $\nu_2(\mathcal{B}_{2k}) = -1$, and $2^{2k}-1$ is odd; thus both coefficients have valuation $-2-\nu_2(k)$. For $n = 2k$ the inequality $-2-\nu_2(k) \ge -2k$ is $2+\nu_2(k) \le 2k$; for $n = 2k-1 \ge 3$ it is $2+\nu_2(k) \le 2k-1$. Both hold because $\nu_2(k) \le \log_2 k$. □

Theorem 10.3. *For every $n \ge 0$,*

$$\nu_2(d_n^{\pm}) = -2n - \nu_2(n!).$$

Equivalently, $\nu_2(d_{2k}^{\pm}) = -\big(5k + \nu_2(k!)\big)$ and $\nu_2(d_{2k+1}^{\pm}) = -\big(5k+2+\nu_2(k!)\big)$ for $k \ge 0$.

Proof. Expand the exponential by partitions,

$$d_n^{\pm} = \sum_{\substack{m_1,\dots,m_n\ge 0 \\ \sum_{j=1}^n j m_j = n}} \prod_{j=1}^{n} \frac{(\pm\lambda_j)^{m_j}}{m_j!}.$$

The partition $m_1 = n$, $m_j = 0$ for $j \ge 2$, contributes $(\pm\lambda_1)^n/n!$, of valuation $-2n-\nu_2(n!)$. Consider any other partition and put $M = \sum_{j\ge 2} j m_j > 0$, so that $m_1 = n - M$. Lemma 10.2 gives

$$\nu_2\left(\prod_j \frac{\lambda_j^{m_j}}{m_j!}\right) \ge -2m_1 - \sum_{j\ge 2} j m_j - \sum_j \nu_2(m_j!) = -2n + M - \sum_j \nu_2(m_j!).$$

The difference with the valuation of the first partition is at least

$$M + \nu_2(n!) - \sum_j \nu_2(m_j!) > 0,$$

because $\prod_j m_j!$ divides $(\sum_j m_j)!$, which divides $n!$, while $M > 0$. Thus exactly one term has minimal valuation, and $\nu_2(d_n^{\pm}) = -2n - \nu_2(n!)$. Finally $\nu_2((2k)!) = k + \nu_2(k!)$ and $\nu_2((2k+1)!) = \nu_2((2k)!)$ give the two other formulas. □

Proposition 10.4. *For every $r, m \ge 1$,*

$$\nu_2\big([v^{2r-1}]\operatorname{Im}\operatorname{Num}\big) = -\big(5r - 4 + \nu_2((r-1)!)\big),$$
$$\nu_2\big([v^{2m}]\operatorname{Im}\operatorname{Num}\big) = -\big(5m - 2 + \nu_2((m-1)!)\big).$$

Proof. By (98) and Theorem 7.36, in which $\operatorname{Im} g_{2k} = 0$,

$$\operatorname{Im}\operatorname{Num} = \operatorname{Den}\cdot \sum_{n\ge 1} \operatorname{Im} g_n\, v^n = \Big(\sum_{n\ge 0} d_n^{+} v^n\Big)\Big(\sum_{k\ge 1} \operatorname{Im} g_{2k-1}\, v^{2k-1}\Big),$$

so that

$$[v^{2r-1}]\operatorname{Im}\operatorname{Num} = \sum_{k=1}^{r} \operatorname{Im} g_{2k-1}\; d^{+}_{2(r-k)}.$$

By Theorems 7.36 and 10.3, the term k has valuation $-H_k - 5(r-k) - \nu_2((r-k)!)$ with $H_k = k + \nu_2((k-1)!)$. For $k = 1$ this is $-(5r - 4 + \nu_2((r-1)!))$, and for $2 \le k \le r$ it exceeds that value by

$$4(k-1) + \nu_2\binom{r-1}{k-1} > 0.$$

Hence the term $k=1$ is the only one of minimal valuation, which gives the first formula. Similarly,

$$[v^{2m}]\,\mathrm{Im\,Num} = \sum_{k=1}^{m} \mathrm{Im}\, g_{2k-1}\; d^{+}_{2(m-k)+1},$$

where the term k has valuation $-H_k - 5(m-k) - 2 - \nu_2((m-k)!)$; for $k=1$ this is $-(5m-2+\nu_2((m-1)!))$, and for $k \geq 2$ it exceeds that value by $4(k-1)+\nu_2\binom{m-1}{k-1} > 0$. □

Proposition 10.5. *For $r \geq 1$ and $1 \leq j \leq 2r-1$ put*

$$c_{r,j} := 2^{\,5r-4+\nu_2((r-1)!)}\,\left[v^{j}\right]\mathrm{Im\,Num}\;\cdot\;\left[v^{2r-1-j}\right]\mathrm{Den}^{-1}.$$

Then $c_{r,j} \in \mathbb{Z}_2$,

$$\nu_2(c_{r,2k-1}) = \nu_2\binom{r-1}{k-1} \qquad (1 \leq k \leq r),$$

$$\nu_2(c_{r,2k}) = 1 + \nu_2\left((r-k)\binom{r-1}{k-1}\right) \qquad (1 \leq k \leq r-1),$$

and

$$\sum_{j=1}^{2r-1} c_{r,j} = 2^{\,5r-4+\nu_2((r-1)!)}\,\mathrm{Im}\, g_{2r-1} \;\in\; 2^{4(r-1)}\,\mathbb{Z}_2^{\times}. \tag{99}$$

Proof. The two valuation formulas follow from Theorem 10.3 and Proposition 10.4 by addition of exponents: for $j = 2k-1$,

$$5r-4+\nu_2((r-1)!) - \big(5k-4+\nu_2((k-1)!)\big) - \big(5(r-k)+\nu_2((r-k)!)\big) = \nu_2\binom{r-1}{k-1},$$

and for $j = 2k$,

$$5r-4+\nu_2((r-1)!) - \big(5k-2+\nu_2((k-1)!)\big) - \big(5(r-k-1)+2+\nu_2((r-k-1)!)\big) = 1+\nu_2\left((r-k)\binom{r-1}{k-1}\right).$$

Since $\mathrm{Im}\, g_{2r-1} = [v^{2r-1}]\big(\mathrm{Im\,Num}\cdot\mathrm{Den}^{-1}\big)$, the sum of the $c_{r,j}$ is $2^{5r-4+\nu_2((r-1)!)}\,\mathrm{Im}\, g_{2r-1}$, whose valuation is $5r-4+\nu_2((r-1)!) - H_r = 4(r-1)$ by Theorem 7.36. □

Remark 10.6. The individual terms $c_{r,j}$ have small valuations (at most of the order of $\log_2 r$), whereas their sum is divisible by exactly $2^{4(r-1)}$. The valuation of $\mathrm{Im}\, g_{2r-1}$ therefore cannot be read off from the individual terms of the convolution $\mathrm{Im\,Num}\cdot\mathrm{Den}^{-1}$.

11. Application: Jacobi zeros for $\alpha, \beta > -1$

Throughout this section $\alpha, \beta > -1$, so that $P_n^{(\alpha,\beta)}$ has n real simple zeros in $(-1,1)$, ordered as in the Introduction. The purpose of the section is to illustrate that the phase expansion (3) can be evaluated in exact arithmetic, with an accuracy away from the endpoints limited only by the truncation order. It is not an optimised algorithm, and no timing comparison with the specialised Gauss–Jacobi solvers [11, 14, 13] is intended. For related asymptotic analyses of Jacobi polynomials and their zeros, see [7, 8].

By (5) the phase constants vanish for $\alpha = \beta$, so the Legendre and Gegenbauer cases need none; for $\alpha \neq \beta$ they are needed. Their denominators are known: by [17], $2^{E_r}\kappa_r \in \mathbb{Z}_2[\mu_\alpha, \mu_\beta]$ with $E_r = 3r-1+\nu_2((r-1)!)$, and the odd part of $\mathrm{den}\,\kappa_r$ divides

$\operatorname{lcm}(1,3,\dots,2r-1)$. The denominators of $\kappa_1,\dots,\kappa_4$ as polynomials in (A,B) are $4=2^2$, $96=2^5\cdot 3$, $7680=2^9\cdot 15$ and $86016=2^{12}\cdot 21$.

Solving (3) by Newton with a per-node least-term gate (the truncation order is the last index before the correction terms stop decreasing), then polishing by a safeguarded Newton on $P_n^{(\alpha,\beta)}$ confined to the cell bounded by the midpoints of the neighbouring reference angles, gives the following maximum errors in θ. The reference zeros are independent of the predictor: they are obtained from $P_n^{(\alpha,\beta)}$ itself in 40-digit multiprecision arithmetic, and each tabulated error is measured against that reference. In the table, "bare" is the leading approximation (1) (no correction terms) over all nodes $k=1,\dots,n$; "gated" is the gated Newton solution of $\Xi(\theta)=(k-\frac{1}{2})\pi$, over all nodes or over the bulk; "polished" is the result after the final Newton step on $P_n^{(\alpha,\beta)}$.

n	α	β	bare	gated, all k	gated, bulk	polished
30	0.3	−0.2	$1.5\cdot 10^{-3}$	$7.6\cdot 10^{-5}$	$1.7\cdot 10^{-18}$	$9.2\cdot 10^{-41}$
30	1.5	0.25	$7.0\cdot 10^{-3}$	$1.2\cdot 10^{-5}$	$2.6\cdot 10^{-19}$	$4.6\cdot 10^{-41}$
60	−0.4	0.7	$5.6\cdot 10^{-4}$	$2.7\cdot 10^{-5}$	$5.3\cdot 10^{-24}$	$9.2\cdot 10^{-41}$
60	0	0	$8.0\cdot 10^{-4}$	$2.1\cdot 10^{-5}$	$1.2\cdot 10^{-23}$	$3.6\cdot 10^{-41}$
100	2.5	−0.5	$5.1\cdot 10^{-3}$	$8.6\cdot 10^{-6}$	$1.3\cdot 10^{-32}$	$3.7\cdot 10^{-40}$

"Bulk" is the middle half of the nodes; the loss of accuracy near $k=1$ and $k=n$ is the known endpoint effect, where an expansion of Bessel type must be used instead.

Computational checks

The identities of this paper have been checked independently by exact computation in finite ranges, in rational, Gaussian-rational or symbolic arithmetic. In particular, the numbers g_n were computed for $n\le 41$ directly from the amplitude (6), and they agree with (13) (real and imaginary parts) and with (98); the relation (88) was checked against these values for $m\le 19$; the valuation formula (87) was checked for $m\le 159$; and the valuation formulas of Section 10 were checked for $r\le 21$. The identities of Section 7 and Appendix A that involve boundary values, lattice sums and their asymptotic expansions were checked numerically to at least 25 digits. None of these computations is used in the proofs, with the exception of the finite verification in Proposition 7.37.

Appendix A. The sectorial estimates

This appendix proves the estimates stated in Lemma 7.27, the vanishing of the constant $c_*(s_0)$ of the quadrature (48) on every admissible lattice (Theorem A.12), and Theorem 1.6. The order is as follows: coefficient bounds and the expansion of the amplitude in a bilateral sector (Subsection A.1); uniform asymptotics of the closed blocks (Subsection A.2); the uniform expansion of the forcing $\widehat{\mathsf{f}}_\Gamma$ and of its lattice sum T (Subsection A.3); the relation between the two boundary determinations, from which the vertical decay of the quotient Q follows (Subsection A.4); a classification lemma for periodic meromorphic functions and the vanishing of $c_*(s_0)$ (Subsection A.5); and the proof of Theorem 1.6 (Subsection A.6). The only classical inputs are Stirling's formula in sectors, the Euler–Maclaurin (Hurwitz) expansion, and the removability of bounded isolated singularities.

A.1. **Coefficient bounds and the bilateral sector expansion.** Throughout, $Z = 1-2s$, and $F(\alpha;Z)$ denotes the specialisation $\beta=0$ of the amplitude (6),

$$F(\alpha;Z)=\sum_{n\ge0}G_n(\alpha)\,e_n(Z),\qquad e_n(Z)=\frac{1}{(Z-1)\cdots(Z-n)},$$

$$G_n(\alpha)=\sum_{l=0}^{n}a_l(\alpha)\,a_{n-l}(0)\,p^l q^{\,n-l},$$

with $p=1+i$, $q=1-i$ and the Hankel coefficients a_k of the Introduction. Two facts proved in the main text are used. First, by Lemma 7.2 the series F is the Kampé de Fériet double series (36) at the midpoint arguments $(x_0,y_0)=(\frac{1+i}{2},\frac{1-i}{2})$, grouped by total degree $\mu+\nu=n$. Second, by (38) (an exact identity for fixed Z),

$$F(\alpha;Z)=y_0^{-Z}(-i)^{1/2}\,\mathcal{H}(\alpha;Z), \tag{100}$$

with an α-independent prefactor, so that

$$\mathcal{A}(s)=\frac12\,\frac{\partial_\alpha^2\mathcal{H}(0;Z)}{\mathcal{H}(0;Z)}=\frac12\,\frac{\partial_\alpha^2F(0;Z)}{F(0;Z)} \tag{101}$$

as an identity of functions, not merely of formal series.

Lemma A.1. *For all $k\ge0$ and all $\alpha\in\mathbb{C}$,*

$$|a_k(\alpha)|\le\cosh(\pi|\alpha|)\,k!\,2^{-k},\qquad\text{and consequently}\qquad |G_n(\alpha)|\le\cosh(\pi|\alpha|)\,(n+1)!\,2^{-n/2}.$$

Proof. From the product form $a_k(\nu)=\prod_{j=0}^{k-1}\bigl(4\nu^2-(2j+1)^2\bigr)/(8^k k!)$,

$$|a_k(\alpha)|\le|a_k(0)|\prod_{j=0}^{k-1}\Bigl(1+\frac{4|\alpha|^2}{(2j+1)^2}\Bigr)\le|a_k(0)|\prod_{j\ge0}\Bigl(1+\frac{4|\alpha|^2}{(2j+1)^2}\Bigr)=|a_k(0)|\cosh(\pi|\alpha|),$$

the last step being the product formula $\cos(\pi z/2)=\prod_{j\ge0}(1-z^2/(2j+1)^2)$ at $z=2i|\alpha|$. For the central value, $|a_k(0)|=\bigl((2k-1)!!\bigr)^2/(8^k k!)$ and $\bigl((2k-1)!!\bigr)^2\le(2k-1)!!\,(2k)!!=(2k)!$, so $|a_k(0)|\le(2k)!/(8^k k!)=k!\binom{2k}{k}8^{-k}\le k!\,4^k/8^k=k!\,2^{-k}$. For G_n: $|p|=|q|=\sqrt2$ and $\sum_{l=0}^{n}l!\,(n-l)!\le(n+1)\,n!=(n+1)!$, hence $|G_n(\alpha)|\le\cosh(\pi|\alpha|)\,2^{n/2}\,2^{-n}\,(n+1)!=\cosh(\pi|\alpha|)(n+1)!\,2^{-n/2}$. □

For $0<\theta_0<1$ and $y_{\min}\ge1/(1-\theta_0)$ put

$$\Omega_{\theta_0}:=\bigl\{Z:\ |\operatorname{Re}Z|\le\theta_0\,|\operatorname{Im}Z|,\ |\operatorname{Im}Z|\ge y_{\min}\bigr\}. \tag{102}$$

Both signs of $\operatorname{Im}Z$ are allowed: the region is symmetric under conjugation, and every vertical strip $|\operatorname{Re}Z|\le h$, $|\operatorname{Im}Z|\ge h/\theta_0$ lies inside it.

Lemma A.2. *Let $Z\in\Omega_{\theta_0}$ and $M\ge0$. Then $(j+1)\le\sqrt2\,|Z-j|$ for every integer $j\ge0$, and*

$$\tau_M(Z):=\sum_{n\ge M+1}(n+1)!\,2^{-n/2}\prod_{j=M+1}^{n}|Z-j|^{-1}\ \le\ (M+1)!\,2^{-M/2}\,\bigl(18|Z|+4\bigr).$$

Proof. Write $a=\operatorname{Re}Z$, $b=|\operatorname{Im}Z|$, so $|a|\le\theta_0 b$ and $b\ge1/(1-\theta_0)$, i.e. $\theta_0 b+1\le b$. Then $2|Z-j|^2-(j+1)^2=2(j-a)^2+2b^2-(j+1)^2$; as a function of j this quadratic has positive leading coefficient and minimum value $2b^2-2(a+1)^2\ge2b^2-2(\theta_0 b+1)^2\ge0$, which proves $(j+1)\le\sqrt2|Z-j|$ for all $j\ge0$. Consequently the terms $t_n:=(n+1)!\,2^{-n/2}\prod_{j=M+1}^{n}|Z-$

$j|^{-1}$ satisfy $t_n/t_{n-1} = (n+1)/(\sqrt{2}\,|Z-n|) \le 1$, so $t_n \le t_M = (M+1)!\,2^{-M/2}$ for every $n > M$. Moreover for $n \ge 18|Z|$ one has $|Z-n| \ge n - |Z| \ge n(1-\frac{1}{18})$ and hence $t_n/t_{n-1} \le \frac{18\,(n+1)}{17\sqrt{2}\,n} \le \frac{4}{5}$ once $n \ge 18|Z| \ge 18$. Splitting the sum at $18|Z|$,

$$\tau_M \le t_M\big(18|Z|\big) + t_M \sum_{r\ge 1} (4/5)^r \le t_M(18|Z|+4). \qquad \square$$

Proposition A.3. *Fix $0 < \theta_0 < 1$. Then:*

(i) *the double series (36) converges absolutely for every $Z \notin \{1,2,3,\dots\}$ and every α; its grouping by $\mu+\nu = n$ is the series $F(\alpha;Z)$ above, which is therefore analytic in (α, Z) on $\mathbb{C} \times (\mathbb{C} \setminus \{1,2,\dots\})$ and satisfies (100) there;*

(ii) *for every N there are $C_N = C_N(\theta_0)$ and $y_{\min} = y_{\min}(\theta_0, N)$ such that, uniformly for $Z \in \Omega_{\theta_0}$ and $|\alpha| \le 1$,*

$$\Big| F(\alpha;Z) - \sum_{m=0}^{N} c_m(\alpha) Z^{-m} \Big| \le C_N\,|Z|^{-N-1},$$

where $c_m(\alpha) = \sum_{n\le m} G_n(\alpha) \left\{ {m \atop n} \right\}$ are the plain coefficients of Lemma 3.1; the same bound holds for $\partial_\alpha^k F$ and $\partial_\alpha^k c_m$, $k = 1, 2$;

(iii) *consequently, uniformly for $Z \in \Omega_{\theta_0}$ (*both *signs of* $\operatorname{Im} Z$*),*

$$\Big| \mathcal{A}(s) - \sum_{n=1}^{N} g_n Z^{-n} \Big| \le C'_N\,|Z|^{-N-1}.$$

Proof. (i) With $|x_0| = |y_0| = 2^{-1/2}$, the modulus of the (μ,ν)-term of (36) is at most $C(\alpha)\,\frac{\mu!\,\nu!}{|(1-Z)_{\mu+\nu}|}\,2^{-(\mu+\nu)/2}\mu^{O(1)}\nu^{O(1)}$ by Lemma A.1 applied to each Pochhammer pair, and $\mu!\,\nu!/(\mu+\nu)! = \binom{\mu+\nu}{\mu}^{-1} \le 1$ while $|(1-Z)_m| \ge c(Z)\,m!\,m^{-\operatorname{Re} Z - 1}$ for m large, by Stirling; the sum over $\mu+\nu = m$ then converges geometrically in m. Grouping the absolutely convergent double series by $\mu+\nu = n$ gives $G_n e_n$ exactly as in the proof of Lemma 7.2. Identity (100) holds for each fixed generic Z by Theorem 7.3 and (38), and both sides are analytic off the indicated sets, so it holds on the whole common domain.

(ii) Split $F = \sum_{n\le N} G_n e_n + \mathrm{Rem}_N$. Each e_n, $n \le N$, is analytic at $Z = \infty$ with $e_n = \sum_{m=n}^{N} \left\{ {m \atop n} \right\} Z^{-m} + O_N(|Z|^{-N-1})$ uniformly for $|Z| \ge 2N+2$ (finitely many geometric expansions of $Z^{-n}\prod_{j\le n}(1-j/Z)^{-1}$), which produces the polynomial part $\sum_{m\le N} c_m Z^{-m}$ with the stated coefficients plus an error $O_N(|Z|^{-N-1})$. For the tail, reserve the first $N+2$ factors: for $n \ge N+2$,

$$\prod_{j=1}^{n} |Z-j| \;\ge\; |\operatorname{Im} Z|^{N+2} \prod_{j=N+3}^{n} |Z-j|,$$

so by Lemmas A.1 and A.2 (with $M = N+2$),

$$|\mathrm{Rem}_N| \le \sum_{n=N+1}^{N+2} \frac{|G_n|}{|\operatorname{Im} Z|^n} + \frac{\cosh \pi}{|\operatorname{Im} Z|^{N+2}}\,\tau_{N+2}(Z)$$
$$\le \frac{C''_N}{|\operatorname{Im} Z|^{N+1}} + \frac{C'''_N\,|Z|}{|\operatorname{Im} Z|^{N+2}} \le C_N\,|Z|^{-N-1},$$

using $|\operatorname{Im} Z| \ge (1+\theta_0^2)^{-1/2}|Z|$ on Ω_{θ_0}. The same argument applies verbatim to $\partial_\alpha^k F = \sum_n \partial_\alpha^k G_n\, e_n$ after bounding $|\partial_\alpha^k G_n(\alpha)| \le C(n+1)!\,2^{-n/2}$ for $|\alpha| \le 2$ by the Cauchy

formula on the circle $|\alpha'-\alpha|=1$ applied to Lemma A.1; alternatively, apply Cauchy's formula directly to the remainder $\mathrm{Rem}_N(\alpha;Z)$, which is analytic in α.

(iii) F is even in α (each a_l is a function of α^2), so $\partial_\alpha F(0;Z)=0$ and, by (101), $\mathcal{A}=\frac{1}{2}\partial_\alpha^2 F(0;Z)/F(0;Z)$. By (ii), $F(0;Z)=1+O(|Z|^{-1})$ uniformly on Ω_{θ_0}, so for $y_{\min}$ large $|F(0;Z)|\geq\frac{1}{2}$ and the quotient of the two uniform expansions is again a uniform expansion, whose coefficients are, by Corollary 7.4, exactly g_n. □

Remark A.4. The cone $|\operatorname{Re} Z|\leq\theta_0|\operatorname{Im} Z|$ with $\theta_0<1$ cannot be widened by the termwise estimate above: for directions closer to the real axis the individual terms $G_n e_n$ with $n\asymp|Z|$ grow exponentially and the series converges only through cancellation. The applications of Lemma 7.27 in this paper do not require a wider aperture: the matching of Subsection A.6 is performed along a single vertical line, the decay statements of Subsection A.5 concern vertical strips, and the coefficients of a Poincaré expansion along one ray are unique. The mirror symmetry used repeatedly below is the Schwarz reflection $\mathcal{A}^{\downarrow}(s)=\overline{\mathcal{A}^{\uparrow}(\bar{s})}$, valid because all four blocks $\mathsf{S}_0,\mathsf{K}_0,\mathsf{S}_2,\mathsf{K}_2$ are real on the real axis.

A.2. **Uniform asymptotics of the closed blocks.**

Lemma A.5. *Uniformly on every closed subsector of $|\arg(\pm is)|<\pi/2$, i.e. off the real axis, and in particular on every vertical strip $|\operatorname{Re} s|\leq h$, $|\operatorname{Im} s|\geq 1$:*

(i) *$\psi'(s)$, $Q_{\mathrm{rat}}(s)$ and the convergent trigamma blocks of (80) are asymptotic to their (Bernoulli) Stirling series, with remainders $O_N(|s|^{-N})$ after N terms, and $|\psi'(s)|+|Q_{\mathrm{rat}}(s)|\leq C/|\operatorname{Im} s|$;*

(ii) $\tan\frac{\pi s}{2}=\pm i+O(e^{-\pi|\operatorname{Im} s|})$ *for* $\operatorname{Im} s\to\pm\infty$, *uniformly in* $\operatorname{Re} s$, *and likewise* $\cot\frac{\pi s}{2}=\mp i+O(e^{-\pi|\operatorname{Im} s|})$, $\sec^2\frac{\pi s}{2}$, $\csc^2(\pi s)=O(e^{-\pi|\operatorname{Im} s|})$;

(iii) $\widehat{\mathsf{r}}$ *of (53) is asymptotic to its Poincaré series (55) uniformly on $|\arg s|\leq\pi-\delta$, for every $\delta>0$.*

Proof. (i) and (iii) are Stirling's formula with remainder in sectors [2, §5.11] applied to the explicit Gamma quotients; the decay bound in (i) follows from $\psi'(z)=O(1/|z|)$ on $|\operatorname{Im} z|\geq\frac{1}{2}$ together with $\psi'(z+1)=\psi'(z)-z^{-2}$ to reach the arguments $(1-s)/2$, $1-s/2$. (ii) is elementary: $\tan\frac{\pi s}{2}\mp i=\mp 2i\,e^{\pm i\pi s}/(1+e^{\pm i\pi s})$, and $|e^{\pm i\pi s}|=e^{-\pi|\operatorname{Im} s|}$ in the indicated half-planes. □

A.3. **The Gamma forcing and its lattice sum.**

Lemma A.6. *Let $\widehat{\mathsf{f}}_\Gamma$ be (56). Then:*

(i) *there is C with $|\widehat{\mathsf{f}}_\Gamma(\varsigma)|\leq C\,|\varsigma|^{-2}$ uniformly on $|\operatorname{Im}\varsigma|\geq 1$, including $\operatorname{Re}\varsigma\to-\infty$;*

(ii) *for every M there is C_M such that, uniformly on $\operatorname{Im}\varsigma\geq 1$,*

$$\Big|\widehat{\mathsf{f}}_\Gamma(\varsigma)-\sum_{j=2}^{M}\varphi_j\,\varsigma^{-j}\Big|\leq C_M\big(|\varsigma|^{-M-1}+|\varsigma|^{-2}e^{-\pi\operatorname{Im}\varsigma}\big),$$

where $\sum_j\varphi_j\varsigma^{-j}$ is the formal series (56); the mirror statement holds for $\operatorname{Im}\varsigma\leq-1$ by $\widehat{\mathsf{f}}_\Gamma(\bar{\varsigma})=\overline{\widehat{\mathsf{f}}_\Gamma(\varsigma)}$.

Proof. Everything reduces to the Gamma quotient in $\widehat{\mathsf{r}}(\varsigma)=\varsigma\,[\Gamma(\varsigma/2)/\Gamma((1+\varsigma)/2)]^2$. On the right regime $|\arg\varsigma|\leq\frac{3}{4}\pi$, Stirling gives $\Gamma(\varsigma/2)/\Gamma((1+\varsigma)/2)=(\varsigma/2)^{-1/2}(1+O(1/\varsigma))$ uniformly, whence $|\widehat{\mathsf{r}}(\varsigma)|\leq C$ and the Poincaré expansion (55) with uniform remainders

(Lemma A.5(iii)). On the left regime $\frac{3}{4}\pi \le |\arg\varsigma| \le \pi$, $\operatorname{Im}\varsigma \ge 1$, apply the two reflection formulas used in the proof of Lemma 7.15:

$$\frac{\Gamma(\varsigma/2)}{\Gamma((1+\varsigma)/2)} = \cot\Big(\frac{\pi\varsigma}{2}\Big)\,\frac{\Gamma\big(\frac{1-\varsigma}{2}\big)}{\Gamma\big(1-\frac{\varsigma}{2}\big)}, \tag{103}$$

whose right-hand Gamma arguments have real part tending to $+\infty$. The Beta-integral bound, for $\operatorname{Re}x > 0$,

$$\Big|\frac{\Gamma(x)}{\Gamma(x+\frac12)}\Big| = \frac{|\mathrm{B}(x,\frac12)|}{\Gamma(\frac12)} \le \frac{\mathrm{B}(\operatorname{Re}x,\frac12)}{\sqrt{\pi}} \le C\,(\operatorname{Re}x)^{-1/2}$$

with $x = \frac{1-\varsigma}{2}$, together with $\cot(\pi\varsigma/2) = -i + O(e^{-\pi\operatorname{Im}\varsigma})$ (Lemma A.5(ii)), gives $|\widehat{\mathsf{r}}(\varsigma)| \le C$ on the left regime as well; (i) follows since $\widehat{\mathsf{f}}_\Gamma = -\frac{(2\varsigma-1)}{4\varsigma^2(\varsigma-1)}\widehat{\mathsf{r}}$. For (ii): on the left regime, (103) squared gives

$$\widehat{\mathsf{r}}(\varsigma) = -\varsigma\Big[\frac{\Gamma\big(\frac{1-\varsigma}{2}\big)}{\Gamma\big(1-\frac{\varsigma}{2}\big)}\Big]^2\big(1+O(e^{-\pi\operatorname{Im}\varsigma})\big),$$

and the squared quotient has a uniform Stirling expansion in ς^{-1} (right half-plane arguments). The resulting Poincaré series coincides with (55): both are asymptotic expansions of the same function $\widehat{\mathsf{r}}$ on the open overlap sector $\frac{3}{4}\pi < |\arg\varsigma| < \pi - \delta$ (where the cot correction is exponentially small in $|\varsigma|$), and the coefficients of a Poincaré expansion on a sector are unique. (Arithmetically the same coincidence is the parity identity of Lemma 7.31.) Combining the two regimes, with the exponentially small cot correction retained additively, gives (ii). □

Lemma A.7. *$\widehat{\mathsf{f}}_\Gamma$ is meromorphic on $\mathbb{C}$ with: double zeros at the negative odd integers; a pole of order 3 at $\varsigma = 0$, a simple pole at $\varsigma = 1$, double poles at the negative even integers, and no other poles. The principal-part coefficients at $\varsigma = -2m$ are $O(m^{-2})$. Consequently $T(s) = \sum_{k\ge0}\widehat{\mathsf{f}}_\Gamma(s-1-2k)$ converges normally on compact sets avoiding $\mathbb{Z}$ and defines a meromorphic function on $\mathbb{C}$ with poles contained in $\mathbb{Z}$; T has (at least) double zeros at the negative even integers, at most double poles at the negative odd integers, and satisfies*

$$|T(s)| \le \frac{C}{|\operatorname{Im}s|}, \qquad |\operatorname{Im}s| \ge 1, \tag{104}$$

uniformly in $\operatorname{Re}s$. Finally $T(s+2) - T(s) = \widehat{\mathsf{f}}_\Gamma(s+1)$.

Proof. $\Gamma((1+\varsigma)/2)^{-2}$ has double zeros at $\varsigma = -1, -3, \dots$; $\Gamma(\varsigma/2)^2$ has double poles at $\varsigma = 0, -2, -4, \dots$; combining with the elementary factor $-(2\varsigma-1)/\big(4\varsigma^2(\varsigma-1)\big)$ of (56) gives the stated orders (at $\varsigma = 0$: $\widehat{\mathsf{r}} \sim 4/(\pi\varsigma)$, so $\widehat{\mathsf{f}}_\Gamma \sim -1/(\pi\varsigma^3)$; at $\varsigma = 1$: $\widehat{\mathsf{r}}(1) = \pi$). At $\varsigma = -2m$ the double-pole coefficient of $\widehat{\mathsf{r}}$ is $-8m\,\Gamma(m+\frac12)^2/\big(\pi^2(m!)^2\big) = O(1)$ by the reflection formula, and the elementary factor contributes an extra $O(m^{-2})$. The term $\widehat{\mathsf{f}}_\Gamma(s-1-2k)$ of T has poles only at integers: at $s = 1+2k$, at $s = 2+2k$, and at the odd integers $s = 1+2k-2m$, $m \ge 1$, where it has a double pole with principal-part coefficients $O(m^{-2})$. At a fixed odd integer these coefficients are therefore summable in k, and away from small discs around the integers the terms are dominated by Lemma A.6(i); this gives normal convergence and meromorphy, and shows that T has at most double poles at the negative odd integers. The double zeros of T at $s = -2m$ hold because every term is then evaluated at a negative odd integer, where $\widehat{\mathsf{f}}_\Gamma$ has a double zero as a function of s. For (104): with $\operatorname{Im}s = Y$, $|Y| \ge 1$, all lattice points $\varsigma_k = s-1-2k$ have

$|\operatorname{Im}\varsigma_k| = |Y|$, so by Lemma A.6(i), $|T| \le C\sum_{k\ge0}\big(Y^2+(\operatorname{Re}s-1-2k)^2\big)^{-1} \le C'/|Y|$ by comparison with $\int_{-\infty}^{\infty}(Y^2+u^2)^{-1}du = \pi/|Y|$. The difference equation is immediate from the definition. □

Proposition A.8. *Let ω_j be the coefficients of the formal series $\widehat{Q}_\Gamma = (2\sinh\partial_s)^{-1}\widehat{\mathsf{f}}_\Gamma \in s^{-1}\mathbb{Q}[[s^{-1}]]$ of (57). For every $\delta>0$ and every N there are $C_{\delta,N}$ and $s_{\min}$ such that, uniformly on the sector $\operatorname{Im}s \ge \delta|s|$, $|s|\ge s_{\min}$,*

$$\Big|T(s) - \sum_{j=1}^{N}\omega_j\, s^{-j}\Big| \;\le\; C_{\delta,N}\,|s|^{-N-1} \;+\; C_\delta\,\frac{e^{-\pi\operatorname{Im}s}}{\operatorname{Im}s}.$$

The mirror statement holds on $\operatorname{Im}s \le -\delta|s|$ with the same (rational) coefficients, by Schwarz reflection.

Proof. Fix $M \ge N+2$. Insert Lemma A.6(ii) termwise:

$$T(s) = \sum_{j=2}^{M}\varphi_j\,\Sigma_j(s) + \mathrm{Err}_1 + \mathrm{Err}_2, \qquad \Sigma_j(s) := \sum_{k\ge0}(s-1-2k)^{-j},$$

with $|\mathrm{Err}_1| \le C_M\sum_{k\ge0}|s-1-2k|^{-M-1}$ and $|\mathrm{Err}_2| \le C\,e^{-\pi\operatorname{Im}s}\sum_{k\ge0}|s-1-2k|^{-2}$. For the error sums split at $k\le|s|$ and $k>|s|$: the near terms number at most $|s|+1$ and each is at most $(\operatorname{Im}s)^{-M-1} \le (\delta|s|)^{-M-1}$, giving $\le 2\delta^{-M-1}|s|^{-M}$; the far terms are dominated by $\sum_{k>|s|}(2k-1-|s|)^{-M-1} \le C|s|^{-M}$. Hence $|\mathrm{Err}_1| \le C'_{\delta,M}|s|^{-M}$, and similarly $|\mathrm{Err}_2| \le C_\delta\,e^{-\pi\operatorname{Im}s}/\operatorname{Im}s$ as in (104). Next, with $a=(1-s)/2$ (which satisfies $|\arg a| \le \pi-\delta'$, $\delta'=\delta'(\delta)>0$, on the sector),

$$\Sigma_j(s) = (-2)^{-j}\,\zeta(j,a),$$

and the Hurwitz zeta function has the classical uniform large-a expansion [2, §25.11(xii)]

$$\zeta(j,a) = \frac{a^{1-j}}{j-1} + \frac{a^{-j}}{2} + \sum_{r=1}^{R}\binom{j+2r-2}{2r-1}\frac{\mathcal{B}_{2r}}{2r}\,a^{-j-2r+1} + O_{j,R}\big(|a|^{-j-2R-1}\big)$$

uniformly on $|\arg a| \le \pi-\delta'$. Substituting these finitely many expansions (each truncated at total order $N+1$ in s^{-1}) and re-expanding $a=(1-s)/2$ in powers of s^{-1} produces a fixed formal series $\widehat{T} = \sum_j \hat\omega_j s^{-j}$, with no constant term (the leading contribution of Σ_j is of order s^{1-j} and φ_j starts at $j=2$), such that $T(s) = \sum_{j\le N}\hat\omega_j s^{-j} + O_{\delta,N}(|s|^{-N-1}) + O_\delta(e^{-\pi\operatorname{Im}s}/\operatorname{Im}s)$ on the sector. It remains to identify $\widehat{T} = \widehat{Q}_\Gamma$. The exact difference equation $T(s+2)-T(s) = \widehat{\mathsf{f}}_\Gamma(s+1)$ (Lemma A.7), the uniform expansion just proved (applied at s and $s+2$, both in the sector), and Lemma A.6(ii) at $\varsigma = s+1$ force the coefficientwise identity $\widehat{T}(s+2)-\widehat{T}(s) = \widehat{\mathsf{f}}_\Gamma$-series$(s+1)$; equivalently, in the shifted variable $u=s+1$, $\widehat{T}(u+1)-\widehat{T}(u-1)$ equals the series (56) at u. By Proposition 7.12 the unique solution of this equation in $s^{-1}\mathbb{Q}[[s^{-1}]]$ is $\widehat{Q}_\Gamma$, so $\widehat{T}=\widehat{Q}_\Gamma$ and $\hat\omega_j=\omega_j$. □

Propositions A.3 and A.8 and Lemma A.5 together prove Lemma 7.27, in the uniform quantitative form stated there.

A.4. **The two-component relation and vertical decay.** With $\mathsf{S}_0 = \mathsf{S}(0;s)$, $\mathsf{S}_2 = \partial_\alpha^2 \mathsf{S}(\alpha;s)|_{\alpha=0}$ and K_0, K_2 as in Theorem 7.9, define the two boundary components

$$\mathcal{A}^{\uparrow}(s) = -\frac{\psi'(s)}{4} + \frac{\mathsf{S}_2 + i\pi\mathsf{K}_2}{2(\mathsf{S}_0 + i\pi\mathsf{K}_0)}, \qquad \mathcal{A}^{\downarrow}(s) = -\frac{\psi'(s)}{4} + \frac{\mathsf{S}_2 - i\pi\mathsf{K}_2}{2(\mathsf{S}_0 - i\pi\mathsf{K}_0)},$$

meromorphic functions of s with $\mathcal{A}^{\uparrow} = \mathcal{A}$ (the upper boundary determination), $\mathcal{A}^{\downarrow}(s) = \overline{\mathcal{A}^{\uparrow}(\bar{s})}$, and $\mathcal{A}^{\downarrow} = \overline{\mathcal{A}}$ on the real axis.

Lemma A.9. *As an identity of meromorphic functions on* $\mathbb{C}$,

$$\Big(\tan\frac{\pi s}{2} - i\Big)\mathcal{A}^{\uparrow}(s) + \Big(\tan\frac{\pi s}{2} + i\Big)\mathcal{A}^{\downarrow}(s) = \tan\frac{\pi s}{2}\,\Big(Q(s) - \frac{\psi'(s)}{2}\Big), \tag{105}$$

with $Q = \mathsf{K}_2/\mathsf{K}_0$ *as in Theorem 7.9.*

Proof. Write $\mathfrak{t} = \tan\frac{\pi s}{2} = \pi\mathsf{K}_0/\mathsf{S}_0$ (Lemma 7.6). From $1 + i\mathfrak{t} = i(\mathfrak{t} - i)$ and $1 - i\mathfrak{t} = -i(\mathfrak{t} + i)$,

$$\frac{\mathfrak{t} - i}{1 + i\mathfrak{t}} = -i, \qquad \frac{\mathfrak{t} + i}{1 - i\mathfrak{t}} = i.$$

Hence

$$(\mathfrak{t} - i)\,\frac{\mathsf{S}_2 + i\pi\mathsf{K}_2}{\mathsf{S}_0(1 + i\mathfrak{t})} + (\mathfrak{t} + i)\,\frac{\mathsf{S}_2 - i\pi\mathsf{K}_2}{\mathsf{S}_0(1 - i\mathfrak{t})} = \frac{-i(\mathsf{S}_2 + i\pi\mathsf{K}_2) + i(\mathsf{S}_2 - i\pi\mathsf{K}_2)}{\mathsf{S}_0} = \frac{2\pi\mathsf{K}_2}{\mathsf{S}_0} = 2\mathfrak{t}\,Q,$$

and adding the $-\psi'/4$ parts, whose coefficients sum to $2\mathfrak{t}$, gives (105). □

Corollary A.10. *For every* $h > 0$ *there are* $C(h)$ *and* $y'_{\min}(h)$ *with*

$$|Q(s)| \le \frac{C(h)}{|\operatorname{Im} s|}, \qquad |\operatorname{Re} s| \le h,\ |\operatorname{Im} s| \ge y'_{\min}(h),$$

uniformly on the strip; in particular $Q \to 0$ *at both vertical ends of every strip, uniformly in* $\operatorname{Re} s$ *on compact sets.*

Proof. Solve (105) for Q:

$$Q = \frac{\psi'(s)}{2} + \frac{(\mathfrak{t} - i)\mathcal{A}^{\uparrow} + (\mathfrak{t} + i)\mathcal{A}^{\downarrow}}{\mathfrak{t}}.$$

The vertical strip $|\operatorname{Re} s| \le h$, $|\operatorname{Im} s| \ge y'_{\min}$ maps under $Z = 1 - 2s$ into $\Omega_{1/2}$ once $y'_{\min} \ge 2(2h+1)$ and $y'_{\min}$ is at least the threshold of Proposition A.3, so Proposition A.3(iii) with $N = 1$ gives $|\mathcal{A}^{\uparrow}(s)| \le C/|Z| \le C'/|\operatorname{Im} s|$ there; the same bound holds for $\mathcal{A}^{\downarrow}(s) = \overline{\mathcal{A}^{\uparrow}(\bar{s})}$ since $\bar{s}$ lies in the mirror strip. By Lemma A.5, $|\psi'(s)| \le C/|\operatorname{Im} s|$, $|\mathfrak{t} \mp i| \le 3e^{-\pi|\operatorname{Im} s|}$ (for $\operatorname{Im} s \to \pm\infty$ respectively) and $|\mathfrak{t}| \ge \frac{1}{2}$ for $|\operatorname{Im} s| \ge 1$. The claim follows. □

A.5. **The classification lemma and the vanishing of the lattice constant.**

Lemma A.11. *Let* f *be meromorphic on* $\mathbb{C}$, *2-periodic, with all poles contained in* $\mathbb{Z}$, *of order at most* k_1 *at the odd integers and at most* k_0 *at the even integers, and suppose that* $f(s) \to 0$ *as* $\operatorname{Im} s \to +\infty$ *and as* $\operatorname{Im} s \to -\infty$, *uniformly for* $\operatorname{Re} s \in [0,2]$. *Then*

$$f(s) = \frac{w\,n_1(w)}{(w+1)^{k_1}\,(w-1)^{k_0}}, \qquad w = e^{i\pi s},$$

with n_1 *a polynomial of degree at most* $k_1 + k_0 - 2$ *(and* $f \equiv 0$ *when* $k_1 + k_0 \le 1$*). In particular, if* $k_1 \le 1$ *and* $k_0 = 0$ *then* $f \equiv 0$; *if* $k_1 = 2$ *and* $k_0 = 0$ *then* $f \in \mathbb{C}\cdot\sec^2\frac{\pi s}{2}$.

Proof. The map $s \mapsto w = e^{i\pi s}$ identifies the cylinder $\mathbb{C}/2\mathbb{Z}$ with $\mathbb{C}^*$; a 2-periodic meromorphic function f is therefore of the form $f(s) = R(e^{i\pi s})$ with R meromorphic on $\mathbb{C}^*$. The poles of f correspond to $w = -1$ (odd integers) and $w = +1$ (even integers), with the stated orders, since $w \mp 1$ vanishes to first order in s at the corresponding integers. The end $\operatorname{Im} s \to +\infty$ is $w \to 0$ and the end $\operatorname{Im} s \to -\infty$ is $w \to \infty$; the hypothesis says that $R(w) \to 0$ as $w \to 0$ and as $w \to \infty$ (uniform decay on one period suffices, every w near 0 or ∞ having a preimage with $\operatorname{Re} s \in [0, 2]$). Hence R is bounded near 0 and ∞, these singularities are removable with value 0, and R is meromorphic on the Riemann sphere, that is, rational, with poles only at -1 and $+1$, of orders at most k_1 and k_0. Thus $R = n_0(w)/((w+1)^{k_1}(w-1)^{k_0})$ with a polynomial n_0 of degree at most $k_1 + k_0 - 1$ (zero at ∞) and $n_0(0) = 0$ (zero at 0): hence $n_0 = w\, n_1$ with $\deg n_1 \le k_1 + k_0 - 2$. For $k_1 = 2$, $k_0 = 0$ this is $R = c\, w/(w+1)^2$, and $4w/(w+1)^2 = \sec^2 \frac{\pi s}{2}$. □

Theorem A.12. *Define*

$$\mathcal{D}(s) := Q(s) - Q_{\mathrm{rat}}(s) + \cot \frac{\pi s}{2}\, T(s).$$

Then $\mathcal{D} \equiv 0$. Equivalently, in the decomposition (60) one has $c_*(s_0) = 0$ for every admissible step-two lattice, real or complex, and (62) holds with $c_*(s_0) = 0$.

Proof. $\mathcal{D}$ is meromorphic with poles in $\mathbb{Z}$. $\mathsf{K}_0 = \Gamma(1+\frac{s}{2})/(\Gamma(1+s)\Gamma(1-\frac{s}{2}))$ is entire, with zeros exactly at the negative odd integers and the positive even integers. K_2 is analytic on $\operatorname{Re} s < \frac{1}{2}$: the defining series converges locally uniformly there; at a nonpositive integer $s = -m$ the only singular inner terms are those with $j = m$, present exactly for $n \ge m+1$, and there the double zero of $((s)_n/n!)^2$ at $s = -m$ cancels the double pole of $(s+m)^{-2}$, while the finitely many terms with $n \le m$ contain no factor $(s+m)^{-2}$. It extends meromorphically to $\mathbb{C}$ by the recurrence (46), whose coefficients vanish only at integers. Hence $Q = \mathsf{K}_2/\mathsf{K}_0$ is meromorphic with poles in $\mathbb{Z}$; so are Q_{rat}, $\cot \frac{\pi s}{2}$ and T (Lemma A.7).

$\mathcal{D}$ is 2-periodic. By (47) at $s+1$, $Q(s+2) - Q(s) = \mathsf{f}(s+1)$; by the trigamma recurrence, $Q_{\mathrm{rat}}(s+2) - Q_{\mathrm{rat}}(s) = \mathsf{f}_{\mathrm{rat}}(s+1)$; by Lemma A.7, $T(s+2) - T(s) = \widehat{\mathsf{f}}_\Gamma(s+1)$; and $\tan \frac{\pi(s+1)}{2} = -\cot \frac{\pi s}{2}$, so $\mathsf{f}_\Gamma(s+1) = -\cot \frac{\pi s}{2}\, \widehat{\mathsf{f}}_\Gamma(s+1)$ (Lemma 7.15). Hence

$$\mathcal{D}(s+2) - \mathcal{D}(s) = \mathsf{f}(s+1) - \mathsf{f}_{\mathrm{rat}}(s+1) + \cot \frac{\pi s}{2}\, \widehat{\mathsf{f}}_\Gamma(s+1) = \mathsf{f}_\Gamma(s+1) + \cot \frac{\pi s}{2}\, \widehat{\mathsf{f}}_\Gamma(s+1) = 0.$$

Pole orders. Because $\mathcal{D}$ is 2-periodic, the pole order on each residue class may be read off at any one representative. At $s = -2m$, $m \ge 1$: $\mathsf{K}_0(-2m) = (-1)^m \binom{2m}{m} \ne 0$ and K_2 is analytic there, so Q is analytic; Q_{rat} is analytic (its trigamma arguments are $\frac{1}{2}+m$ and $1+m$); and the simple pole of $\cot \frac{\pi s}{2}$ is cancelled by the double zero of T (Lemma A.7). Hence $\mathcal{D}$ is analytic on the even class. At $s = -(2m+1)$: K_0 has a simple zero and K_2 is analytic, so Q has at most a simple pole; Q_{rat} is analytic; and $\cot \frac{\pi s}{2}$ has a simple zero, while T has at most a double pole. Hence $\mathcal{D}$ has at most a simple pole on the odd class.

Decay at the ends. On the strip $\operatorname{Re} s \in [0, 2]$, $|\operatorname{Im} s| \ge y'_{\min}$, there is a constant c such that $|Q| \le c/|\operatorname{Im} s|$ by Corollary A.10, $|Q_{\mathrm{rat}}| \le c/|\operatorname{Im} s|$ by Lemma A.5(i), and $|T| \le c/|\operatorname{Im} s|$ by (104), while $\cot \frac{\pi s}{2}$ is bounded. Hence $\mathcal{D}(s) \to 0$ uniformly on the strip as $\operatorname{Im} s \to \pm\infty$.

By Lemma A.11 with $k_1 = 1$, $k_0 = 0$, $\mathcal{D} \equiv 0$. On any step-two lattice $\{s_0 - 2k\}$ avoiding the poles, comparison with (60) gives $c_*(s_0) = \mathcal{D}(s_0) = 0$. □

Corollary A.13. *In Proposition 7.17 the remainder (62) is* $\mathcal{E}(s) = \frac{1}{2}Q_{\mathrm{rat}}(s) - \operatorname{Re}\mathcal{A} - \frac{1}{4}\psi'(s)$*, with no lattice term; and the two-component relation takes the closed form*

$$\Big(\tan\frac{\pi s}{2} - i\Big)\mathcal{A}^{\uparrow} + \Big(\tan\frac{\pi s}{2} + i\Big)\mathcal{A}^{\downarrow} = -T(s) + \tan\frac{\pi s}{2}\left(Q_{\mathrm{rat}}(s) - \frac{\psi'(s)}{2}\right) =: X(s). \tag{106}$$

Proof. Insert $Q = Q_{\mathrm{rat}} - \cot\frac{\pi s}{2}\,T$ (Theorem A.12) into (105) and use $\tan\cdot\cot = 1$. □

A.6. **Proof of Theorem 1.6.** One more formal lemma is needed. Let $S_\psi(s) = s^{-1} + \frac{1}{2}s^{-2} + \sum_{k\ge1}\mathcal{B}_{2k}s^{-2k-1}$ denote the Stirling series of ψ'.

Lemma A.14. *As an identity in* $\mathbb{Q}[[s^{-1}]]$*, with the left side re-expanded in powers of* s^{-1}*,*

$$S_\psi(1-s) = -S_\psi(s).$$

Proof. Work in the Borel (Watson) picture, in which $\sum_n c_n s^{-n-1}$ corresponds to the kernel $\sum_n c_n t^n/n!$ and a shift $s \mapsto s-a$ corresponds, exactly and coefficientwise, to multiplication of the kernel by e^{at}. The kernel of S_ψ is $\varphi_\psi(t) = t/(1-e^{-t})$. First, $\varphi_\psi(t)(1-e^t) = -te^t$ is the kernel of $-\sum_{n\ge1} n s^{-n-1} = -(s-1)^{-2}$, i.e.

$$S_\psi(s) - S_\psi(s-1) = -(s-1)^{-2} \tag{107}$$

coefficientwise. Second, directly from the parity of the coefficients, $S_\psi(-x)+S_\psi(x) = x^{-2}$. Applying this at $x = s-1$ and then (107),

$$S_\psi(1-s) = -S_\psi(s-1) + (s-1)^{-2} = -\big[S_\psi(s) + (s-1)^{-2}\big] + (s-1)^{-2} = -S_\psi(s).$$

□

Proof of Theorem 1.6. All expansions below are along the vertical line $s = -\frac{1}{2} + iY$, $Y \to -\infty$; recall that the coefficients of a Poincaré expansion along a single ray are unique, so establishing (13) there establishes it as an identity of formal series. The line lies, for $Z = 1-2s$, in $\Omega_{1/2}$, and in the sector $\operatorname{Im} s \le -\frac{1}{2}|s|$ for $|Y| \ge 1$.

By Proposition A.3(iii),

$$\mathcal{A}^{\uparrow}(s) = \sum_{n=1}^{N} g_n Z^{-n} + O(|Z|^{-N-1}), \tag{108}$$

and $\mathcal{A}^{\downarrow}(s) = \overline{\mathcal{A}^{\uparrow}(\bar s)} = O(|Z|^{-1})$ since $\bar s$ lies in the mirror line. By Lemma A.5(ii), $\tan\frac{\pi s}{2} + i = O(e^{-\pi|Y|})$ and $\tan\frac{\pi s}{2} - i = -2i + O(e^{-\pi|Y|})$. The right side of (106) is $O(1/|Y|)$ by (104) and Lemma A.5(i). Hence, solving (106) for $\mathcal{A}^{\uparrow}$,

$$\mathcal{A}^{\uparrow}(s) = \frac{X(s)}{-2i} + O\big(e^{-\pi|Y|}\big) = \frac{i}{2}X(s) + O\big(e^{-\pi|Y|}\big). \tag{109}$$

Now expand X. By the mirror statement of Proposition A.8 (with $\delta = \frac{1}{2}$), $T(s) = \sum_{j\le N}\omega_j s^{-j} + O(|s|^{-N-1}) + O(e^{-\pi|Y|})$, with $\sum\omega_j s^{-j} = \widehat{Q}_\Gamma$; by Lemma A.5(i) and Corollary 7.13, Q_{rat} and ψ' are asymptotic to their Stirling series, the former to the unique formal solution of its central-difference equation; and $\tan\frac{\pi s}{2} = -i + O(e^{-\pi|Y|})$. Therefore

$$X(s) = -\widehat{Q}_\Gamma(s) - i\Big(Q_{\mathrm{rat}}^{\mathrm{ser}}(s) - \tfrac{1}{2}S_\psi(s)\Big) + O(|s|^{-N-1}) + O(e^{-\pi|Y|})$$

in the obvious truncated sense, where $Q_{\mathrm{rat}}^{\mathrm{ser}}$ denotes the (rational-coefficient) Stirling series of (52). Combining with (108)–(109) and equating coefficients along the line,

$$\sum_{n\geq 1} g_n Z^{-n} = \frac{1}{2}\, Q_{\mathrm{rat}}^{\mathrm{ser}}(s) - \frac{1}{4}\, S_\psi(s) - \frac{i}{2}\, \widehat{Q}_\Gamma(s), \tag{110}$$

all three series on the right being re-expanded in Z^{-1} through the exact substitution $s = (1-Z)/2$. The three series have rational coefficients, so the splitting of (110) into real and imaginary parts is coefficientwise: $\operatorname{Im} g_n = -\frac{1}{2}\widehat{q}_n$, which is (14), and $\sum \operatorname{Re} g_n Z^{-n} = \frac{1}{2}Q_{\mathrm{rat}}^{\mathrm{ser}} - \frac{1}{4}S_\psi(s)$. For the real part, $\frac{1}{2}Q_{\mathrm{rat}}^{\mathrm{ser}} = \frac{1}{16}\psi'\big(\frac{Z+1}{4}\big) - \frac{1}{16}\psi'\big(\frac{Z+3}{4}\big)$ as Bernoulli series (the arguments $(1-s)/2$ and $1-s/2$ equal $(Z+1)/4$ and $(Z+3)/4$), while Lemma A.14 gives $-\frac{1}{4}S_\psi(s) = \frac{1}{4}S_\psi(1-s) = \frac{1}{4}\psi'\big(\frac{Z+1}{2}\big)$ as Bernoulli series ($1 - s = (Z+1)/2$); so the real part of (110) is exactly (79), in agreement with Theorem 7.26. This proves (13). □

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